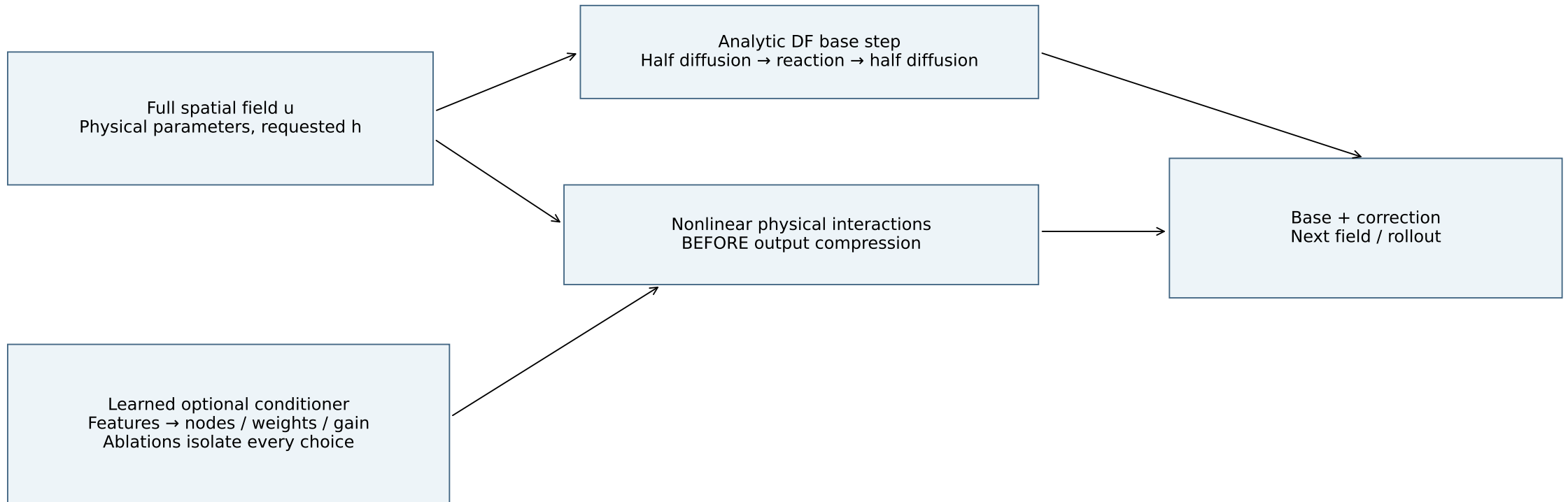


Architecture and information flow



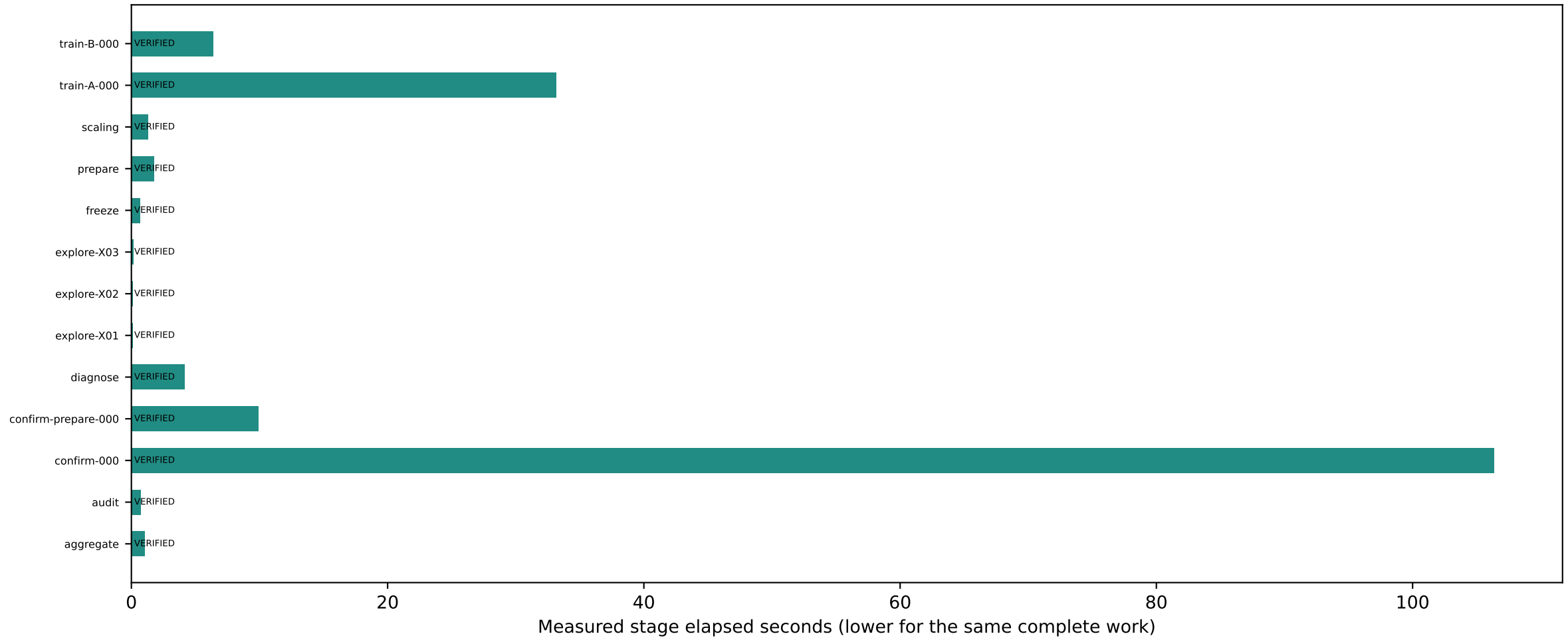
A: normalized controls and attribution B: grounded changes C: independent exploratory formulations

HOW TO READ: Arrows show information and computation, not an accuracy ranking.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Schematic for the interaction-correction family. Competitors and high-risk kernels/closures have different computational graphs; exact configurations and parameter reports remain in chart data.

Stage completion and complete campaign cost

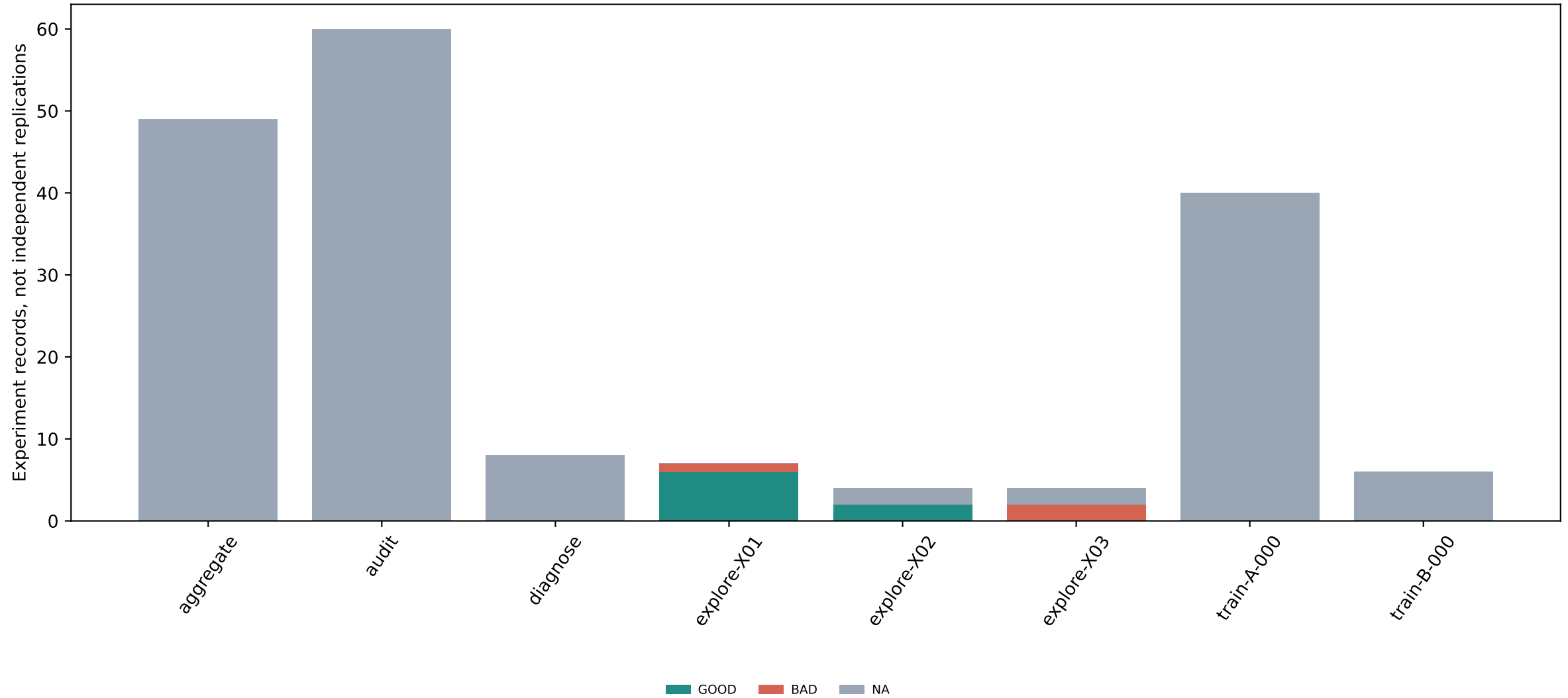


HOW TO READ: Shorter measured time is cheaper only for the same completed work. NA/failed stages are not zero-cost successes.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Measured numerical stage time is separate from scheduler allocation and accounting. Failed/interrupted attempt costs may be unavailable; no monetary rate is assumed.

Every experiment outcome including failures

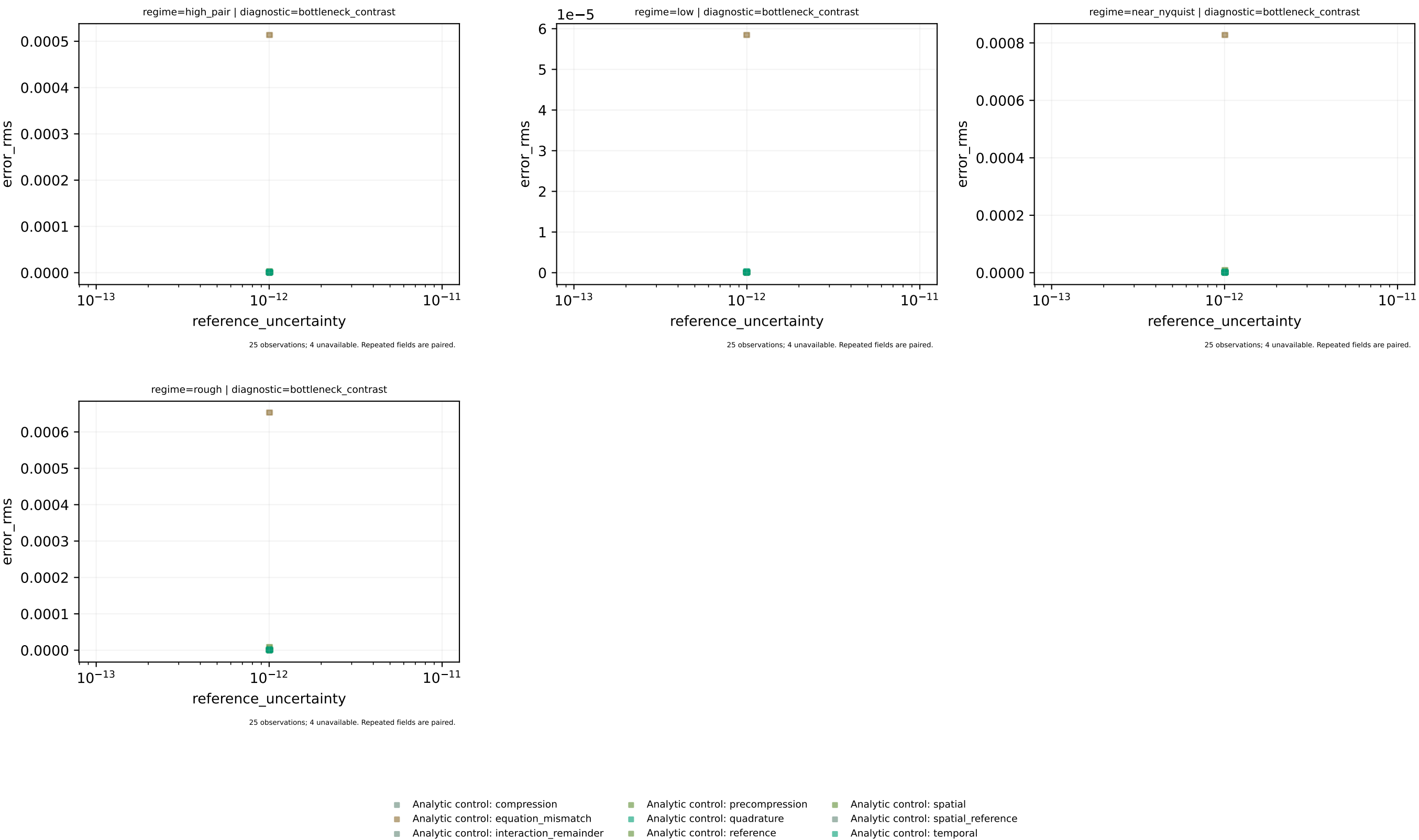


HOW TO READ: GOOD means declared checks met; BAD means failed checks; NA means unresolved. Taller bars are not better models.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

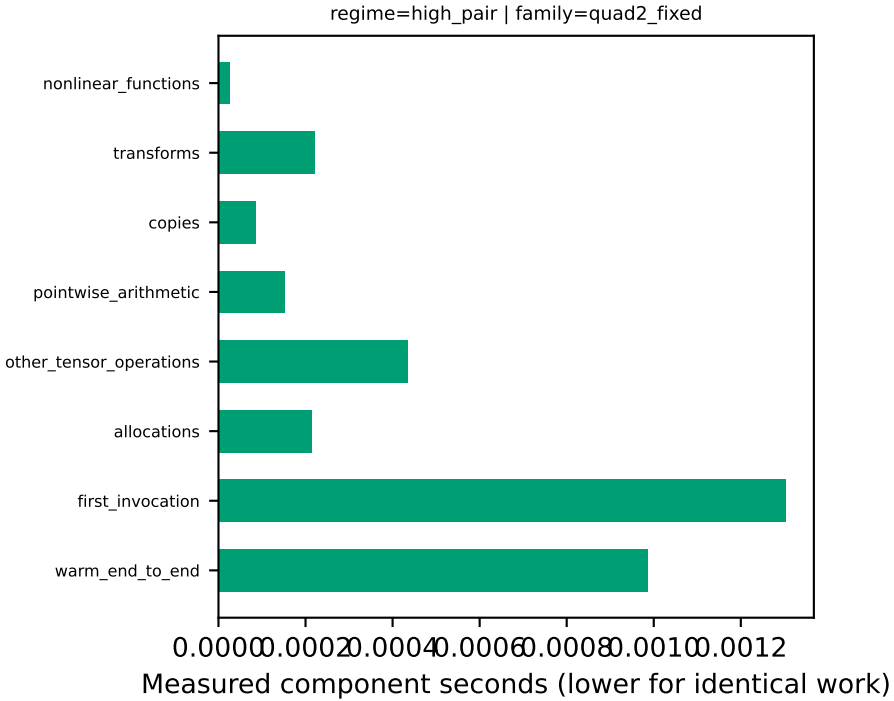
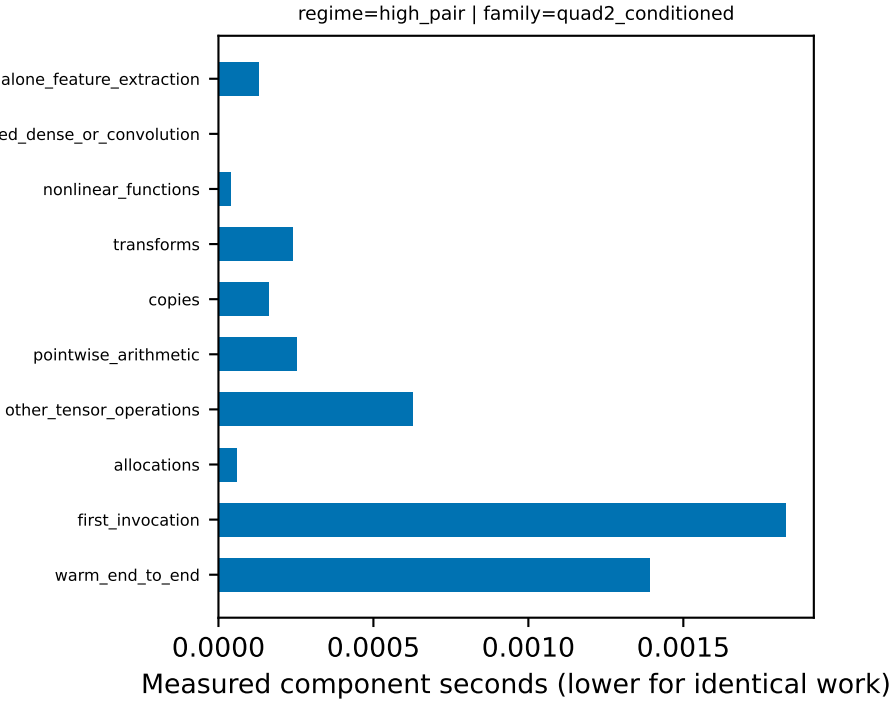
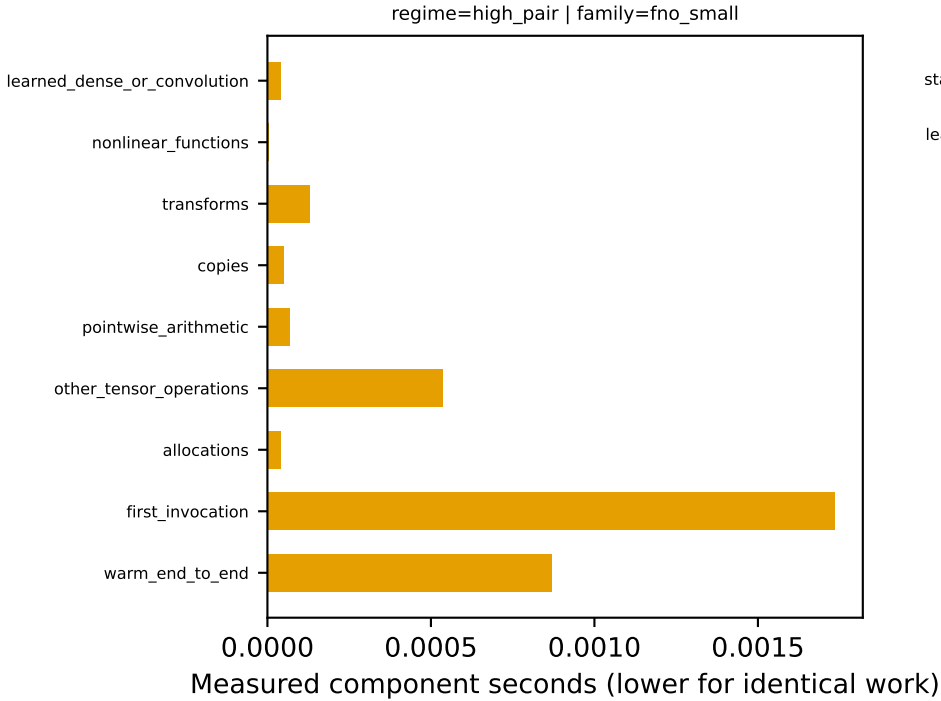
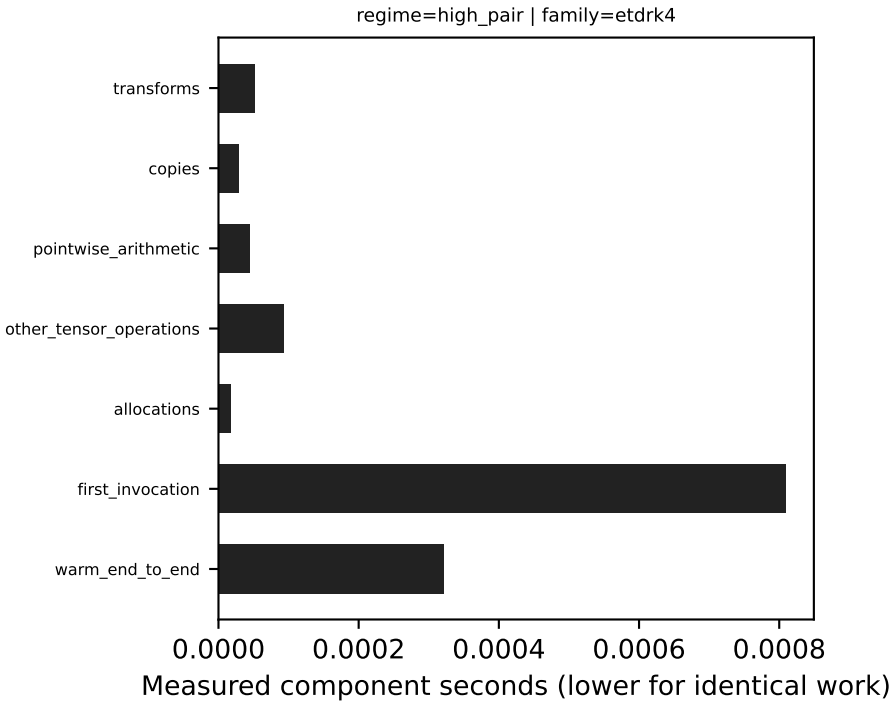
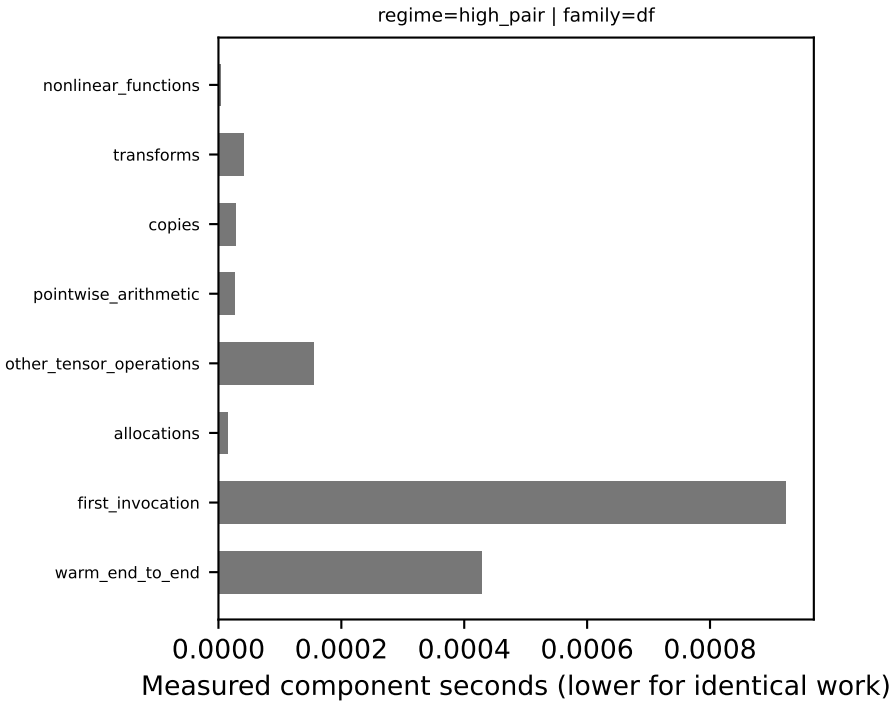
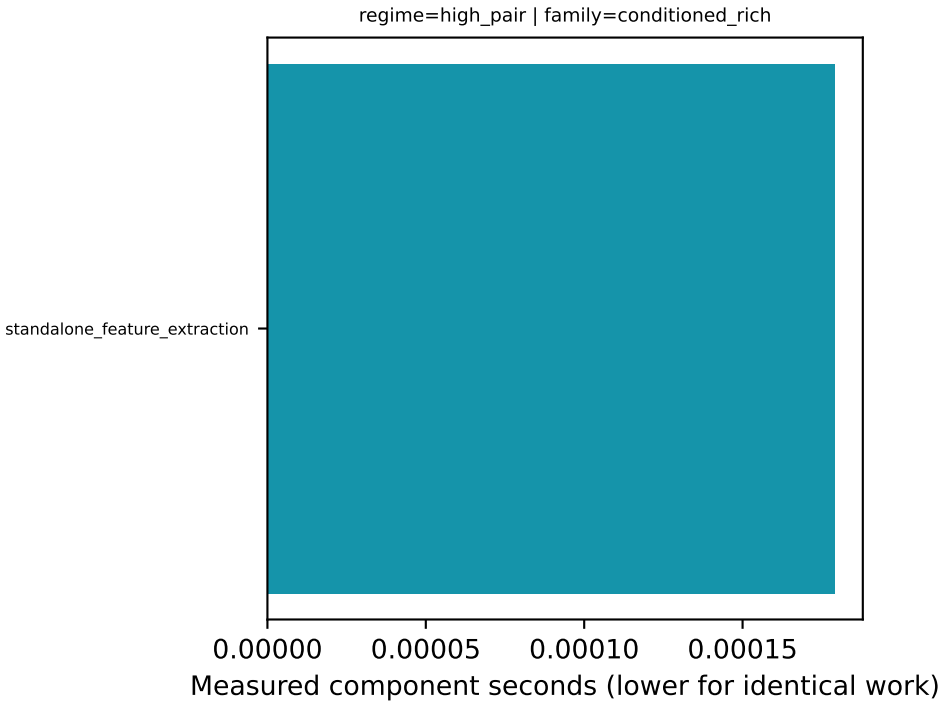
SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Diagnosing the dominant error source



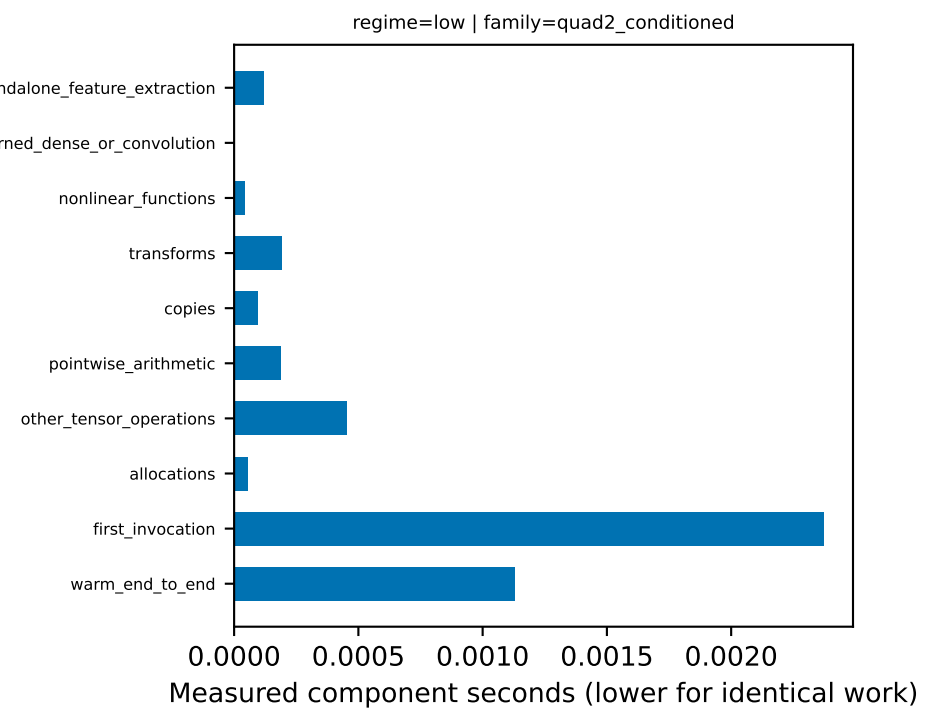
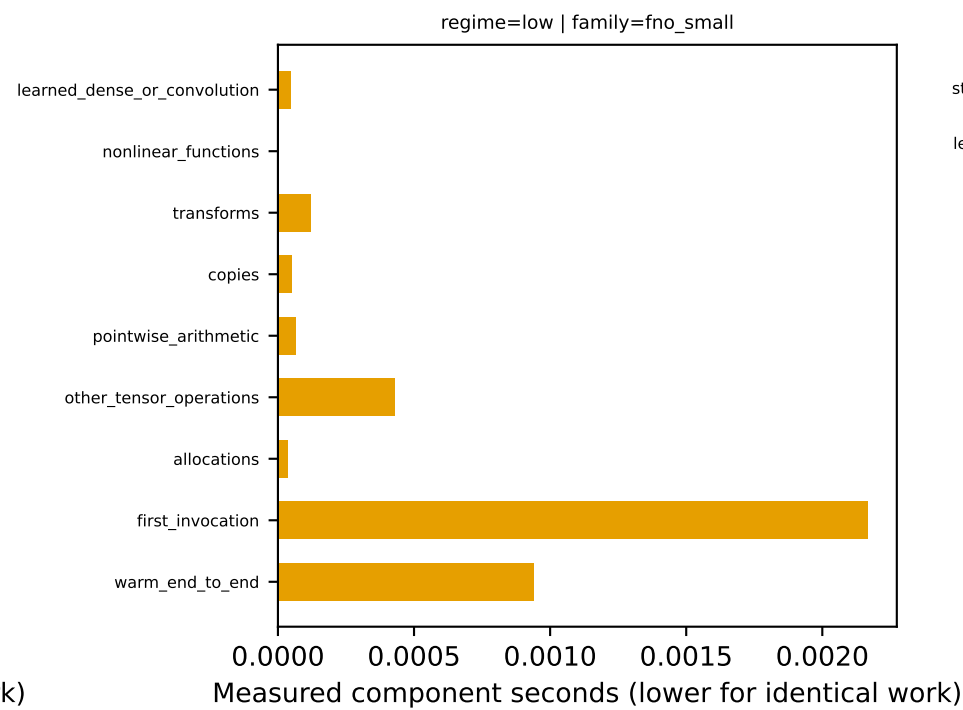
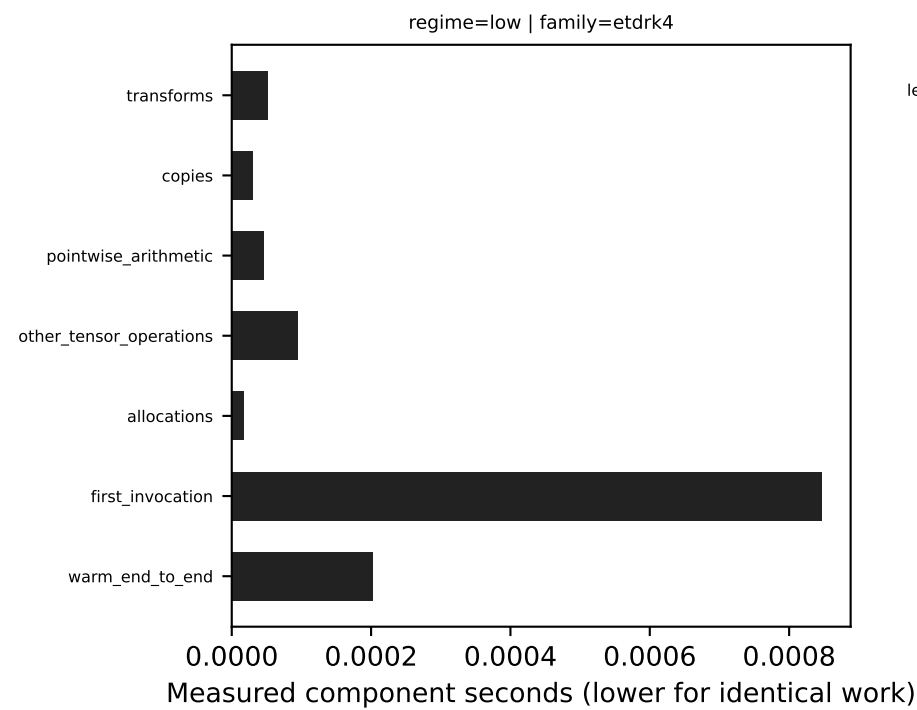
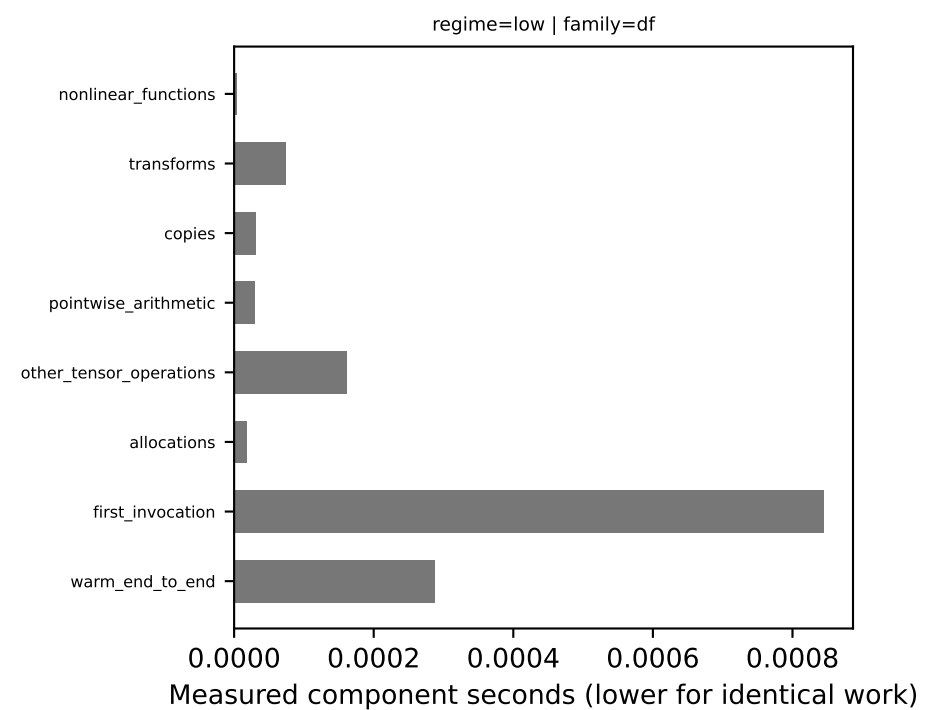
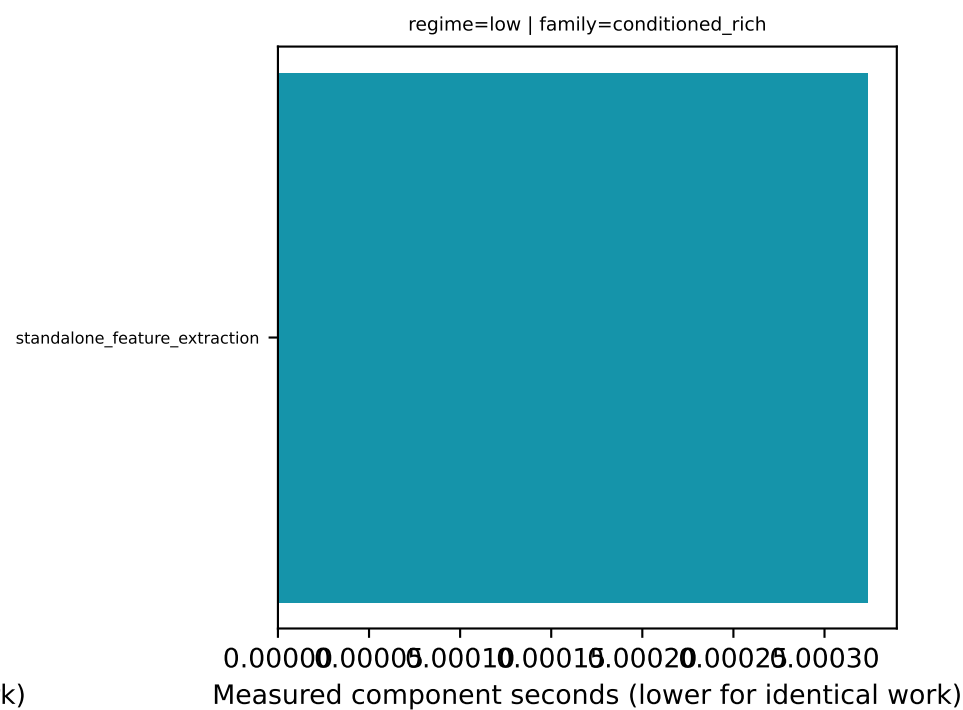
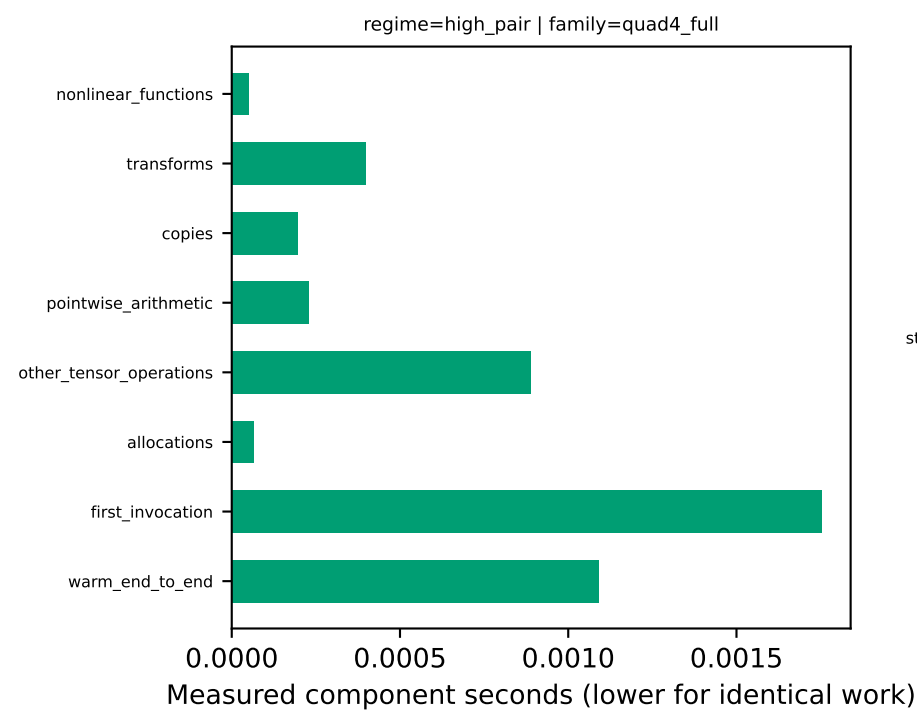
HOW TO READ: Error is favorable when smaller, but agreement below reference uncertainty is unresolved. Compare the explicitly named component and regime.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Discrete dynamics and continuum compensation are separate targets. Temporal, quadrature, compression and reference-refinement probes are diagnostic, not independent final-solver comparisons.

Runtime component profiling (1)



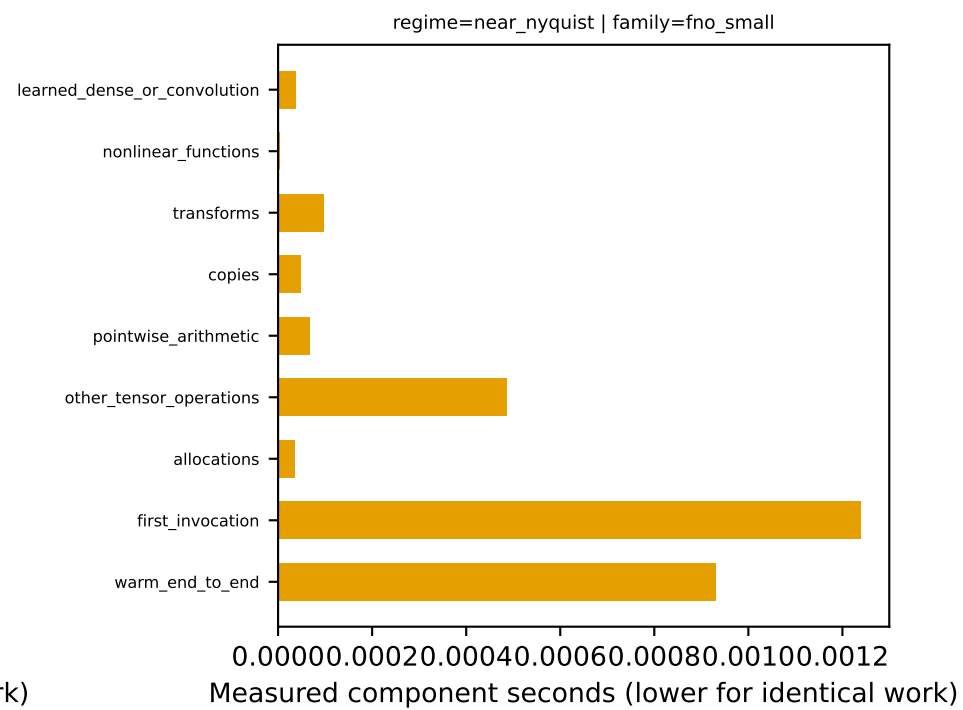
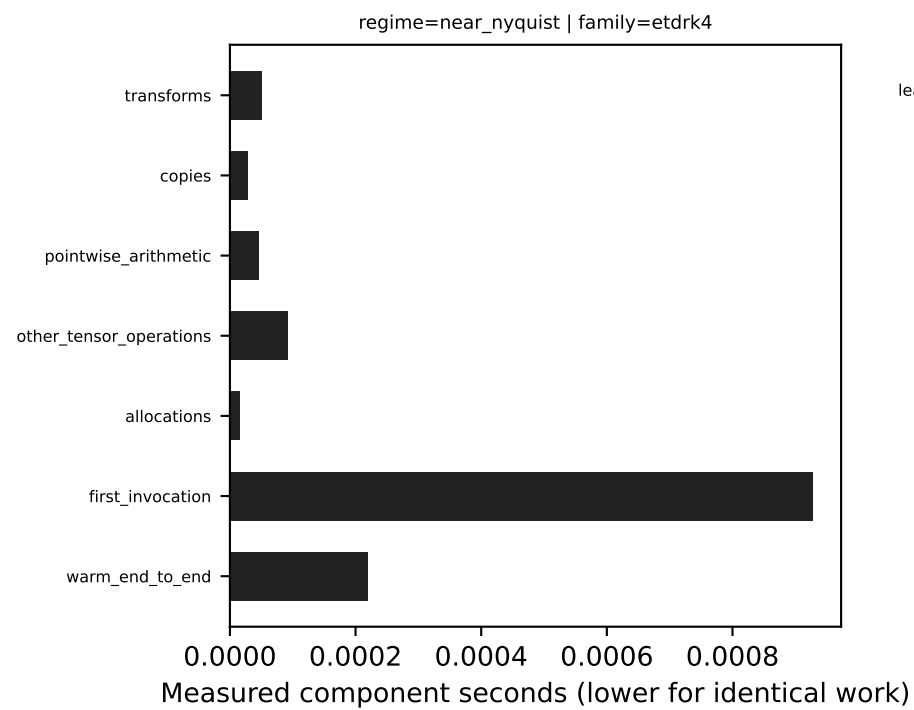
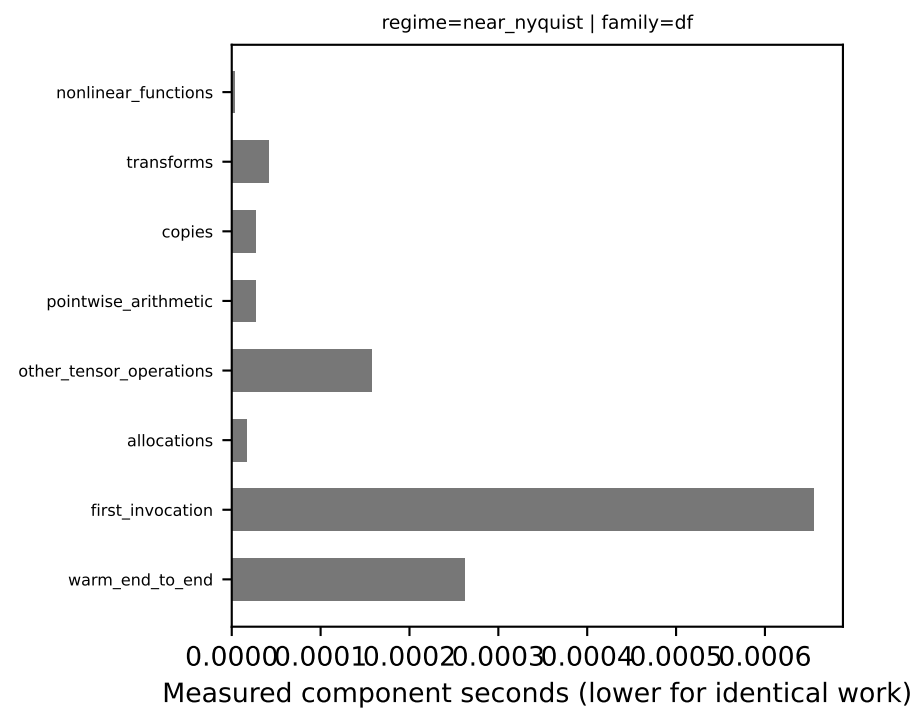
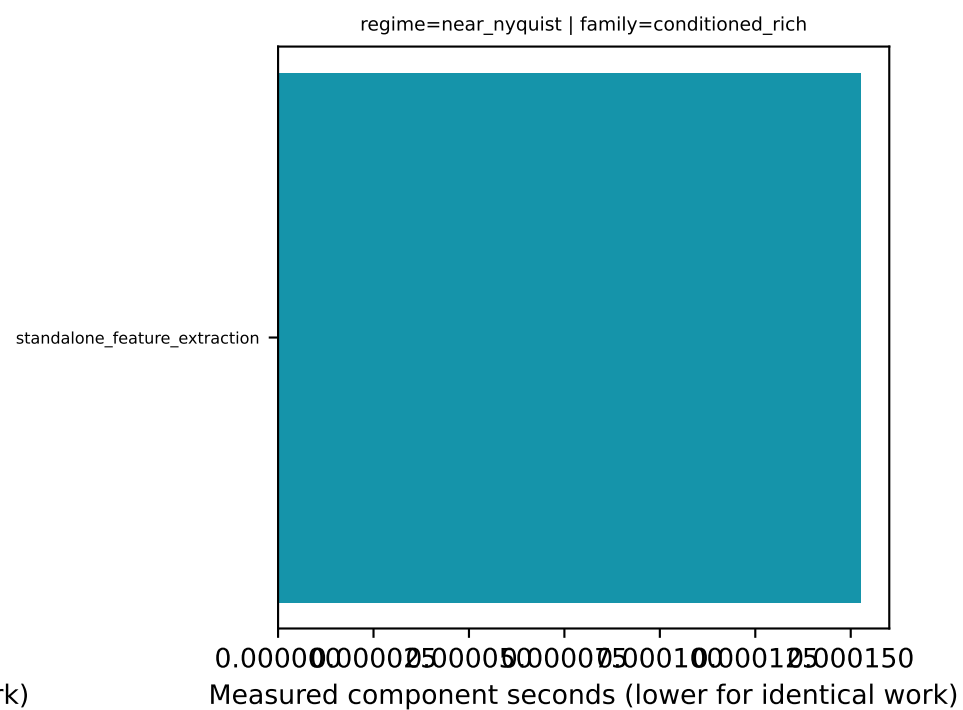
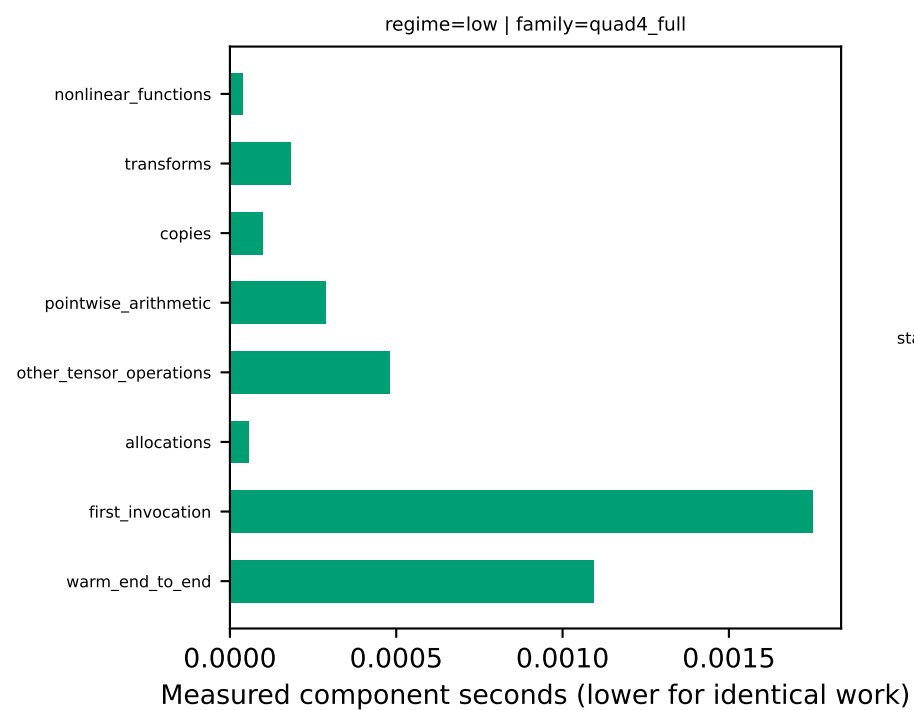
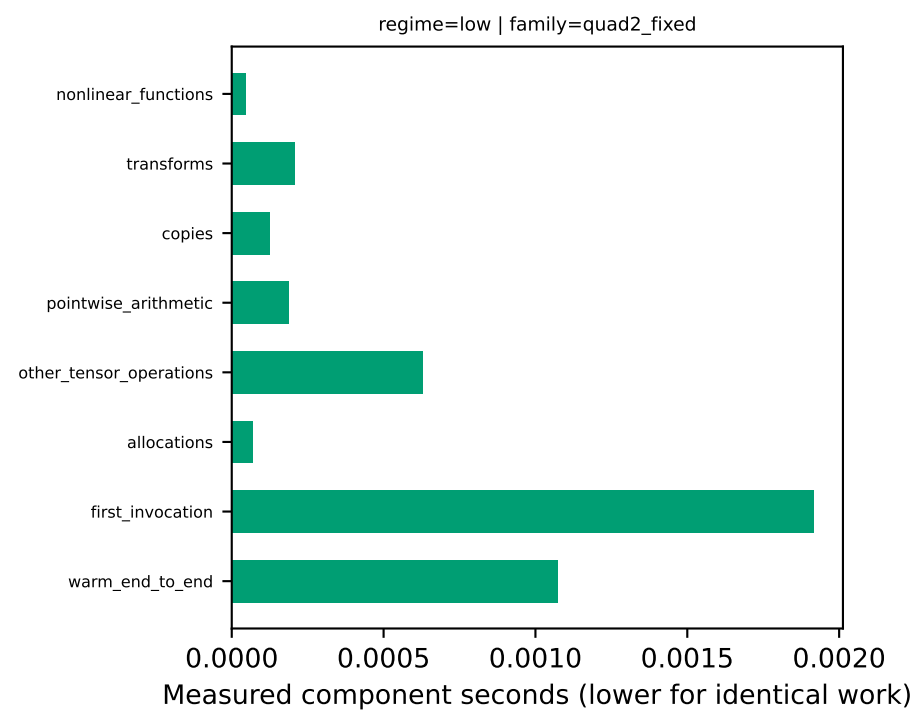
HOW TO READ: Less time is favorable for the same operation and accuracy. Profiling overhead is excluded from deployment comparisons.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Transforms, transport, products, features, synchronization and allocation are measured only when instrumented; a missing category is NA, never zero. Nested profiler events must not be summed as independent costs.

Runtime component profiling (2)



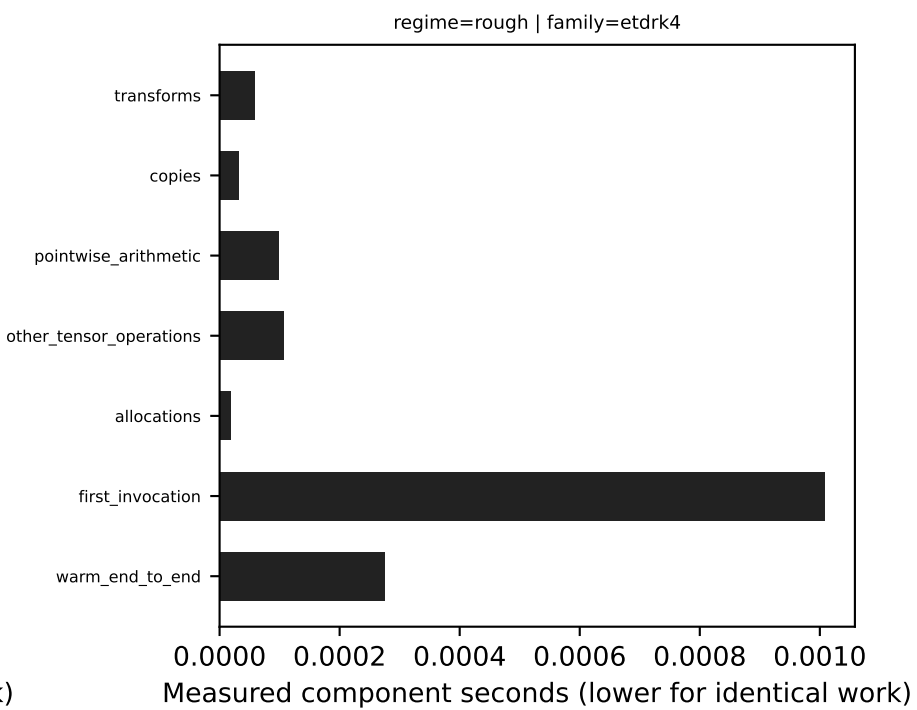
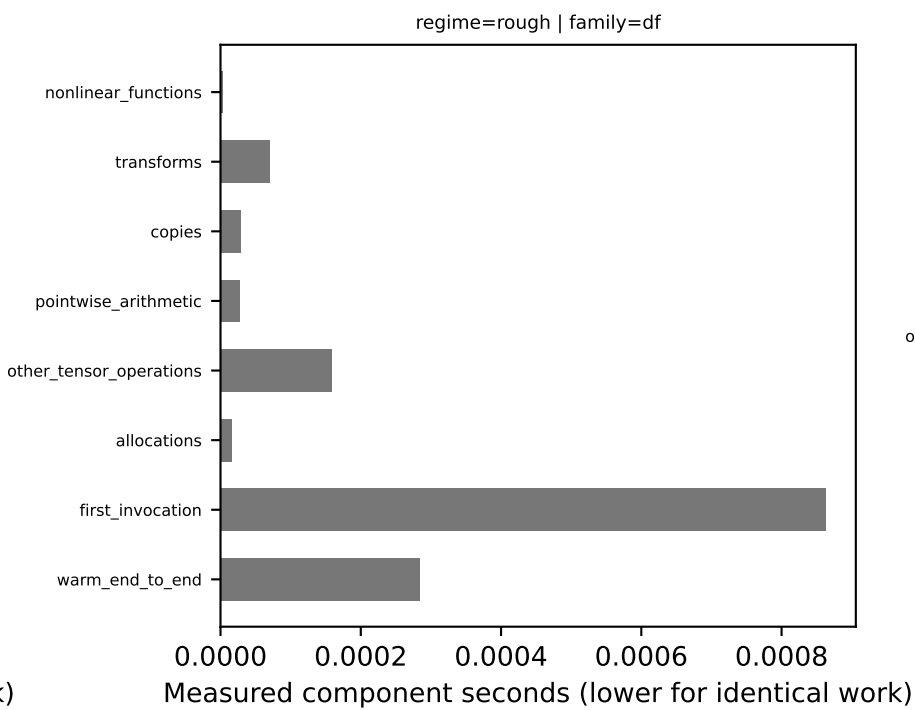
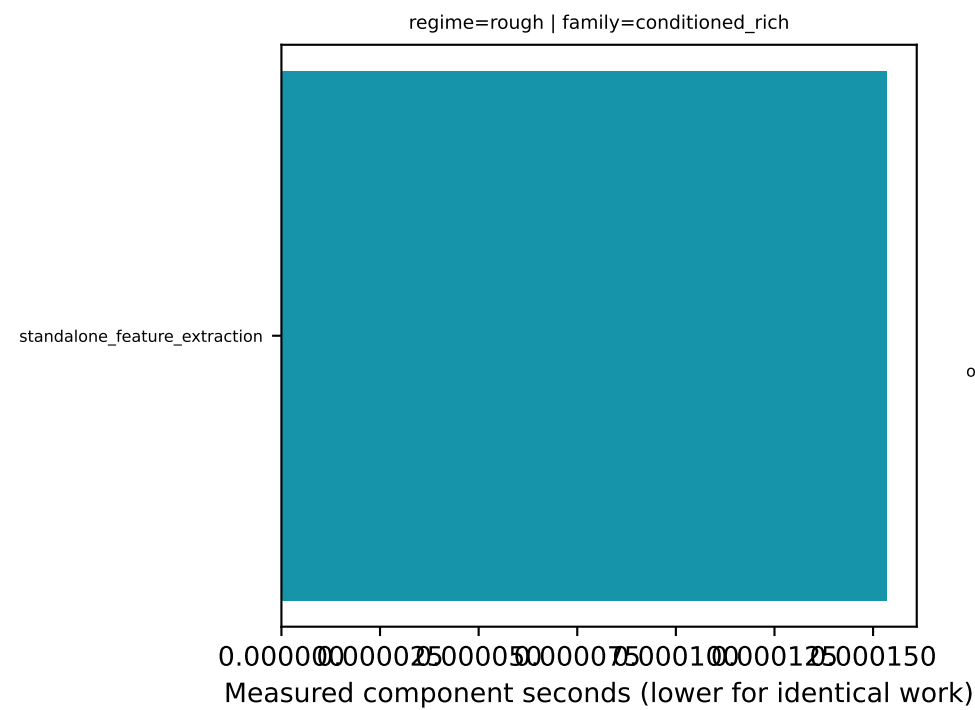
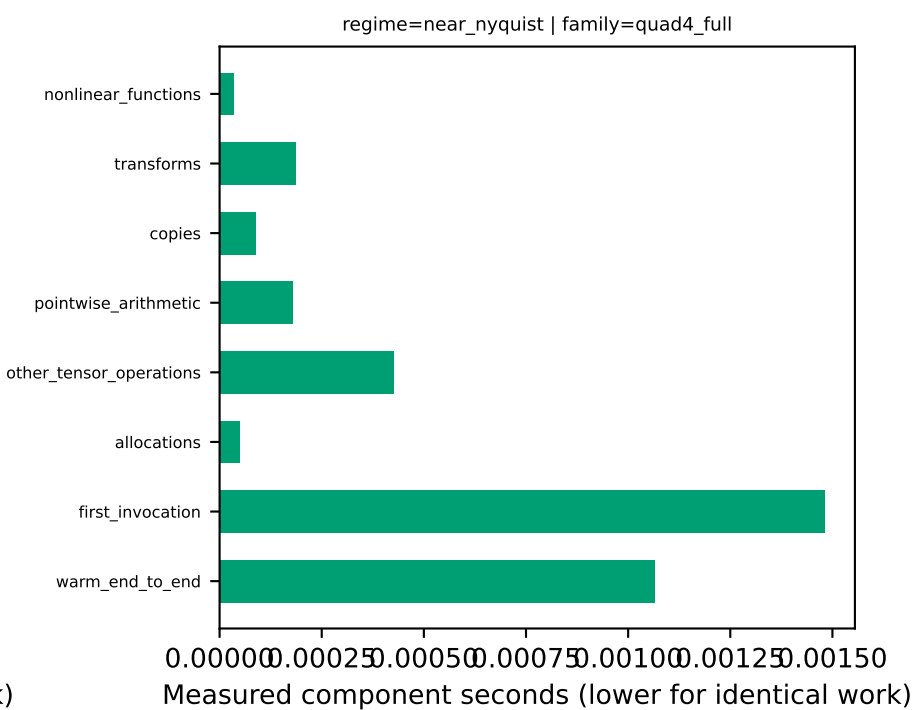
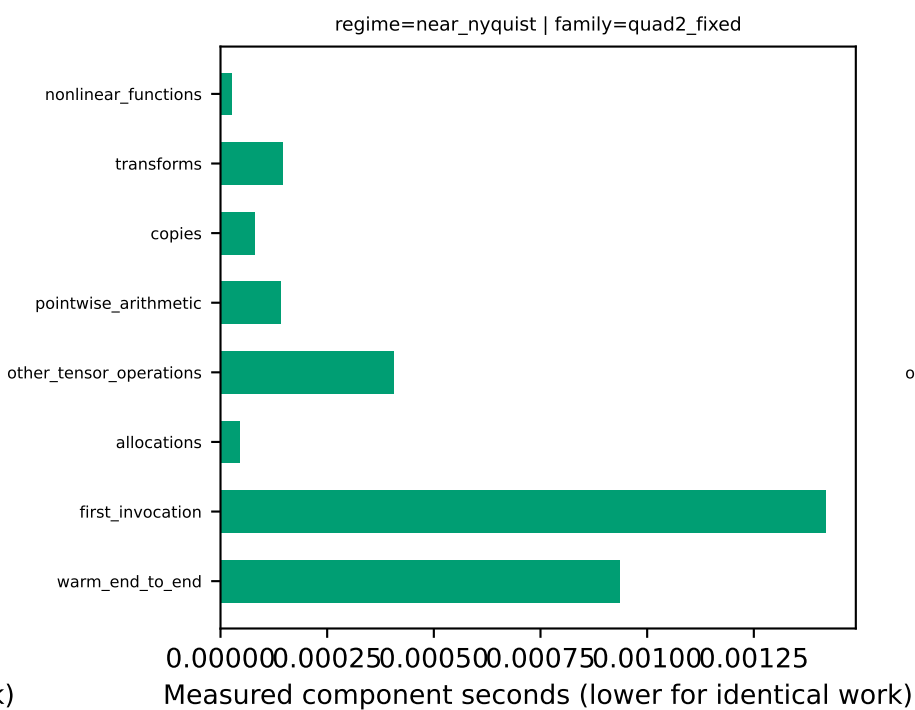
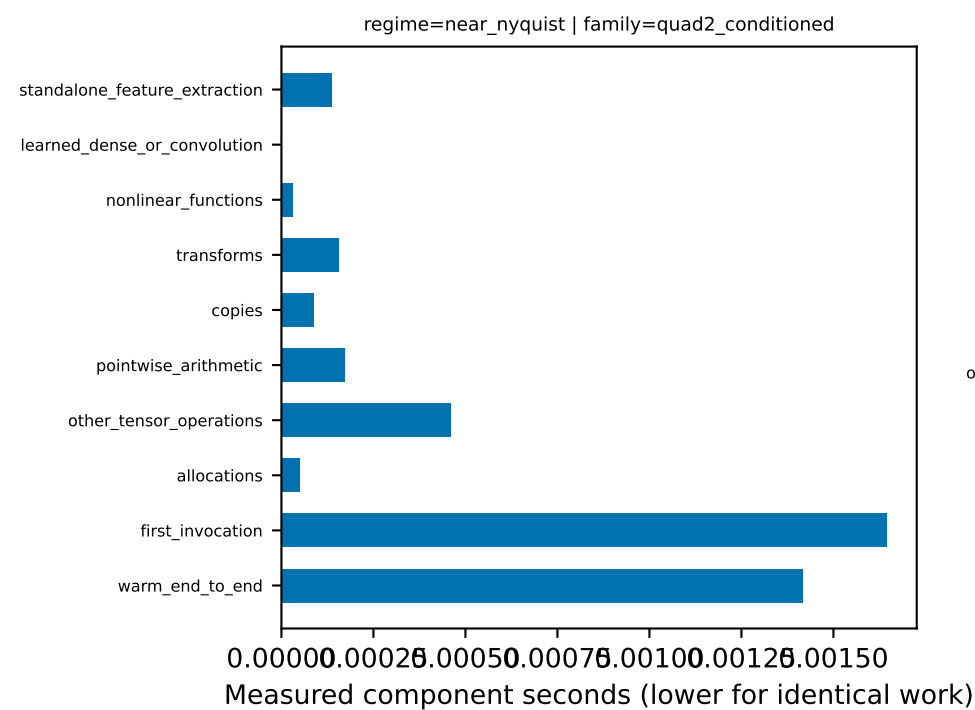
HOW TO READ: Less time is favorable for the same operation and accuracy. Profiling overhead is excluded from deployment comparisons.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Transforms, transport, products, features, synchronization and allocation are measured only when instrumented; a missing category is NA, never zero. Nested profiler events must not be summed as independent costs.

Runtime component profiling (3)



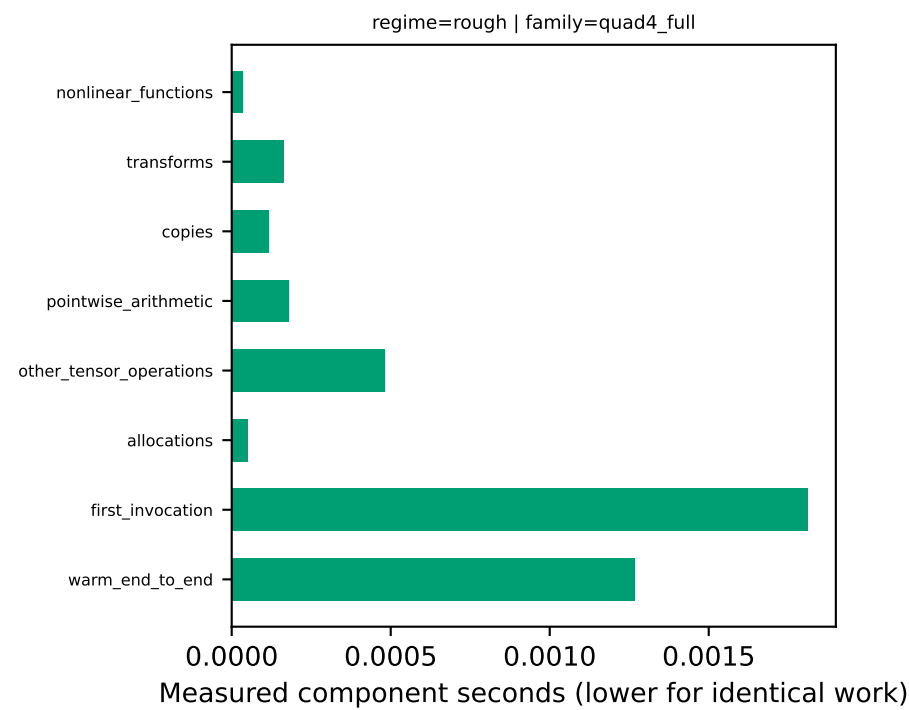
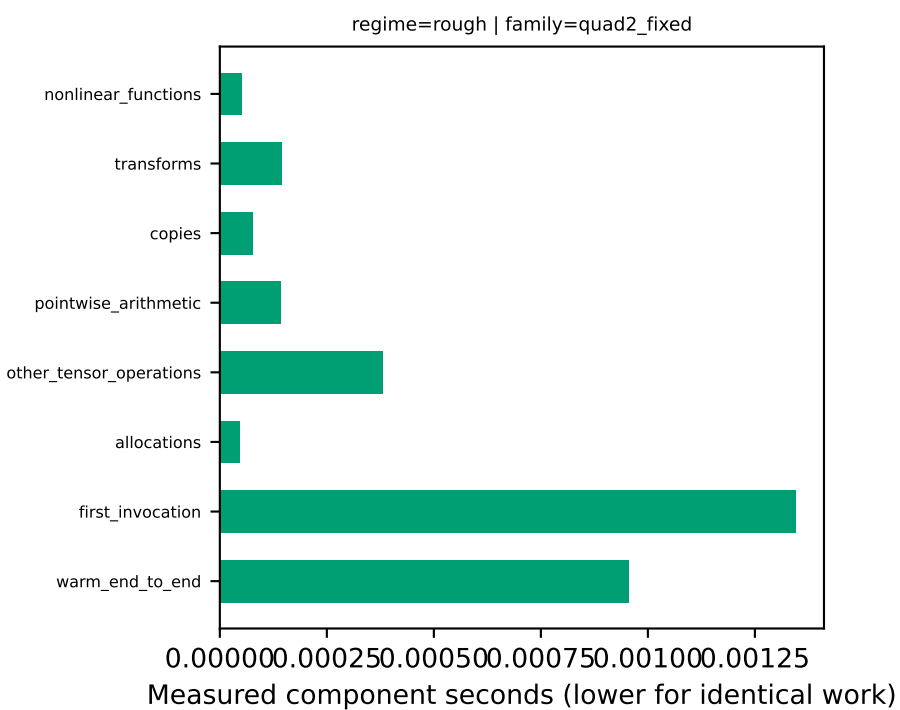
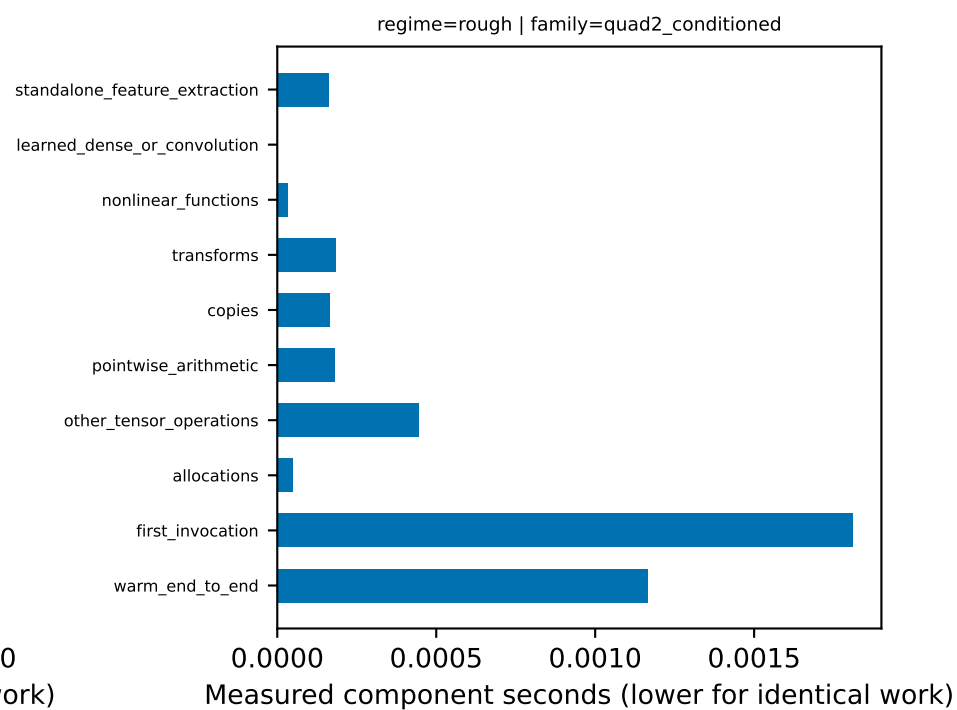
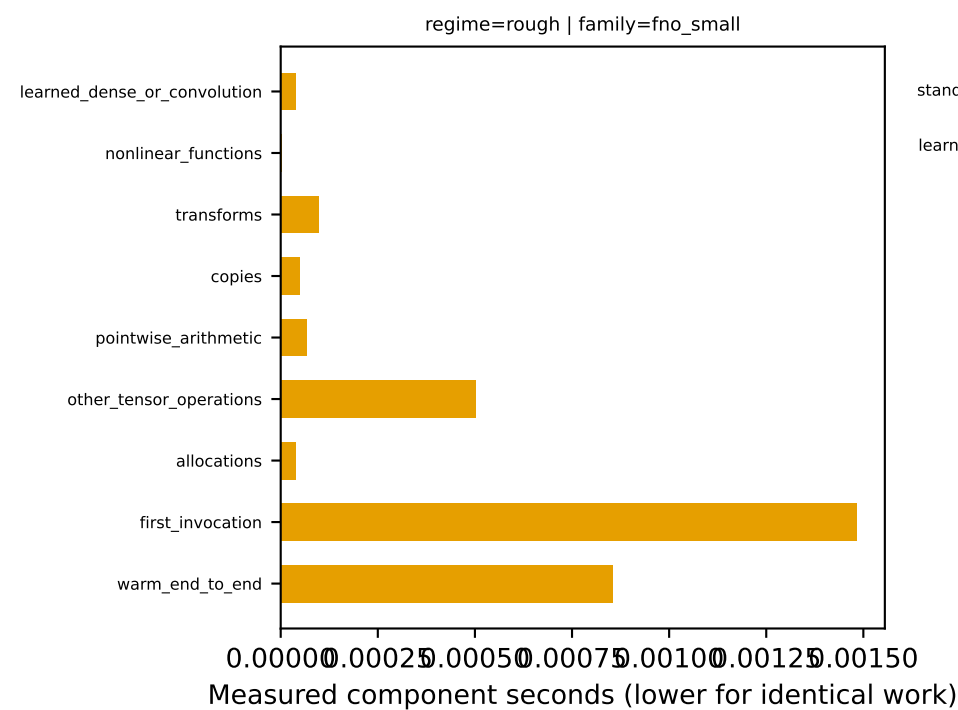
HOW TO READ: Less time is favorable for the same operation and accuracy. Profiling overhead is excluded from deployment comparisons.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Transforms, transport, products, features, synchronization and allocation are measured only when instrumented; a missing category is NA, never zero. Nested profiler events must not be summed as independent costs.

Runtime component profiling (4)



HOW TO READ: Less time is favorable for the same operation and accuracy. Profiling overhead is excluded from deployment comparisons.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Transforms, transport, products, features, synchronization and allocation are measured only when instrumented; a missing category is NA, never zero. Nested profiler events must not be summed as independent costs.

Runtime component profiling (5)



HOW TO READ: Less time is favorable for the same operation and accuracy. Profiling overhead is excluded from deployment comparisons.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Transforms, transport, products, features, synchronization and allocation are measured only when instrumented; a missing category is NA, never zero. Nested profiler events must not be summed as independent costs.

Primary learned attribution and neural accuracy gates

track=continuum | horizon=0.06 | train_count=2

track=continuum | horizon=0.12 | train_count=2

track=discrete | horizon=0.06 | train_count=2

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	1	NA	2	24	0
quad2_amplitude	NA	1	NA	2	24	0
quad2_nodes	NA	1	NA	2	24	0
quad2_joint	NA	1	NA	2	24	0
quad2_linear	NA	1	NA	2	24	0
fno_small	NA	105	NA	2	24	0.5
fno_standard	NA	122	NA	2	24	0.5

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	1	NA	2	20	0
quad2_amplitude	NA	1	NA	2	20	0
quad2_nodes	NA	1	NA	2	20	0
quad2_joint	NA	1	NA	2	20	0
quad2_linear	NA	1	NA	2	20	0
fno_small	NA	515	NA	2	20	0.8
fno_standard	NA	629	NA	2	20	0.8

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	1	NA	2	24	0
quad2_amplitude	NA	1	NA	2	24	0
quad2_nodes	NA	1	NA	2	24	0
quad2_joint	NA	1	NA	2	24	0
quad2_linear	NA	1	NA	2	24	0
fno_small	NA	111	NA	2	24	0.5
fno_standard	NA	132	NA	2	24	0.5

track=discrete | horizon=0.12 | train_count=2

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	1	NA	2	20	0
quad2_amplitude	NA	1	NA	2	20	0
quad2_nodes	NA	1	NA	2	20	0
quad2_joint	NA	1	NA	2	20	0
quad2_linear	NA	1	NA	2	20	0
fno_small	NA	701	NA	2	20	0.8
fno_standard	NA	857	NA	2	20	0.8

HOW TO READ: GOOD means the declared bounded gates met; BAD means insufficient effect/coverage or failures; NA means unresolved. Control/candidate ratios above one favor Ours. More repeated rows are not more independent fields.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Primary contrasts were declared before confirmation. Lower CI is a descriptive field/seed interval, unadjusted for multiplicity. Coverage* is candidate-minus-control pass fraction for accuracy, candidate all-query field coverage for cost. Smoke/development verdicts remain development diagnostics, not confirmation. Exact thresholds, unresolved pairs and full interval metadata are in claims tables.

Primary accuracy qualified solver cost gates

track=continuum | horizon=0.06 | train_count=2

track=continuum | horizon=0.12 | train_count=2

track=discrete | horizon=0.06 | train_count=2

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	0.988	NA	2	4	1
df	NA	0.576	NA	2	4	1
etdrk4	NA	0.198	NA	2	4	1
fno_small	NA	1.48	NA	2	4	1
fno_standard	NA	1.61	NA	2	4	1

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	0.95	NA	2	4	1
df	NA	0.614	NA	2	4	1
etdrk4	NA	0.251	NA	2	4	1
fno_small	NA	5.89	NA	2	4	1
fno_standard	NA	6.68	NA	2	4	1

Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	0.826	NA	2	4	1
df	NA	0.226	NA	2	4	1
etdrk4	NA	0.36	NA	2	4	1
fno_small	NA	1.49	NA	2	4	1
fno_standard	NA	2.39	NA	2	4	1

track=discrete | horizon=0.12 | train_count=2

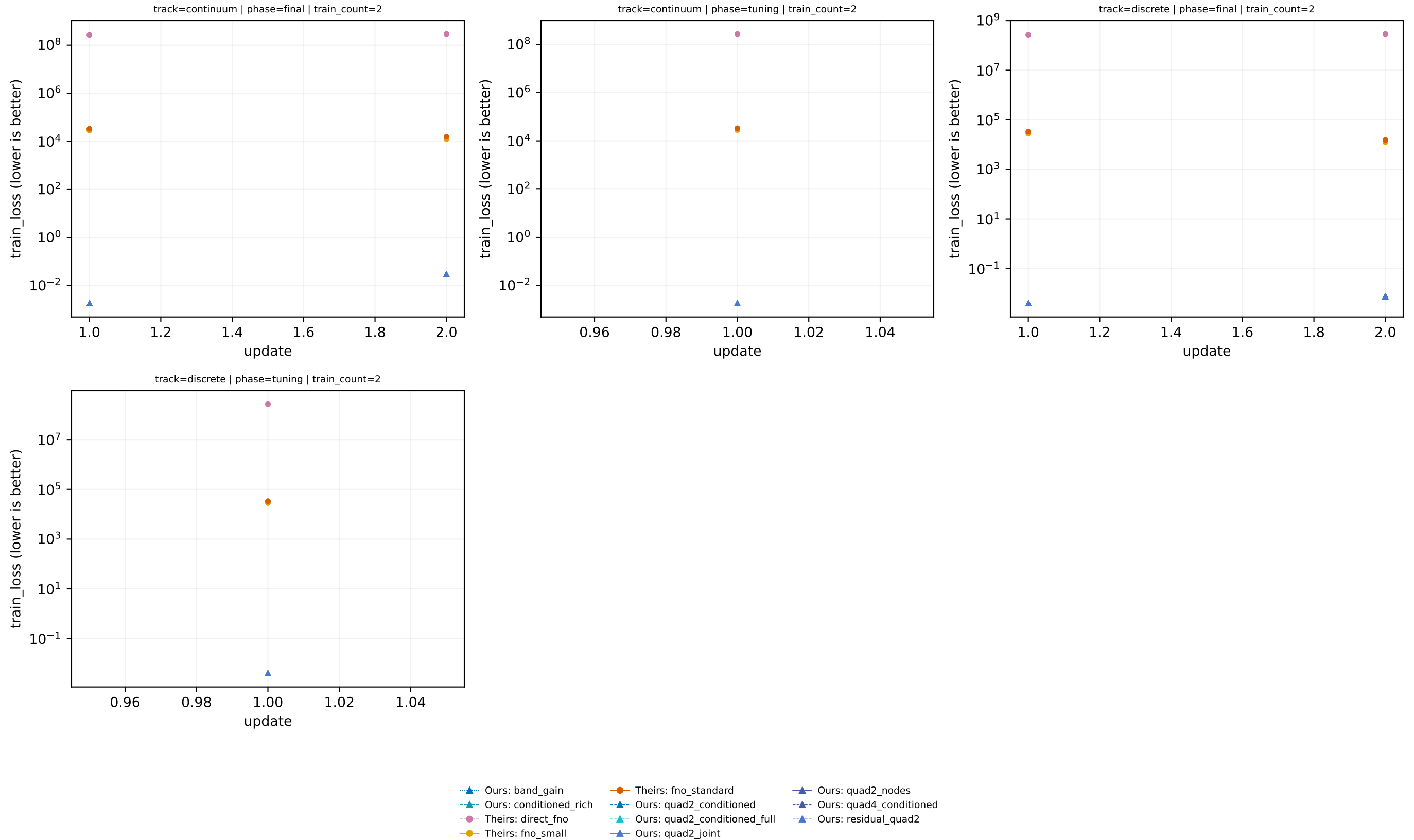
Control	Verdict	Ratio	Lower CI	Fields	Rows	Coverage*
quad2_fixed	NA	0.748	NA	2	4	1
df	NA	0.201	NA	2	4	1
etdrk4	NA	0.249	NA	2	4	1
fno_small	NA	5.49	NA	2	4	1
fno_standard	NA	8.08	NA	2	4	1

HOW TO READ: GOOD means the declared bounded gates met; BAD means insufficient effect/coverage or failures; NA means unresolved. Control/candidate ratios above one favor Ours. More repeated rows are not more independent fields.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Primary contrasts were declared before confirmation. Lower CI is a descriptive field/seed interval, unadjusted for multiplicity. Coverage* is candidate-minus-control pass fraction for accuracy, candidate all-query field coverage for cost. Smoke/development verdicts remain development diagnostics, not confirmation. Exact thresholds, unresolved pairs and full interval metadata are in claims tables.

Training loss observed ranges

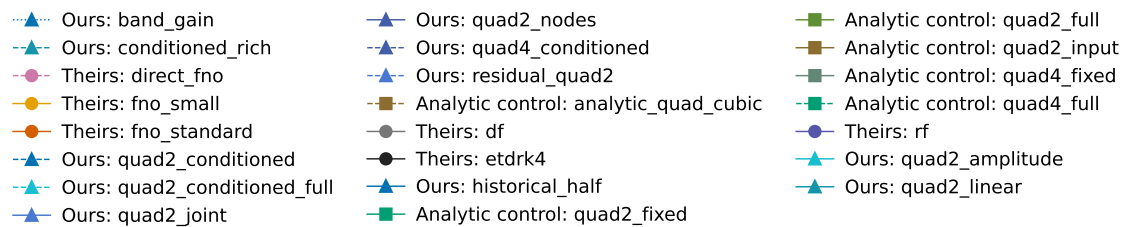
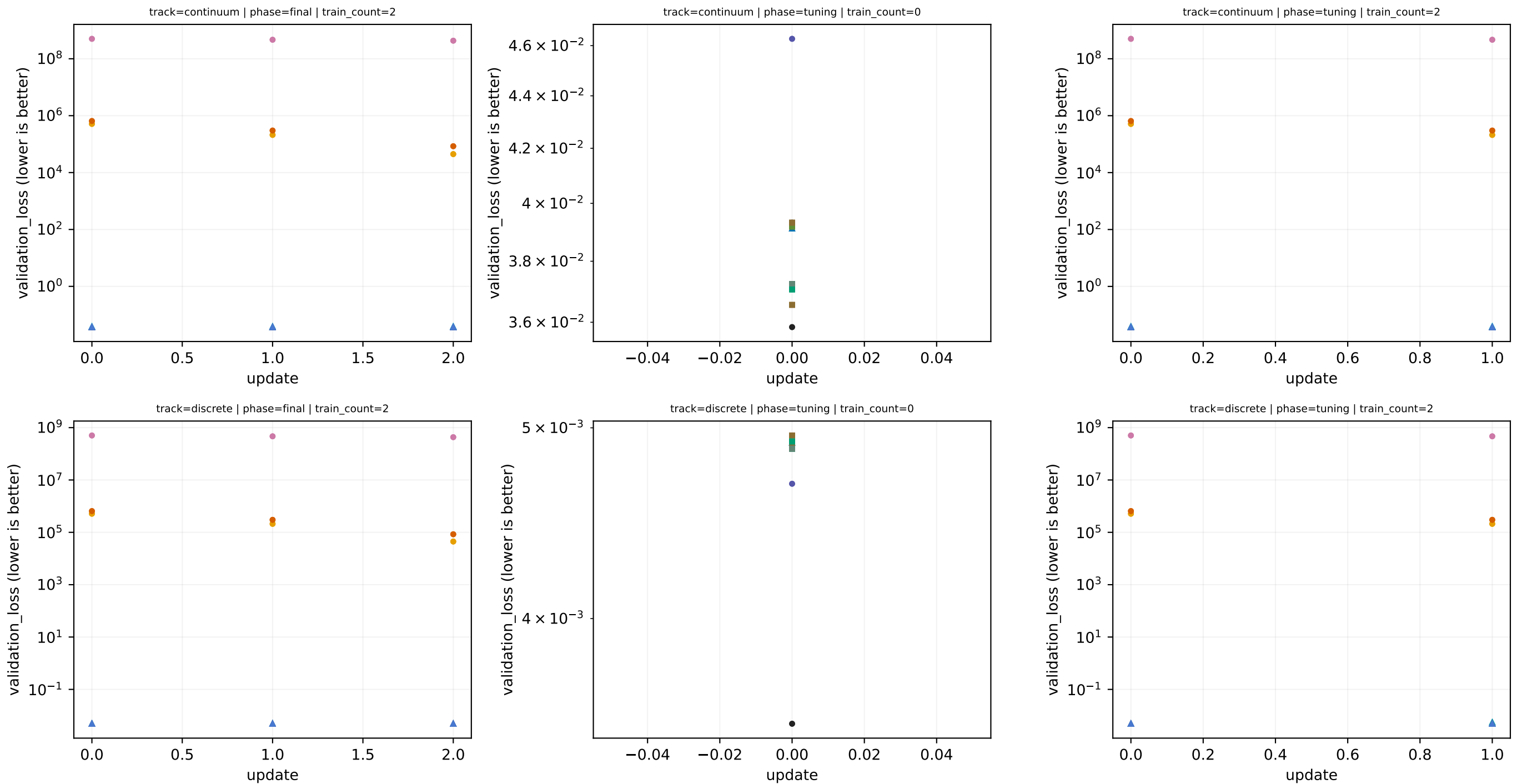


HOW TO READ: Lower loss/error is better for the same objective. Left is less training work, not necessarily a better selected model.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Continuous translucent observed min/max ranges in shared x-windows; not confidence intervals. Straight boundary connections interpolate summaries, not observations. Empty windows and isolated checkpoints break bands; tuning configurations remain development evidence.

Validation loss observed ranges

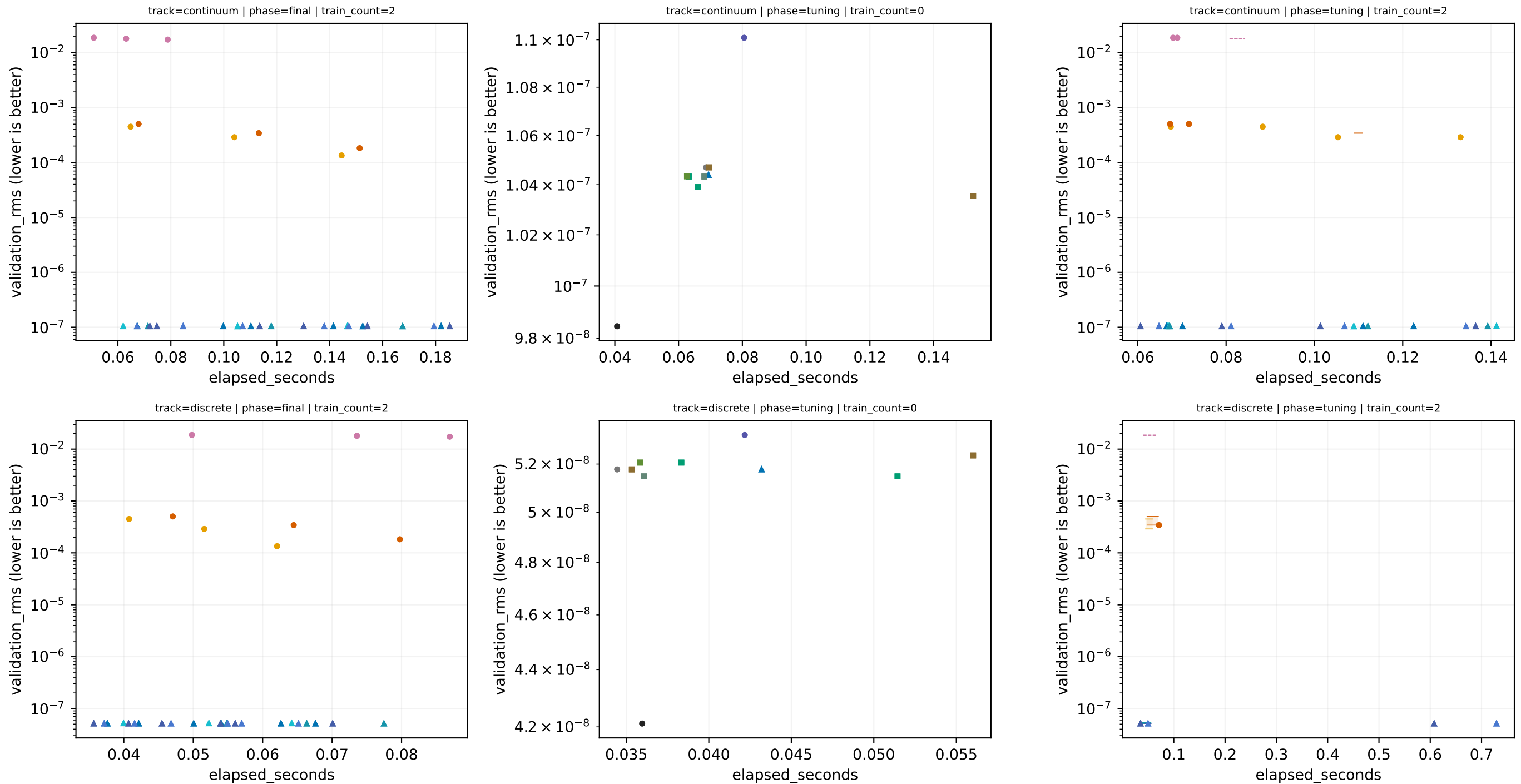


HOW TO READ: Lower loss/error is better for the same objective. Left is less training work, not necessarily a better selected model.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Continuous translucent observed min/max ranges in shared x-windows; not confidence intervals. Straight boundary connections interpolate summaries, not observations. Empty windows and isolated checkpoints break bands; tuning configurations remain development evidence.

Validation error versus training compute



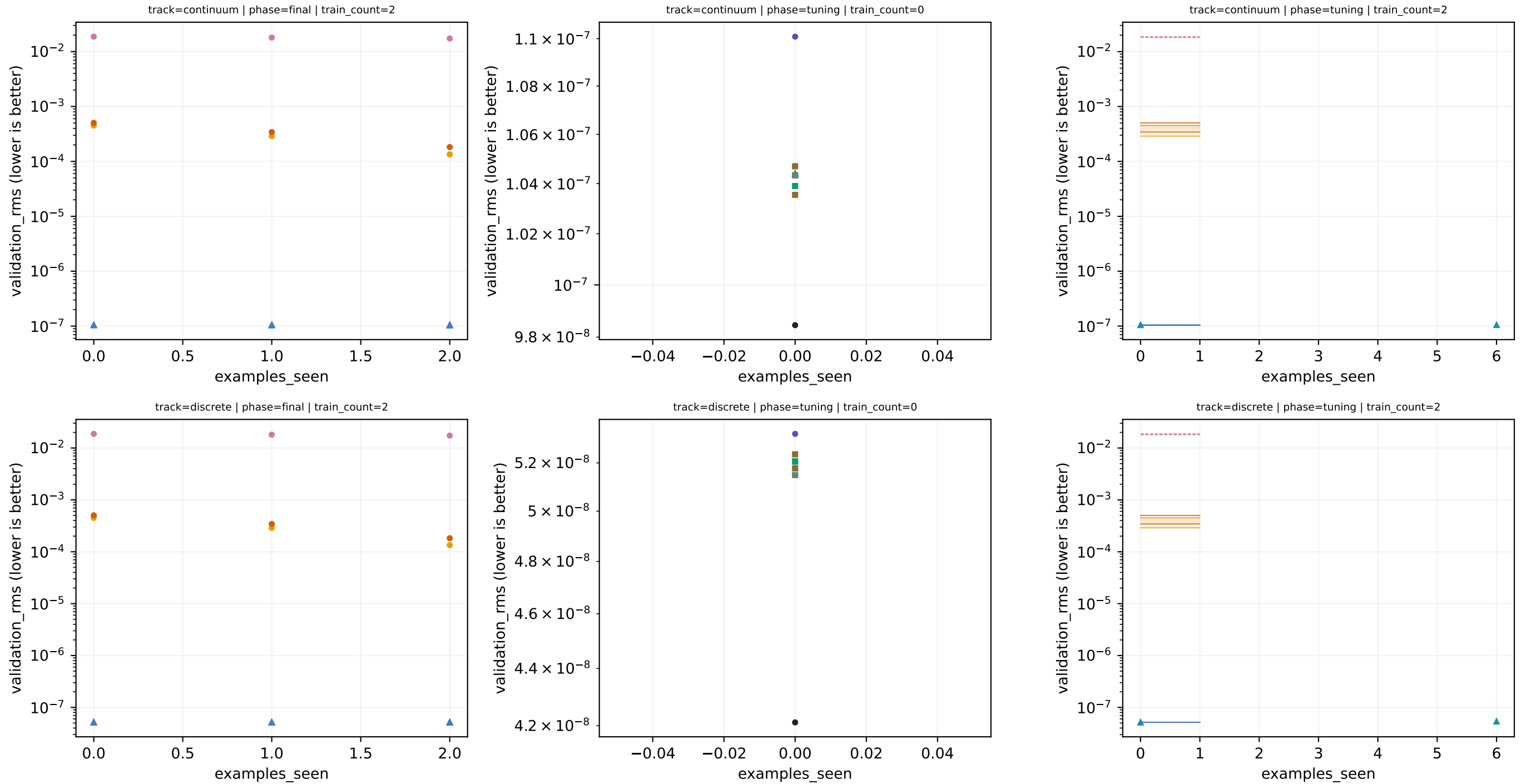
- Ours: band_gain
 - Ours: conditioned_rich
 - Theirs: direct_fno
 - Theirs: fno_small
 - Theirs: fno_standard
 - Ours: quad2_conditioned
 - Ours: quad2_conditioned_full
 - Ours: quad2_joint
 - Ours: quad2_nodes
 - Ours: quad4_conditioned
 - Ours: residual_quad2
 - Analytic control: analytic_quad_cubic
 - Theirs: df
 - Theirs: etdrk4
 - Ours: historical_half
 - Analytic control: quad2_fixed
 - Analytic control: quad2_full
 - Analytic control: quad2_input
 - Analytic control: quad4_fixed
 - Analytic control: quad4_full
 - Theirs: rf
 - Ours: quad2_amplitude
 - Ours: quad2_linear

HOW TO READ: Lower loss/error is better for the same objective. Left is less training work, not necessarily a better selected model.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Continuous translucent observed min/max ranges in shared x-windows; not confidence intervals. Straight boundary connections interpolate summaries, not observations. Empty windows and isolated checkpoints break bands; tuning configurations remain development evidence.

Validation error versus examples seen



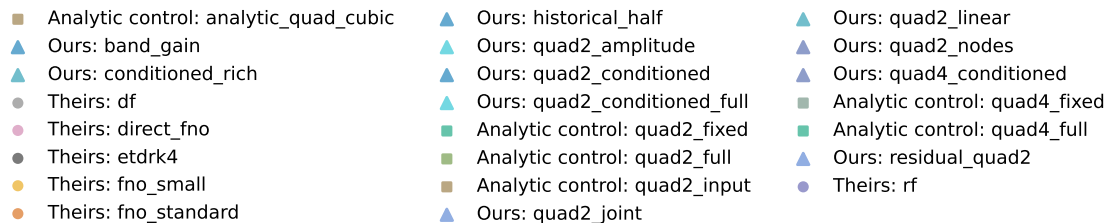
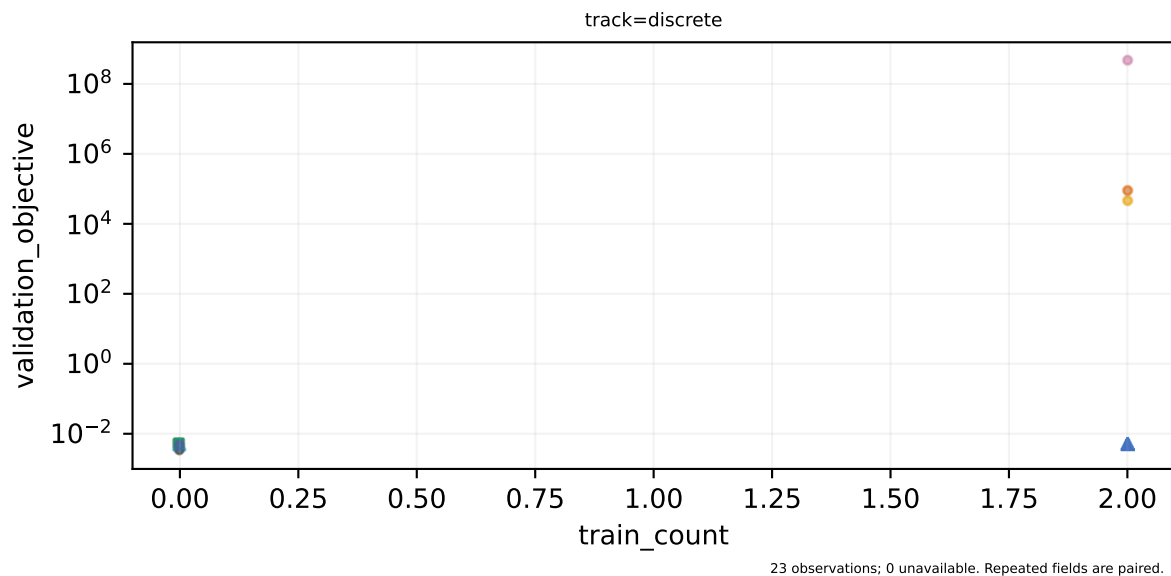
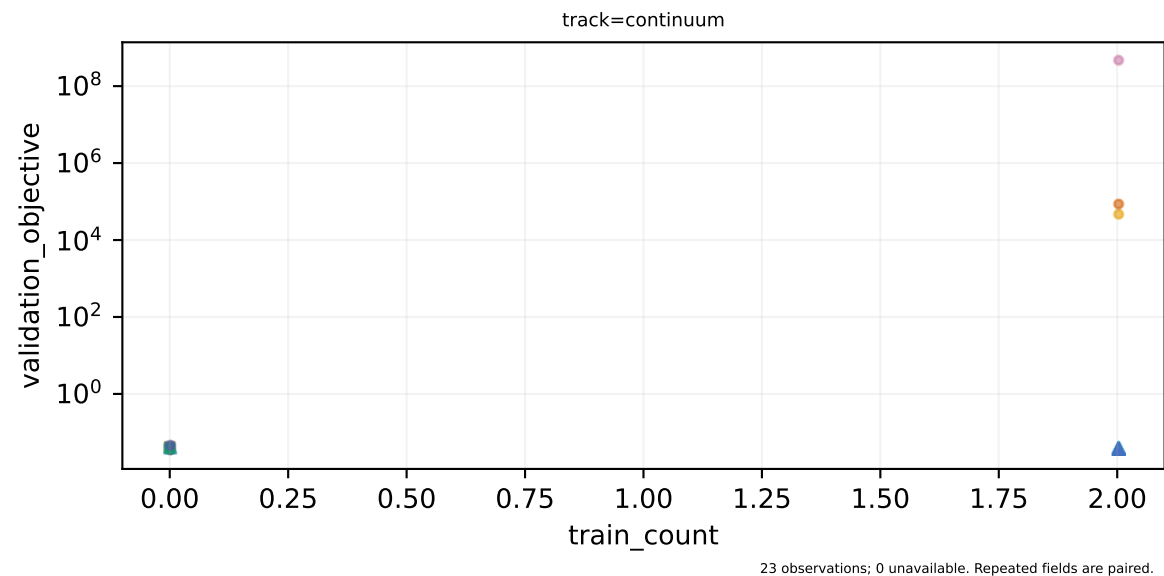
- | | | |
|--|---|--|
| <ul style="list-style-type: none"> ...▲ Ours: band_gain ...▲ Ours: conditioned_rich ...● Theirs: direct_fno ...● Theirs: fno_small ...● Theirs: fno_standard ...▲ Ours: quad2_conditioned ...▲ Ours: quad2_conditioned_full ...▲ Ours: quad2_joint | <ul style="list-style-type: none"> ...▲ Ours: quad2_nodes ...▲ Ours: quad4_conditioned ...▲ Ours: residual_quad2 ...■ Analytic control: analytic_quad_cubic ...● Theirs: df ...● Theirs: etdrk4 ...▲ Ours: historical_half ...■ Analytic control: quad2_fixed | <ul style="list-style-type: none"> ...■ Analytic control: quad2_full ...■ Analytic control: quad2_input ...■ Analytic control: quad4_fixed ...■ Analytic control: quad4_full ...● Theirs: rf ...▲ Ours: quad2_amplitude ...▲ Ours: quad2_linear |
|--|---|--|

HOW TO READ: Lower loss/error is better for the same objective. Left is less training work, not necessarily a better selected model.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

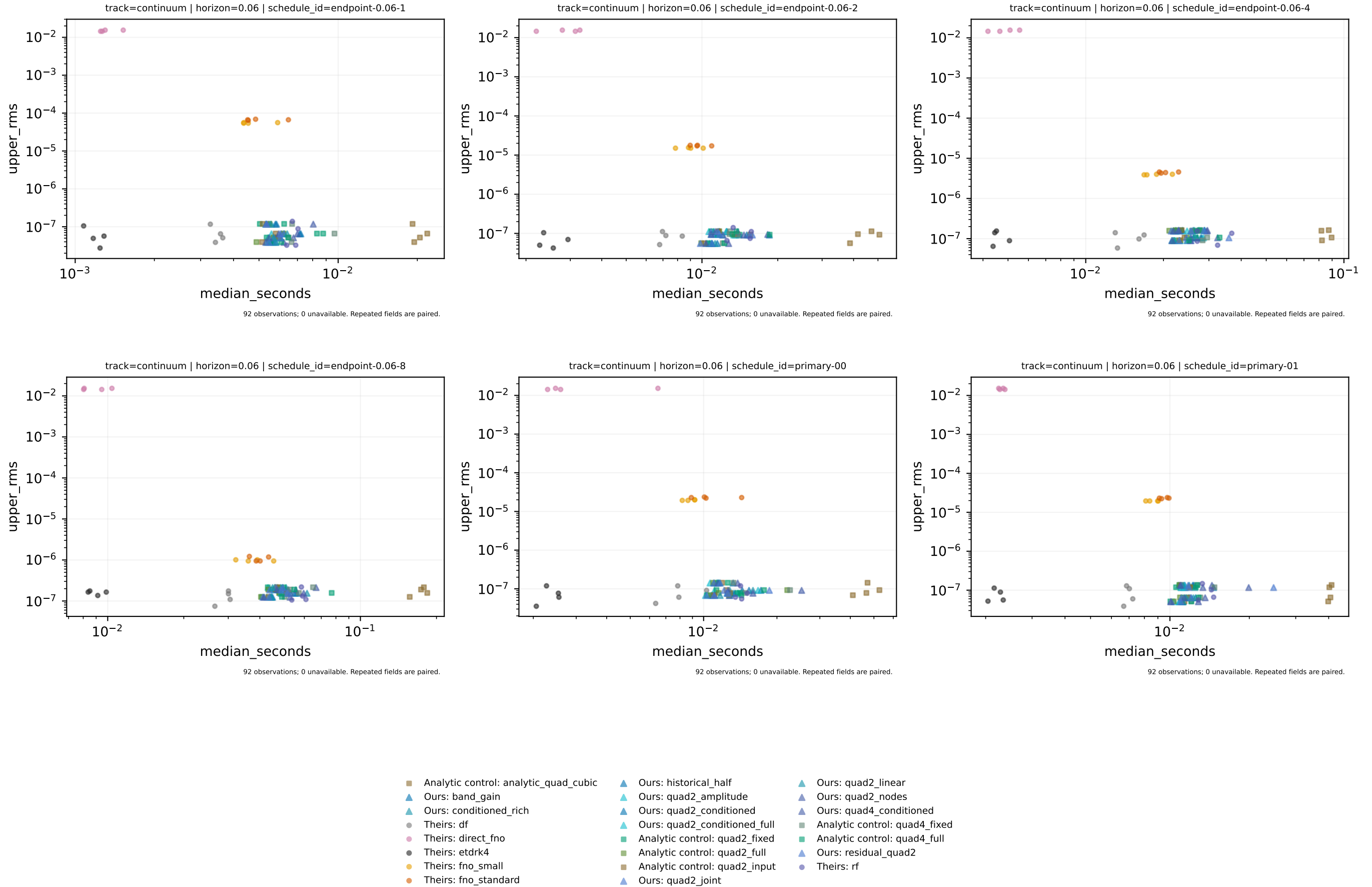
SCOPE: Continuous translucent observed min/max ranges in shared x-windows; not confidence intervals. Straight boundary connections interpolate summaries, not observations. Empty windows and isolated checkpoints break bands; tuning configurations remain development evidence.

Selected models and data efficiency



HOW TO READ: Lower-left is favorable: fewer independent training fields and lower held-out objective. Objective definitions must match.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Validation-selected checkpoints, including initialization-selected and failed trials. Selection is not untouched confirmation; every record remains in catalog.csv.

Matched schedule accuracy and inference cost (1)

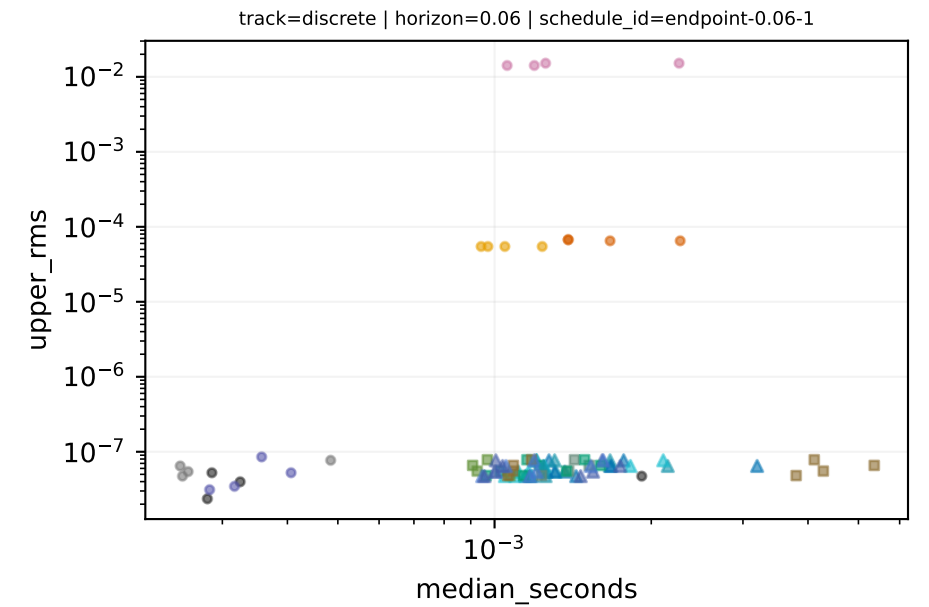
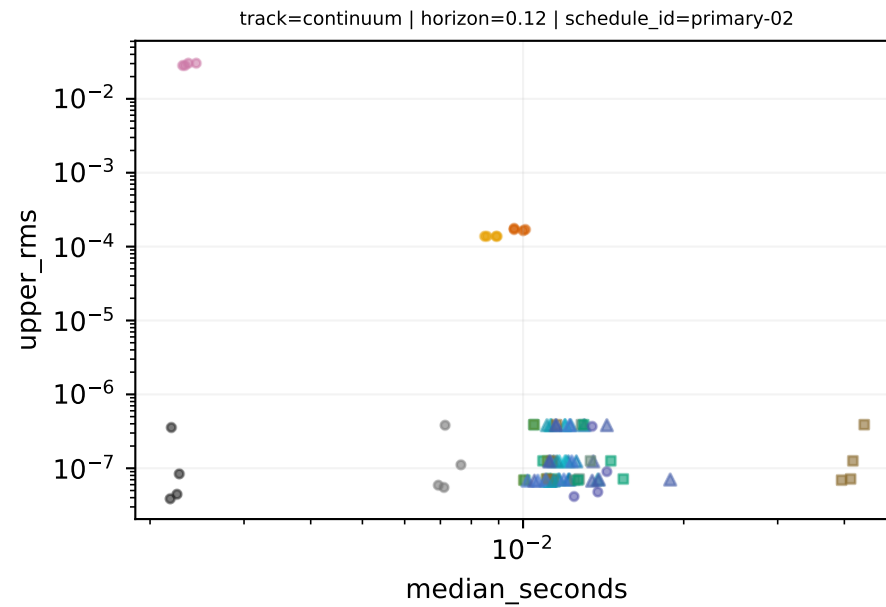
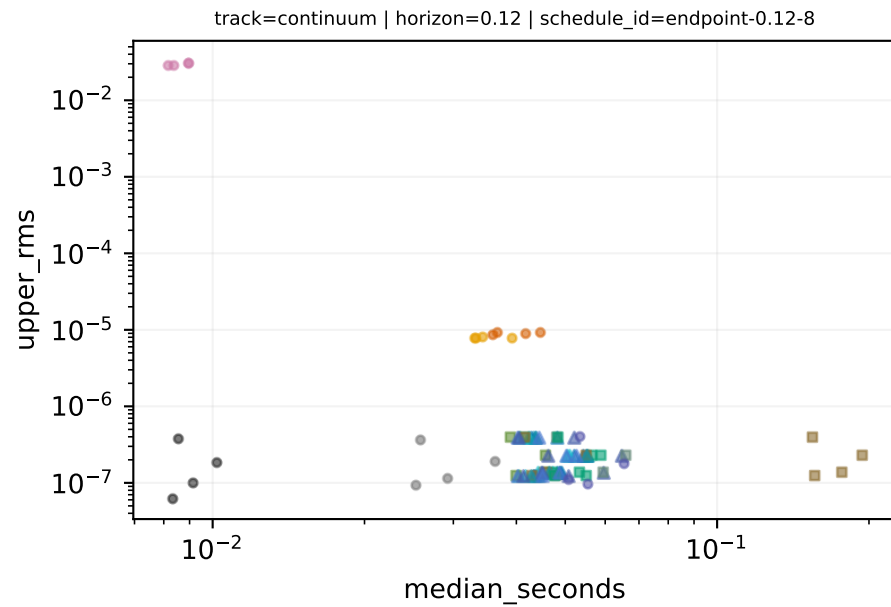
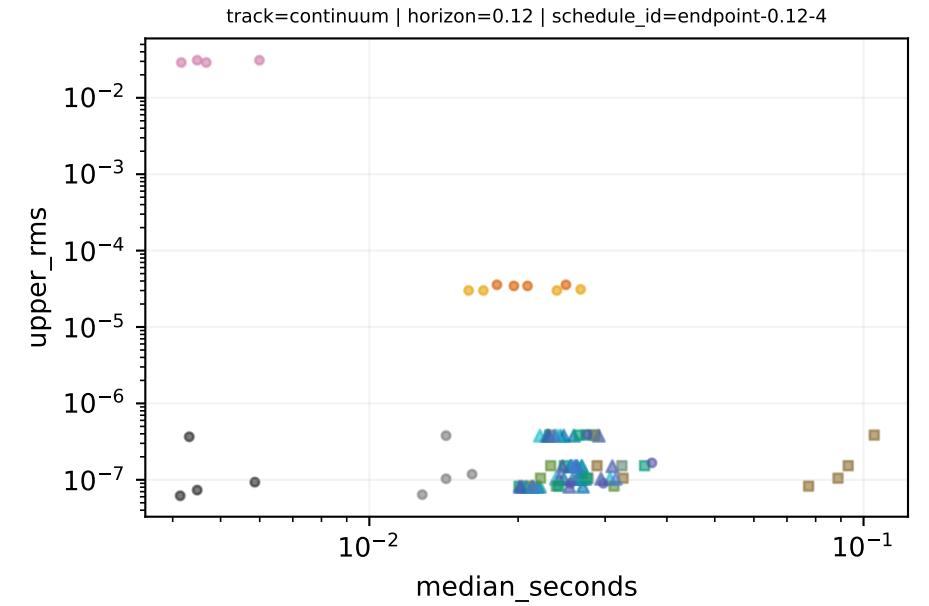
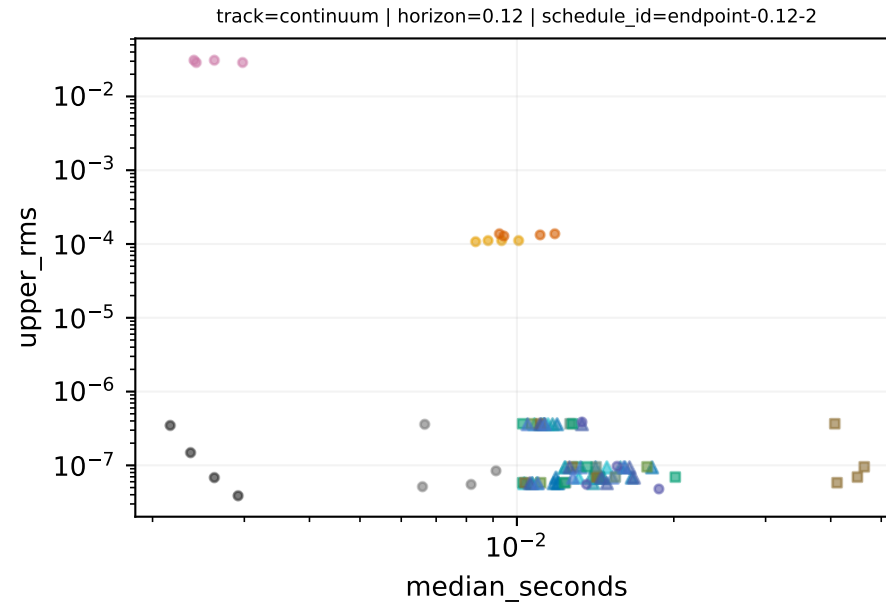
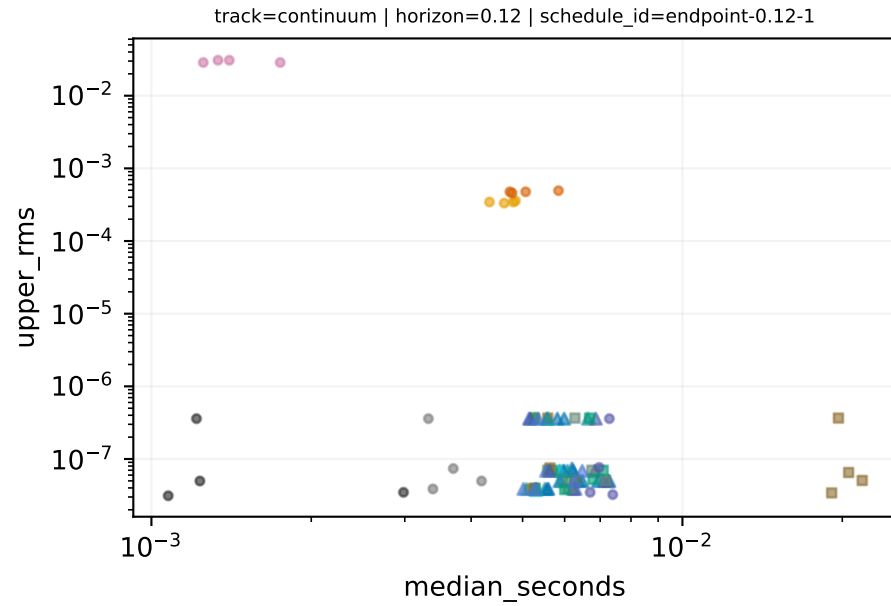


HOW TO READ: Lower-left is favorable: less complete endpoint time and smaller RMS plus reference uncertainty.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Exact schedule, final time and target are separated. Points repeat independent fields across methods, grids and seeds. Maximum-error requirements are retained separately in raw rows.

Matched schedule accuracy and inference cost (2)



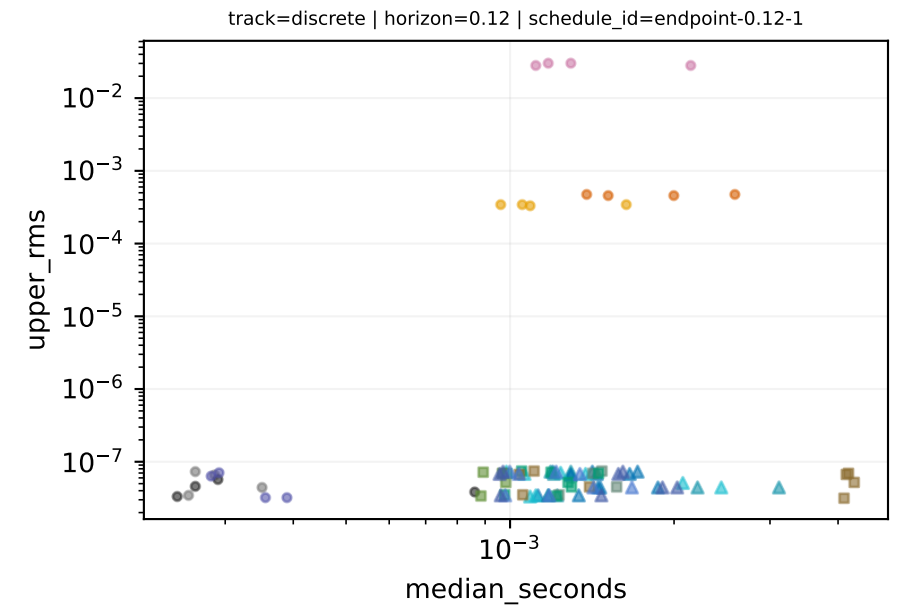
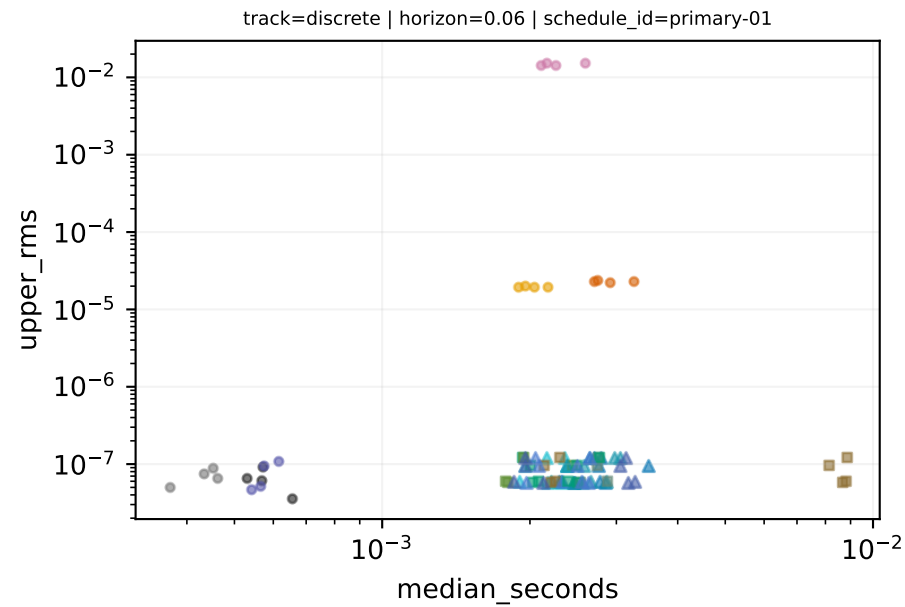
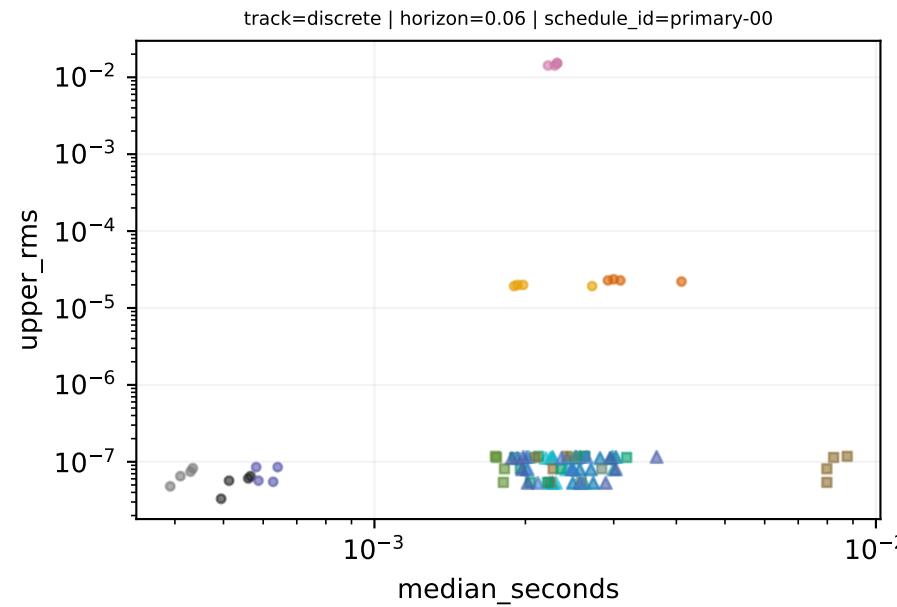
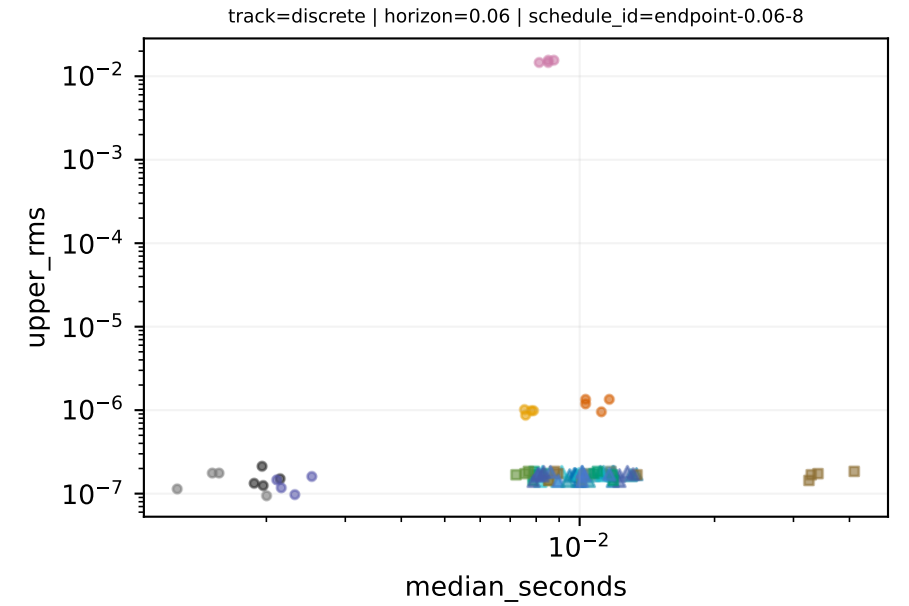
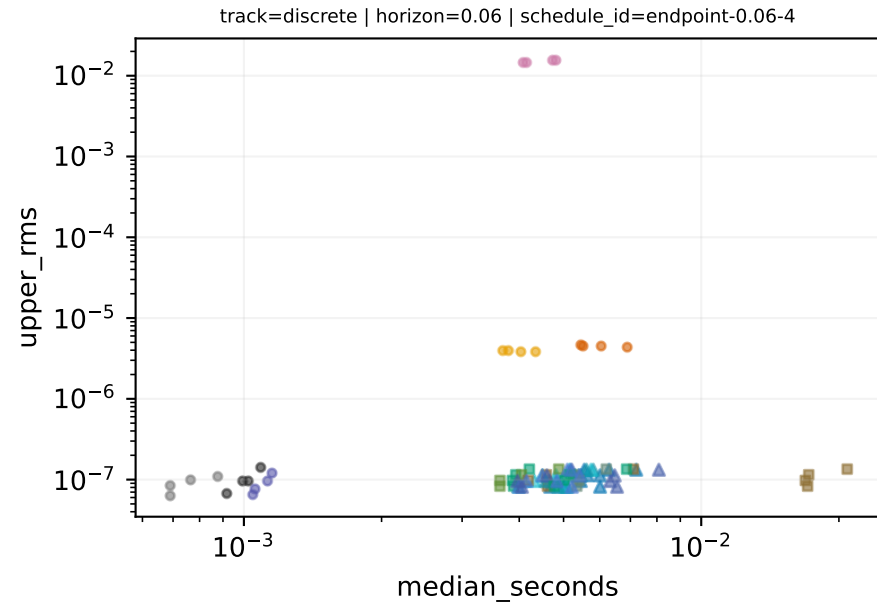
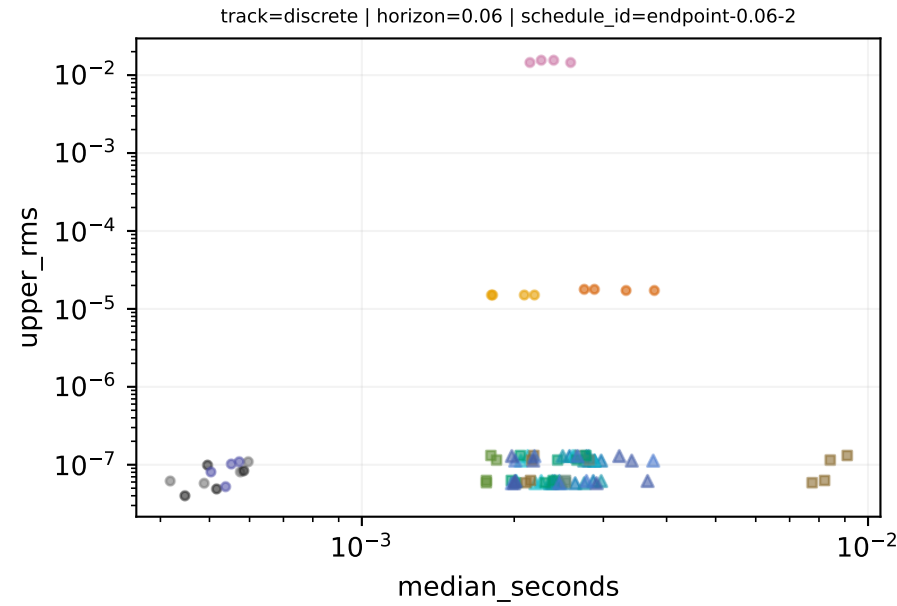
- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: historical_half | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ▲ Ours: quad2_conditioned_full | ■ Analytic control: quad4_fixed |
| ● Theirs: direct_fno | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_full |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ▲ Ours: residual_quad2 |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ● Theirs: rf |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | |

HOW TO READ: Lower-left is favorable: less complete endpoint time and smaller RMS plus reference uncertainty.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Exact schedule, final time and target are separated. Points repeat independent fields across methods, grids and seeds. Maximum-error requirements are retained separately in raw rows.

Matched schedule accuracy and inference cost (3)



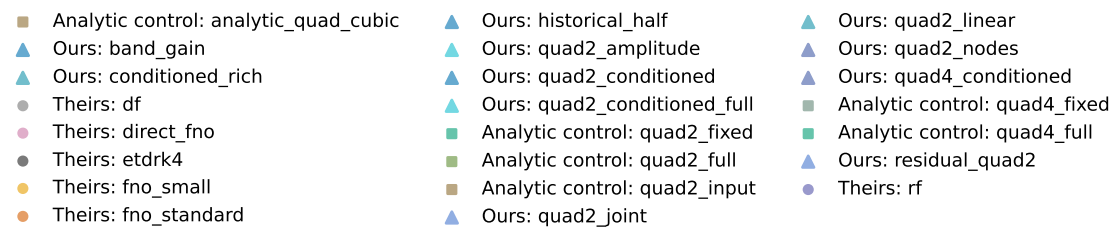
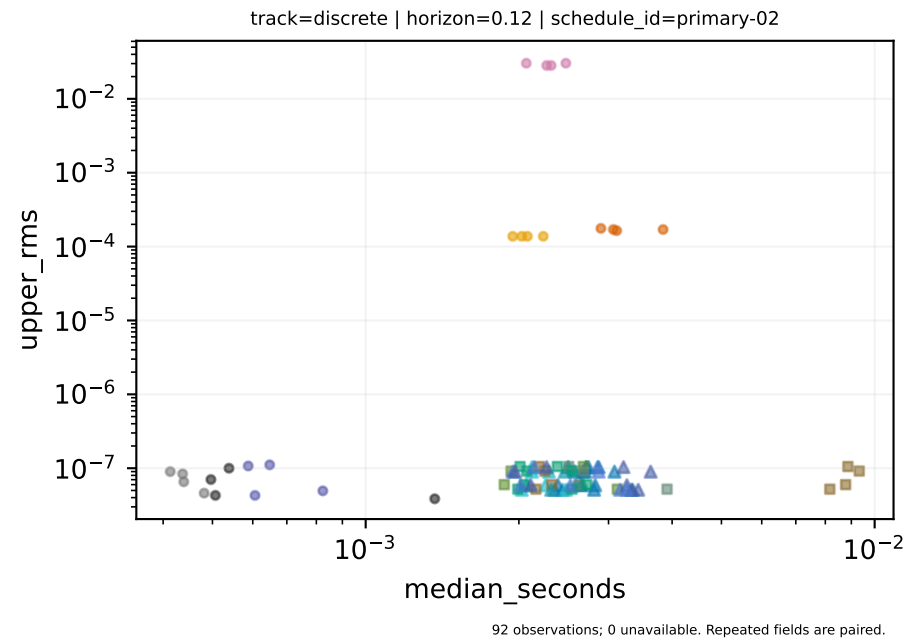
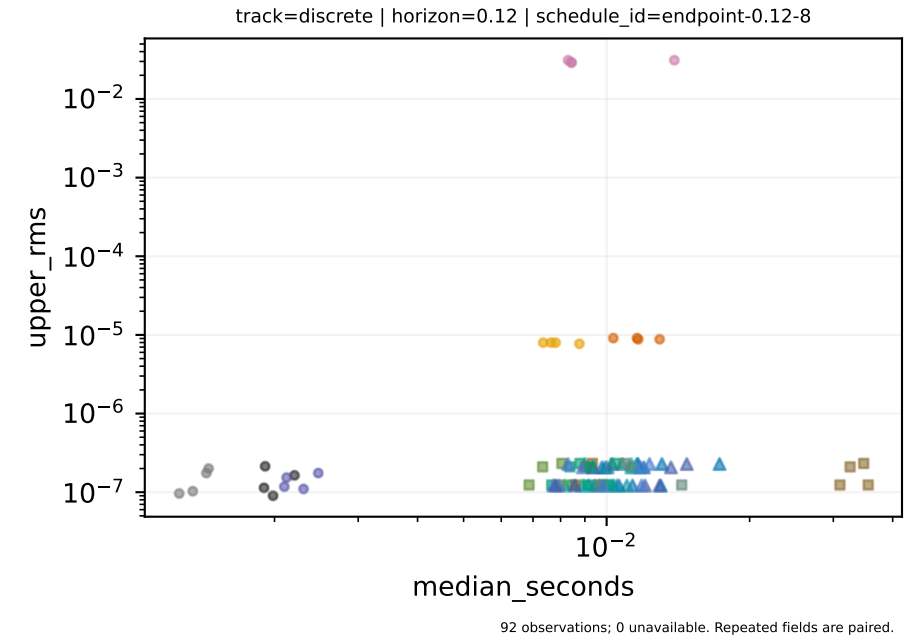
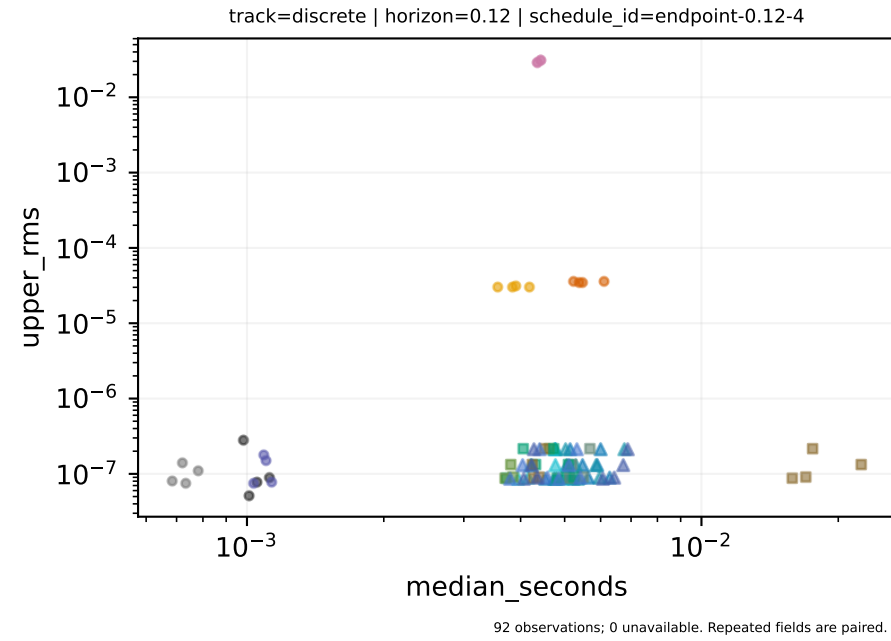
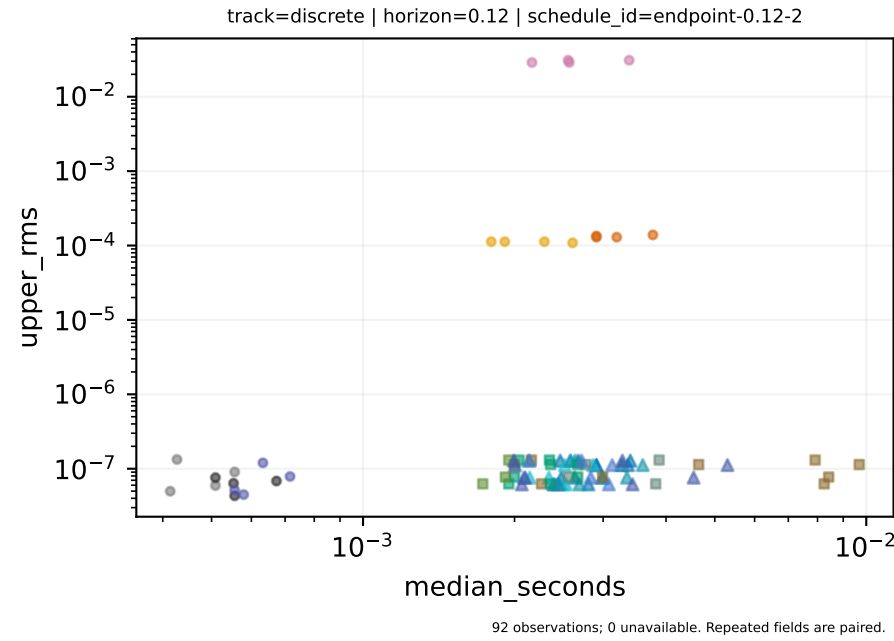
- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: historical_half | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ▲ Ours: quad2_conditioned_full | ■ Analytic control: quad4_fixed |
| ● Theirs: direct_fno | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_full |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ▲ Ours: residual_quad2 |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ● Theirs: rf |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | |

HOW TO READ: Lower-left is favorable: less complete endpoint time and smaller RMS plus reference uncertainty.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Exact schedule, final time and target are separated. Points repeat independent fields across methods, grids and seeds. Maximum-error requirements are retained separately in raw rows.

Matched schedule accuracy and inference cost (4)



HOW TO READ: Lower-left is favorable: less complete endpoint time and smaller RMS plus reference uncertainty.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Exact schedule, final time and target are separated. Points repeat independent fields across methods, grids and seeds. Maximum-error requirements are retained separately in raw rows.

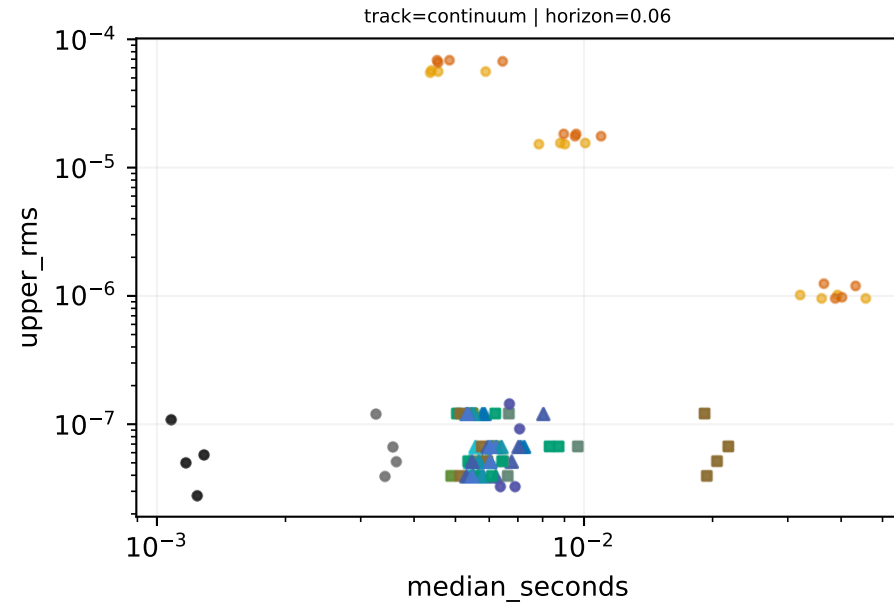
Partial confirmation evidence

track=NA | horizon=NA

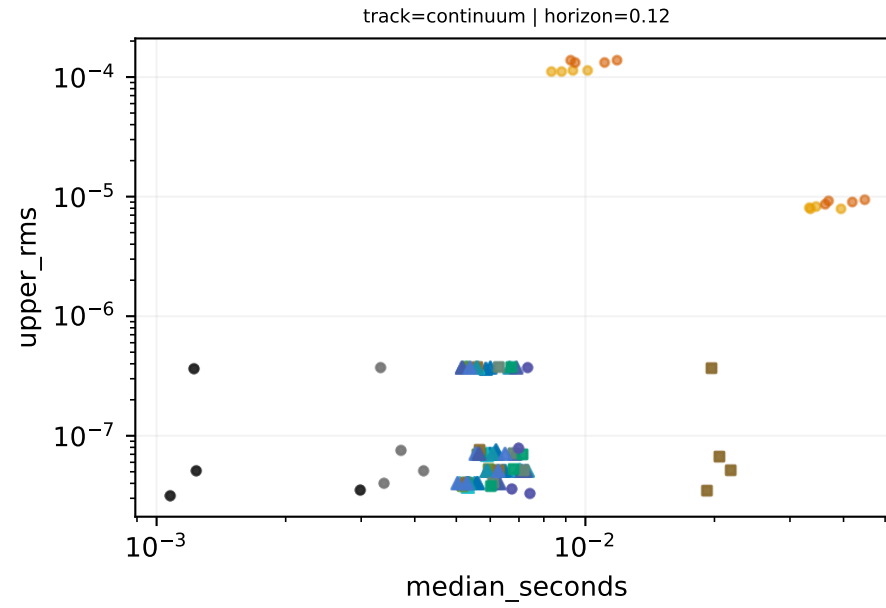
NA: no verified measurements for this panel

HOW TO READ: Lower-left is favorable within a matched workload; incomplete cohorts cannot establish confirmation.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Only sealed available partitions, shown when the full aggregate is absent. Missing partitions remain NA; no complete-cohort claims or best-model selection are made.

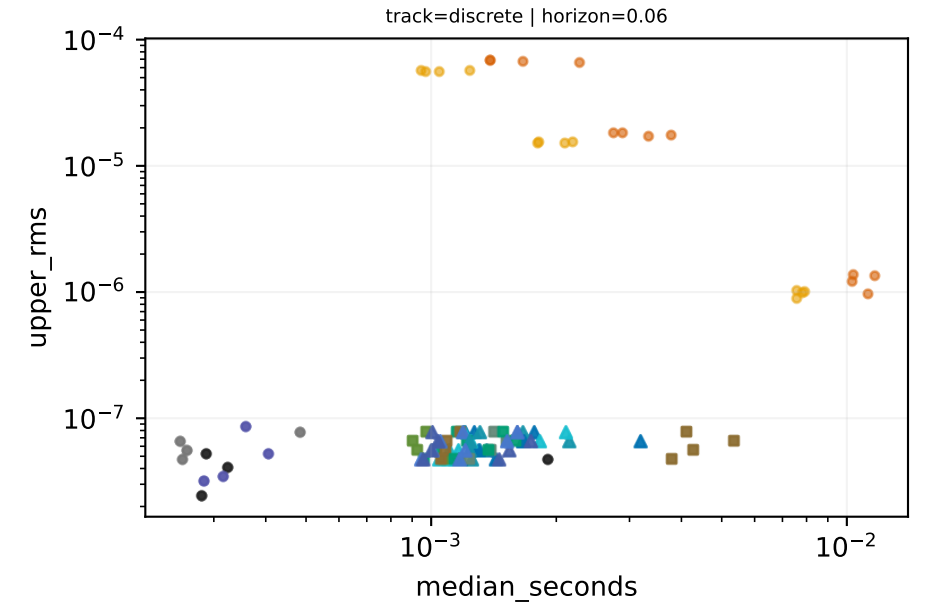
Validation locked accuracy cost frontier



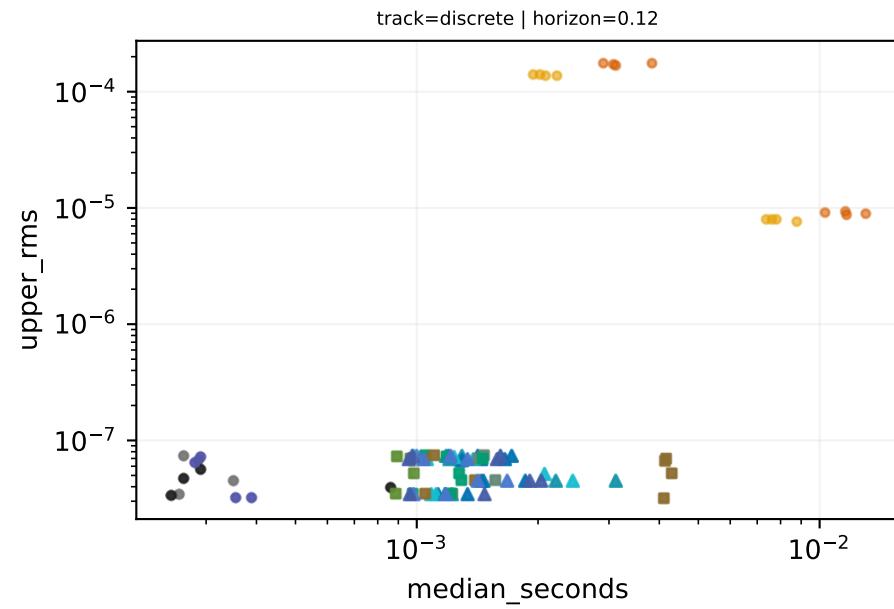
264 observations; 12 unavailable. Repeated fields are paired.



256 observations; 20 unavailable. Repeated fields are paired.



264 observations; 12 unavailable. Repeated fields are paired.



256 observations; 20 unavailable. Repeated fields are paired.

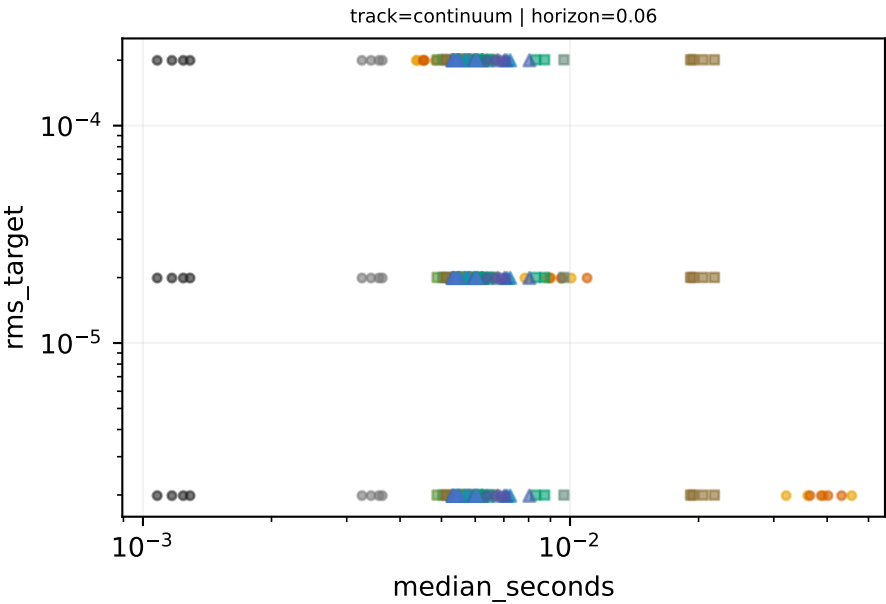
- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_conditioned | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned_full | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_fixed |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ■ Analytic control: quad4_full |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ▲ Ours: residual_quad2 |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | ● Theirs: rf |
| ▲ Ours: historical_half | | |

HOW TO READ: Lower-left means less inference time and error on the identical target. Missing columns produce NA.

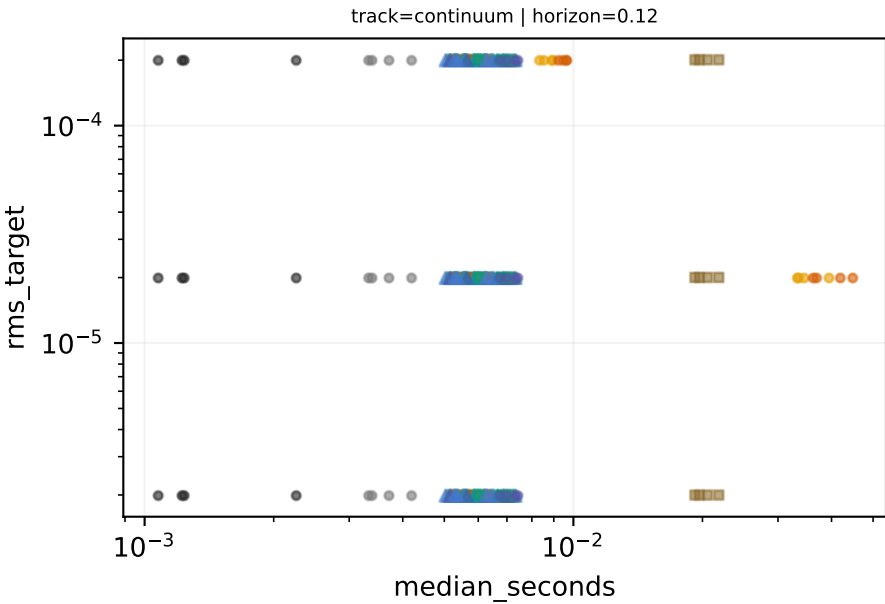
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Locked choices were made on validation; post-hoc minima inspect confirmation truth and are descriptive only. Exact tolerances, infeasible candidates and schedule identities remain in the table.

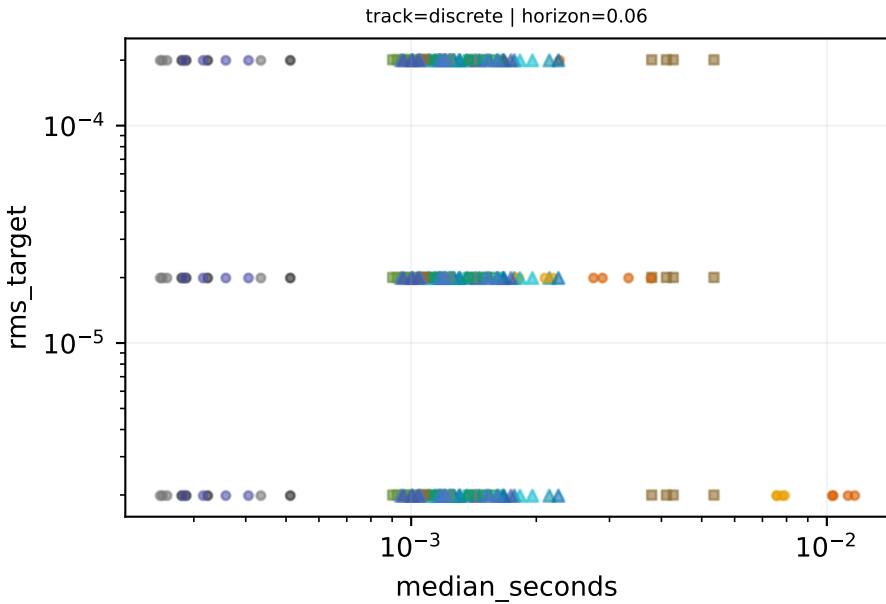
Posthoc descriptive accuracy cost frontier



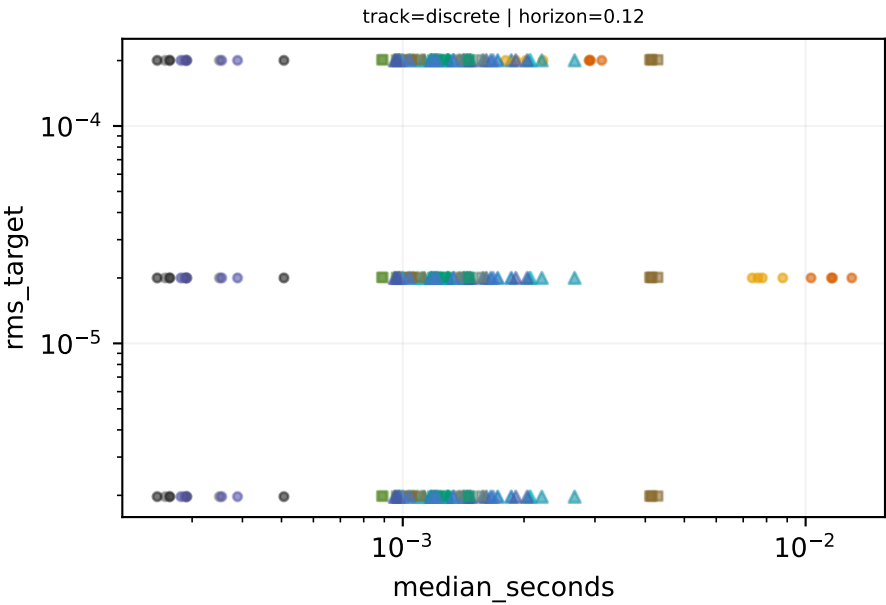
264 observations; 12 unavailable. Repeated fields are paired.



256 observations; 20 unavailable. Repeated fields are paired.



264 observations; 12 unavailable. Repeated fields are paired.

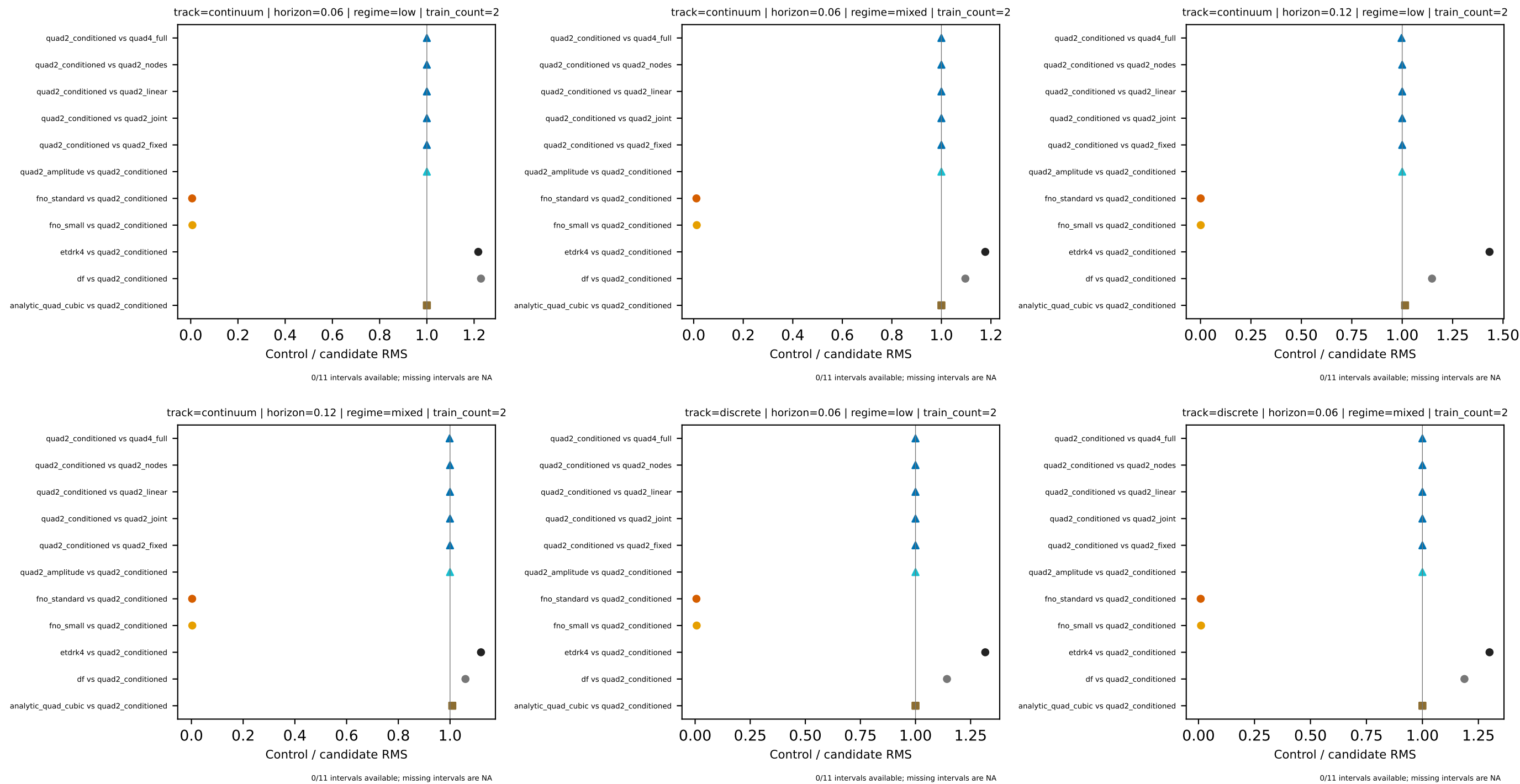


256 observations; 20 unavailable. Repeated fields are paired.

- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_conditioned | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned_full | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_fixed |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ■ Analytic control: quad4_full |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ▲ Ours: residual_quad2 |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | ● Theirs: rf |
| ▲ Ours: historical_half | | |

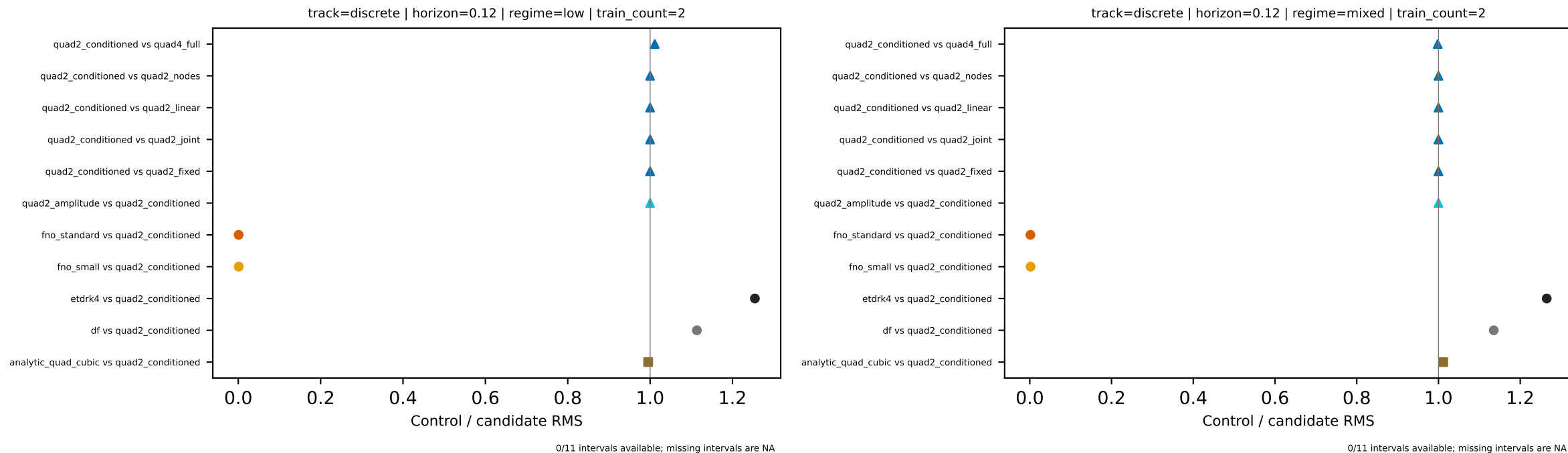
HOW TO READ: Lower-left means less inference time and error on the identical target. Missing columns produce NA.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Locked choices were made on validation; post-hoc minima inspect confirmation truth and are descriptive only. Exact tolerances, infeasible candidates and schedule identities remain in the table.

Primary paired field statistical uncertainty (1)



HOW TO READ: Control/candidate RMS ratio: above one favors the first listed candidate, below one favors its control. Field-bootstrap intervals are separate from observed training ranges.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Plot shows the declared primary family against normalization, learned-attribution and main solver controls for legibility. Every other measured pair, denominator and interval is preserved in paired_field_summaries.csv; no selection by observed effect.

Primary paired field statistical uncertainty (2)

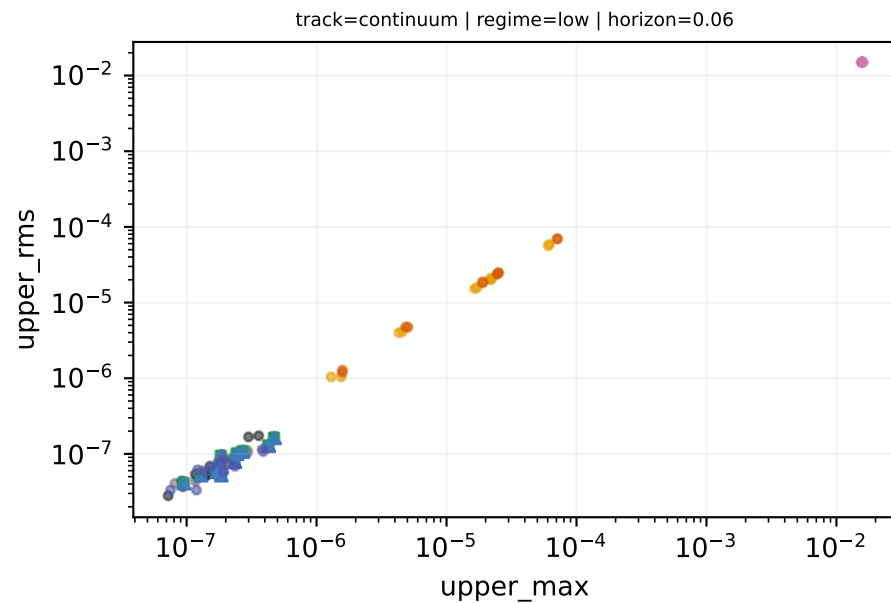


HOW TO READ: Control/candidate RMS ratio: above one favors the first listed candidate, below one favors its control. Field-bootstrap intervals are separate from observed training ranges.

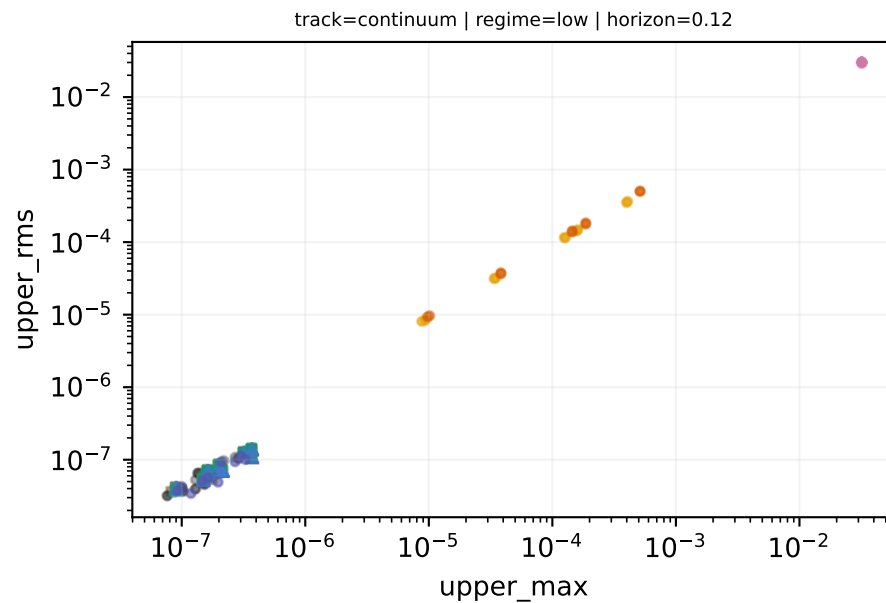
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Plot shows the declared primary family against normalization, learned-attribution and main solver controls for legibility. Every other measured pair, denominator and interval is preserved in paired_field_summaries.csv; no selection by observed effect.

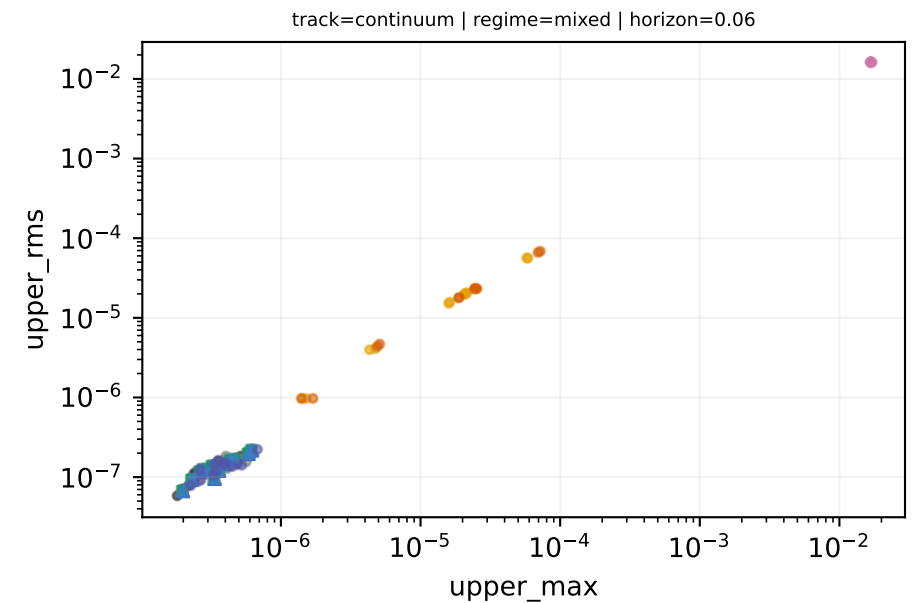
Generalization and failure map (1)



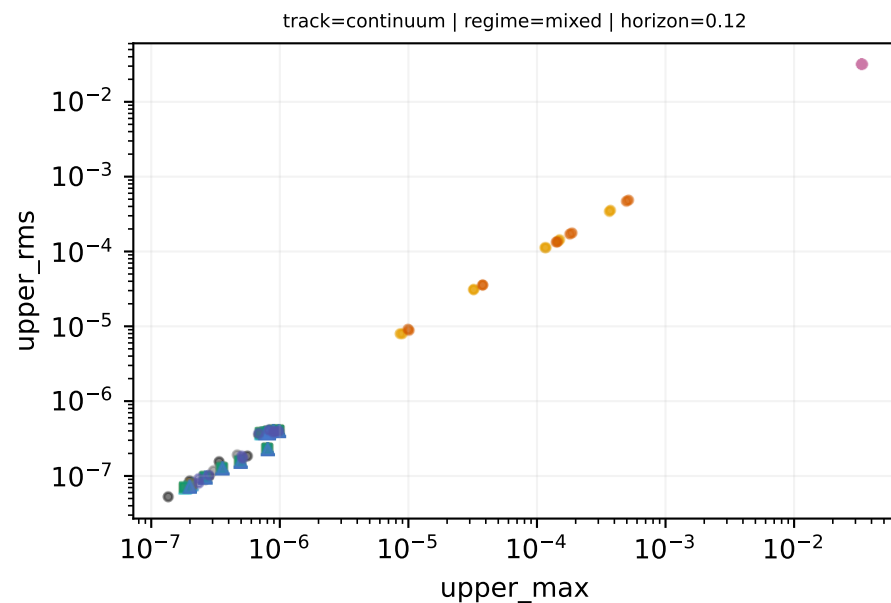
276 observations; 0 unavailable. Repeated fields are paired.



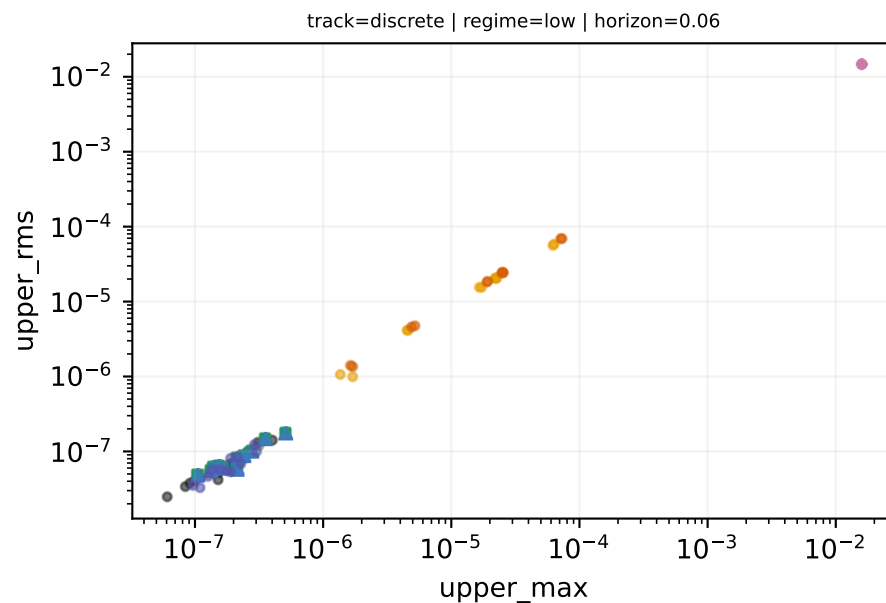
230 observations; 0 unavailable. Repeated fields are paired.



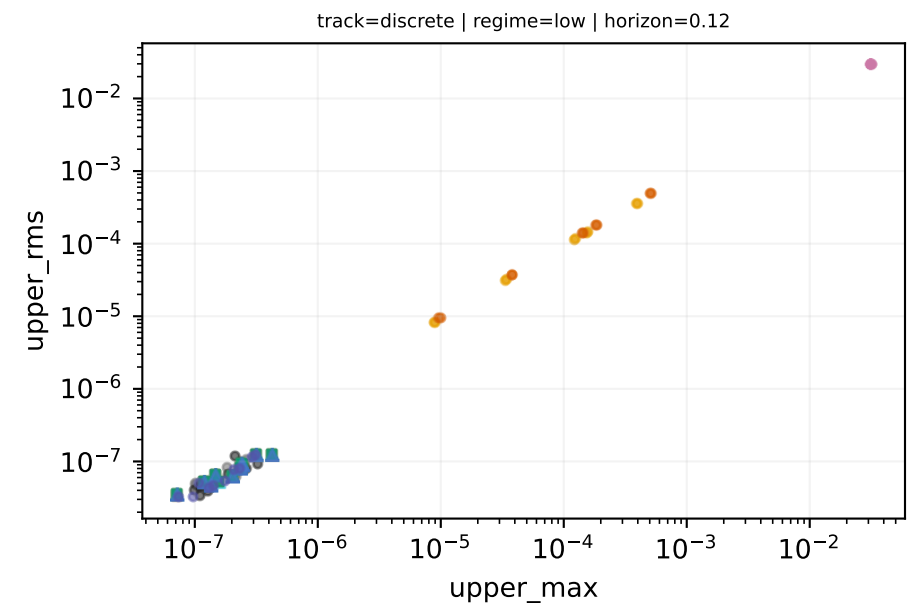
276 observations; 0 unavailable. Repeated fields are paired.



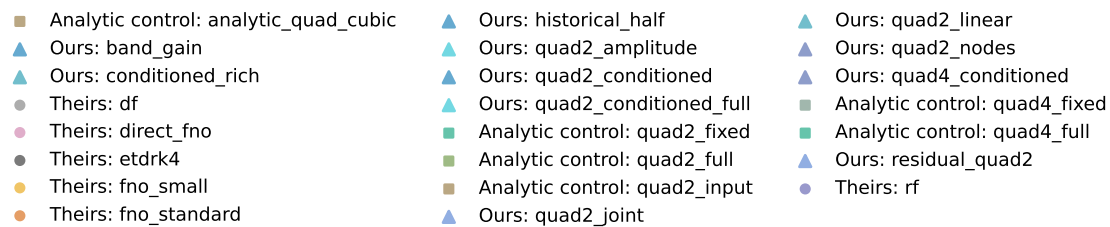
230 observations; 0 unavailable. Repeated fields are paired.



276 observations; 0 unavailable. Repeated fields are paired.



230 observations; 0 unavailable. Repeated fields are paired.



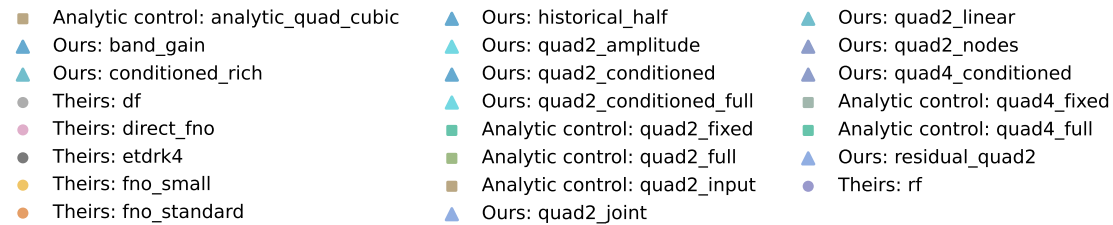
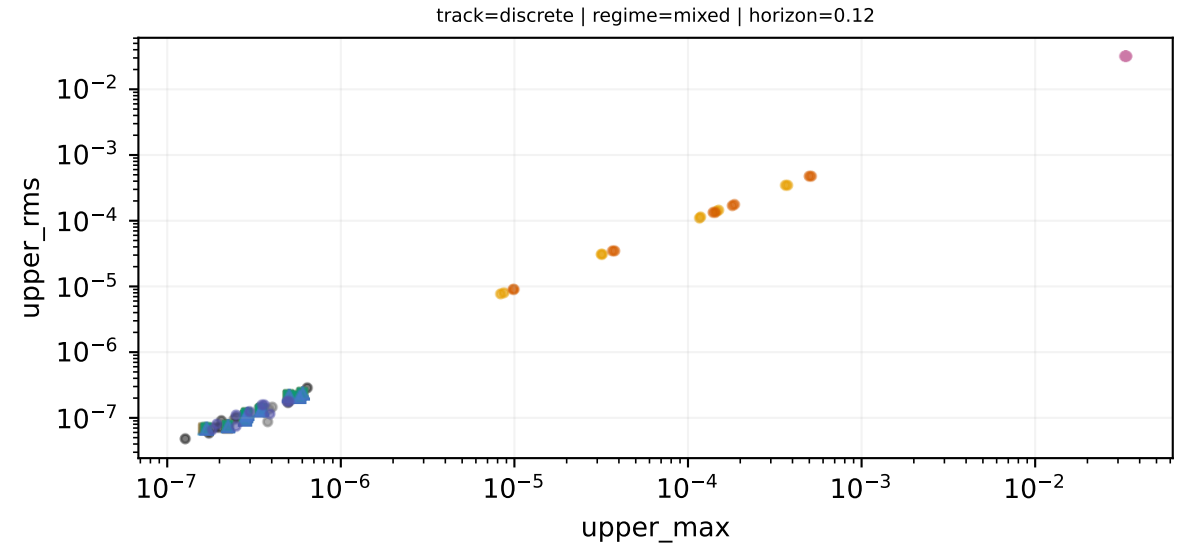
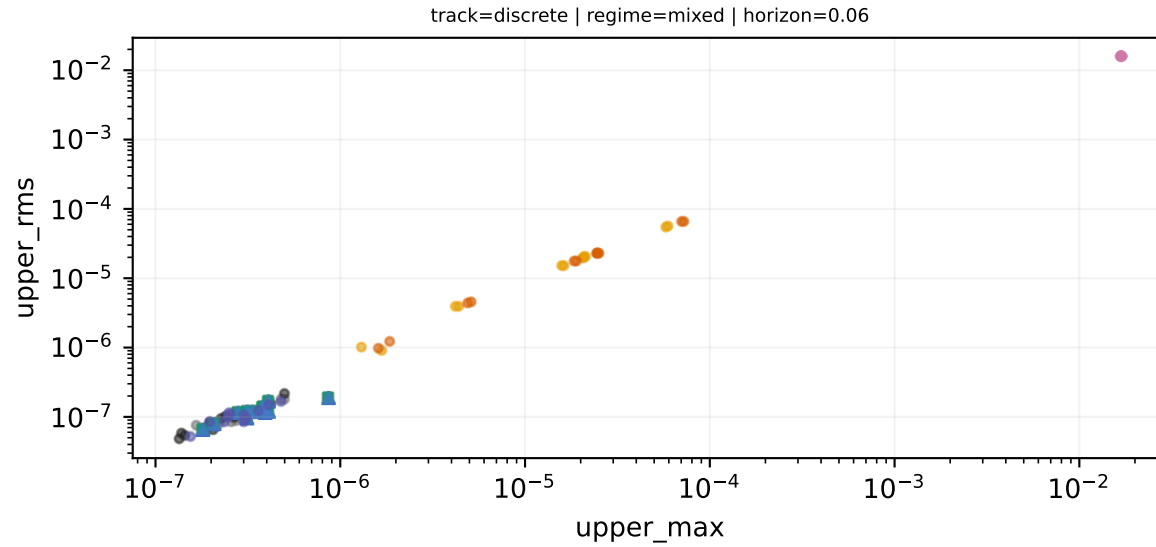
HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals.

No universal leaderboard.

Generalization and failure map (2)

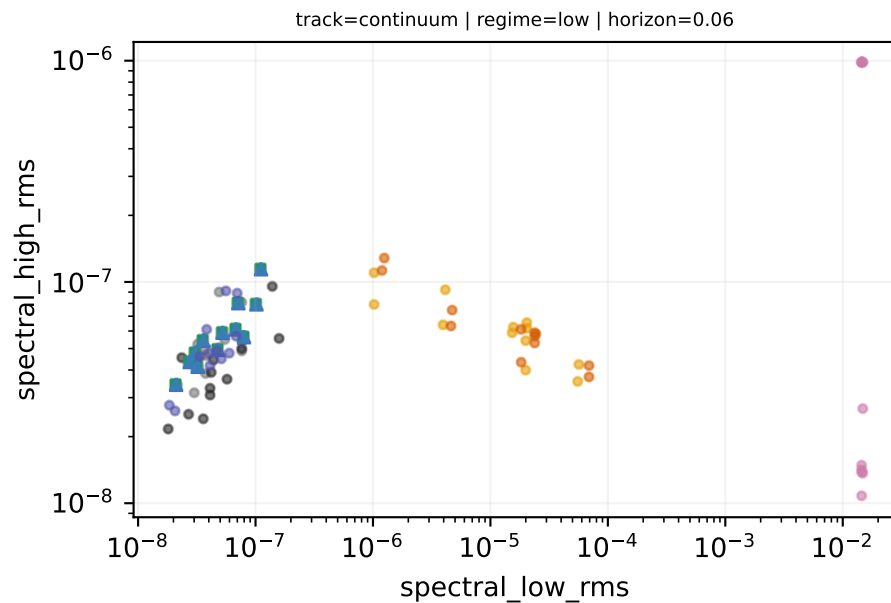


HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

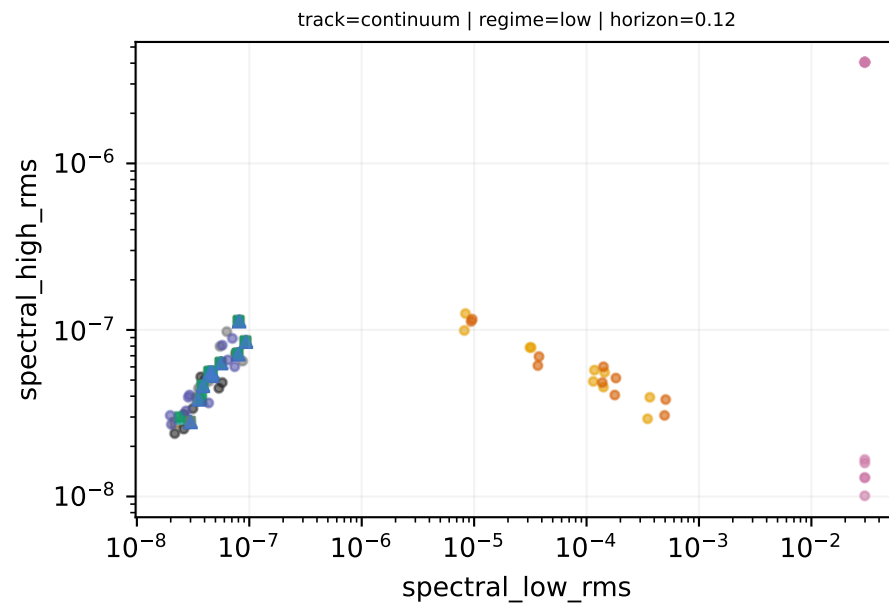
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

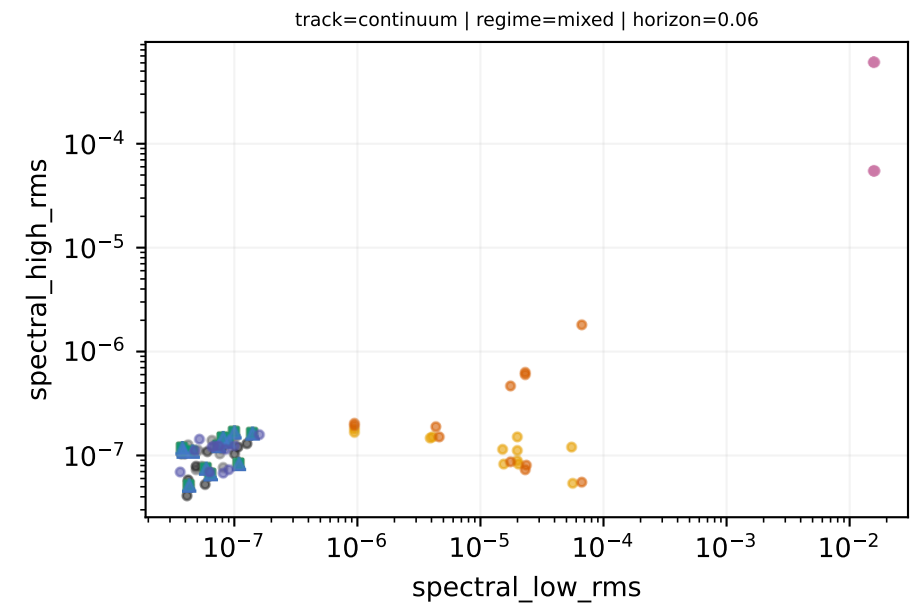
Spectral error decomposition (1)



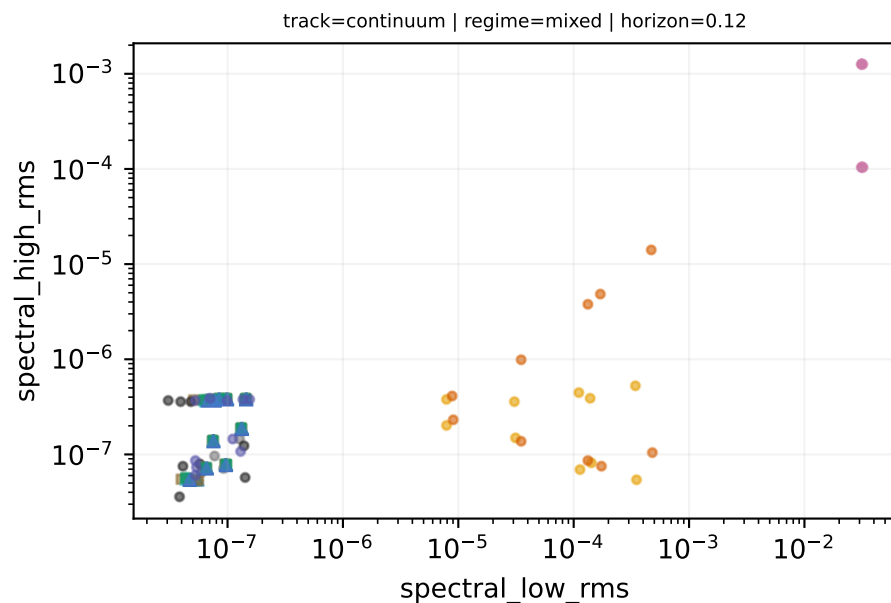
276 observations; 0 unavailable. Repeated fields are paired.



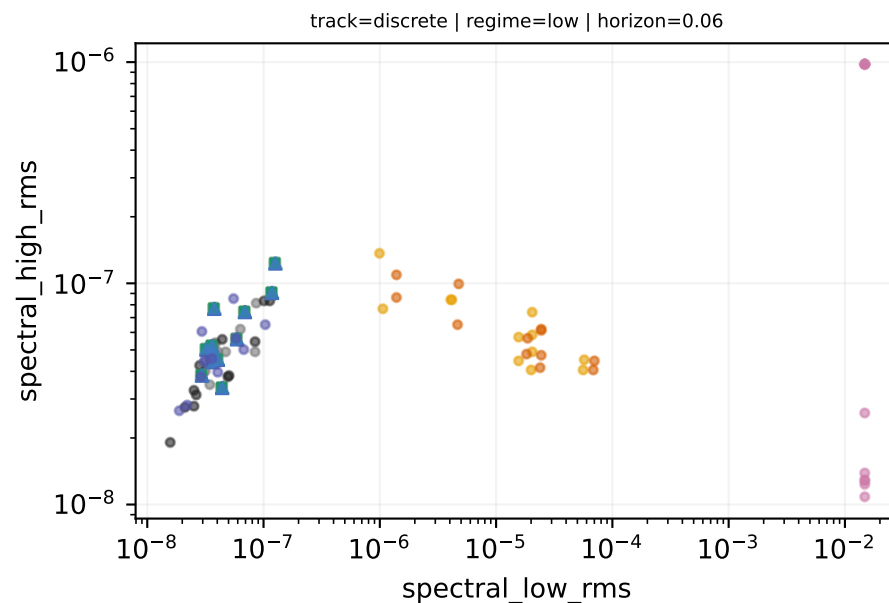
230 observations; 0 unavailable. Repeated fields are paired.



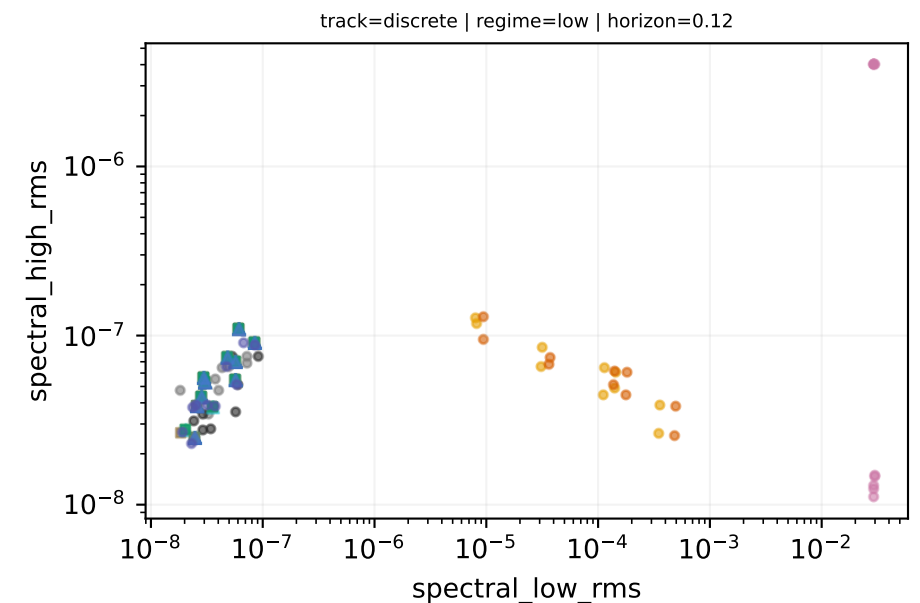
276 observations; 0 unavailable. Repeated fields are paired.



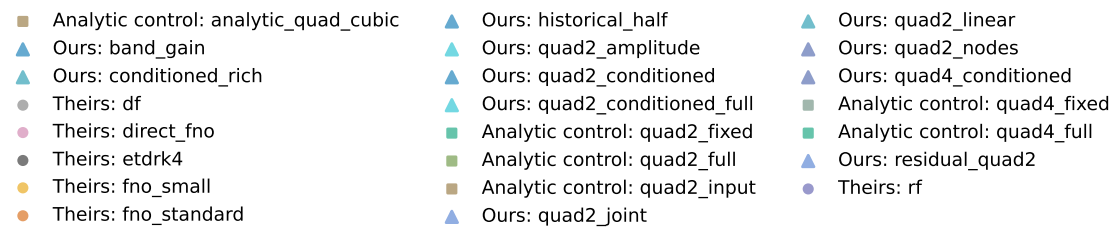
230 observations; 0 unavailable. Repeated fields are paired.



276 observations; 0 unavailable. Repeated fields are paired.



230 observations; 0 unavailable. Repeated fields are paired.

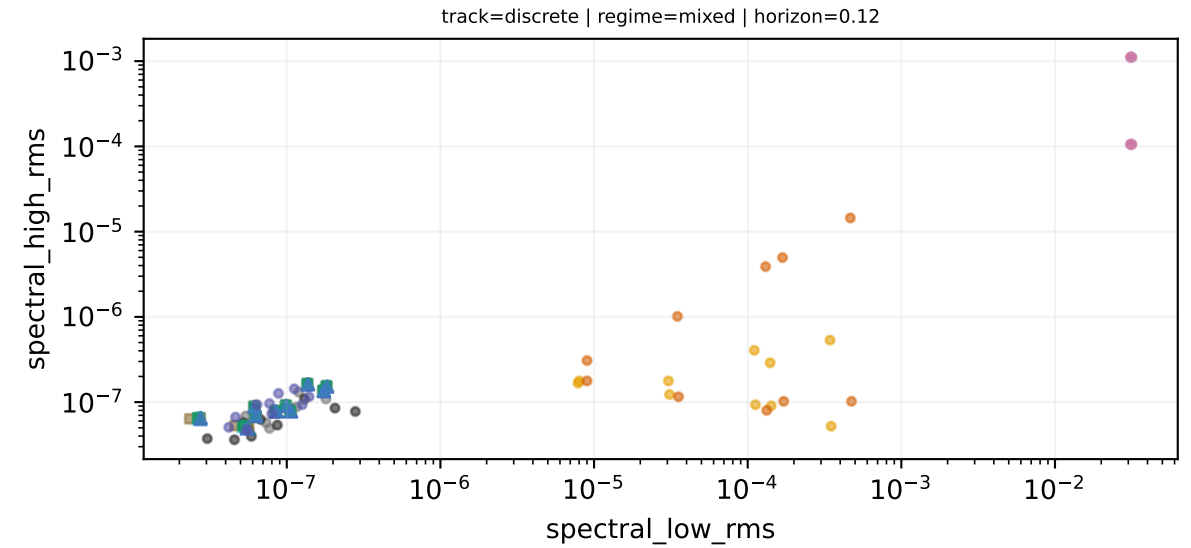
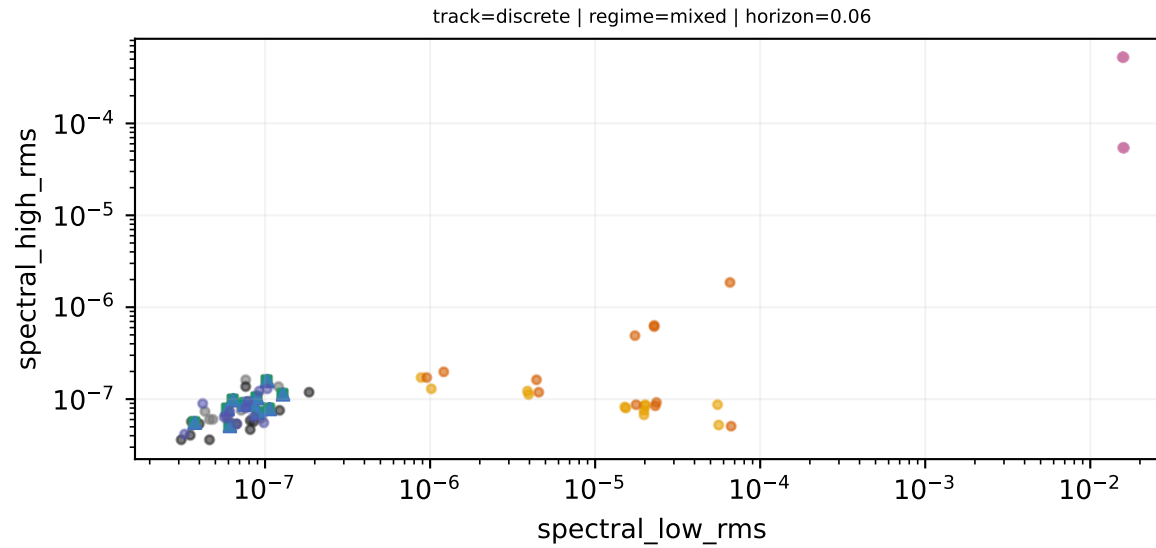


HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Spectral error decomposition (2)



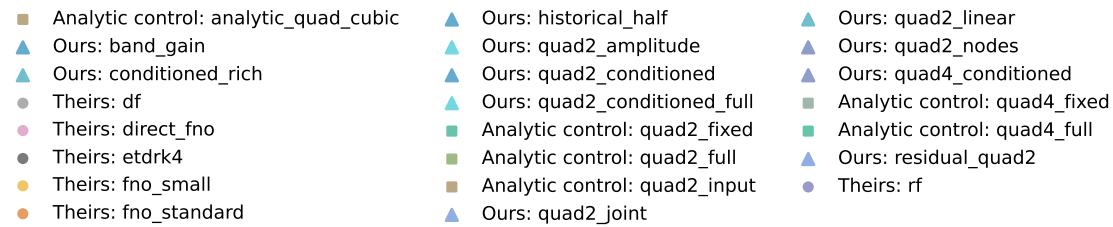
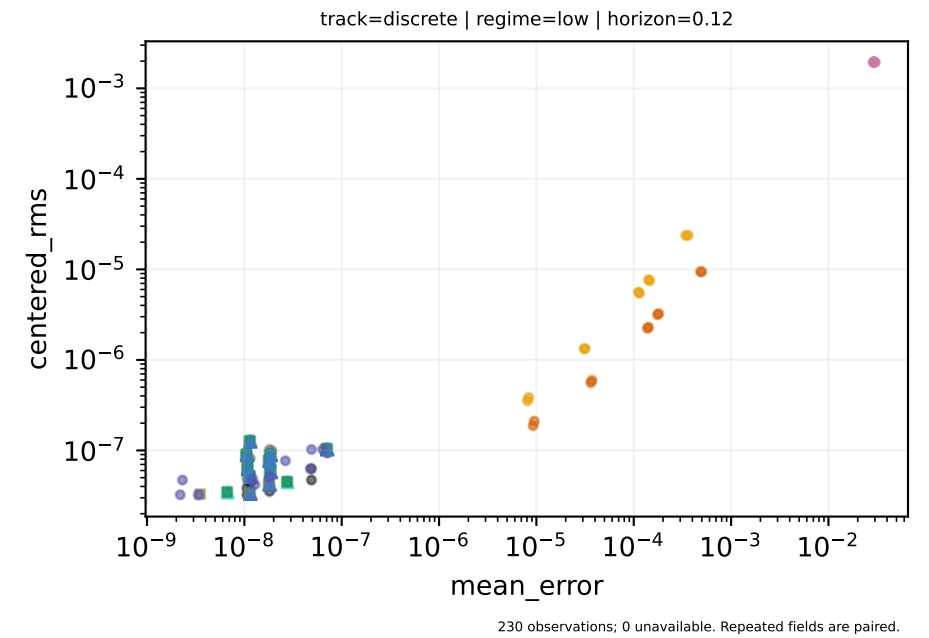
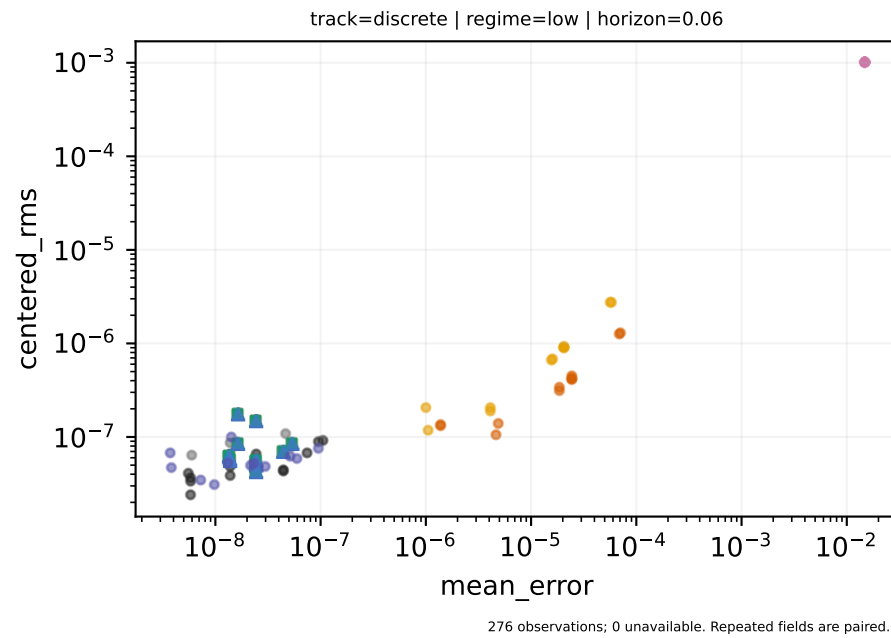
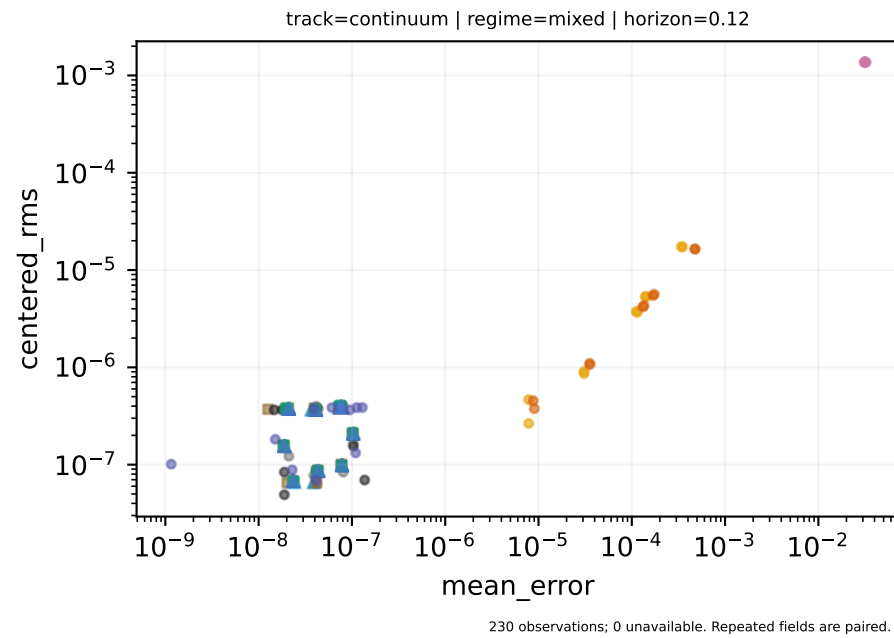
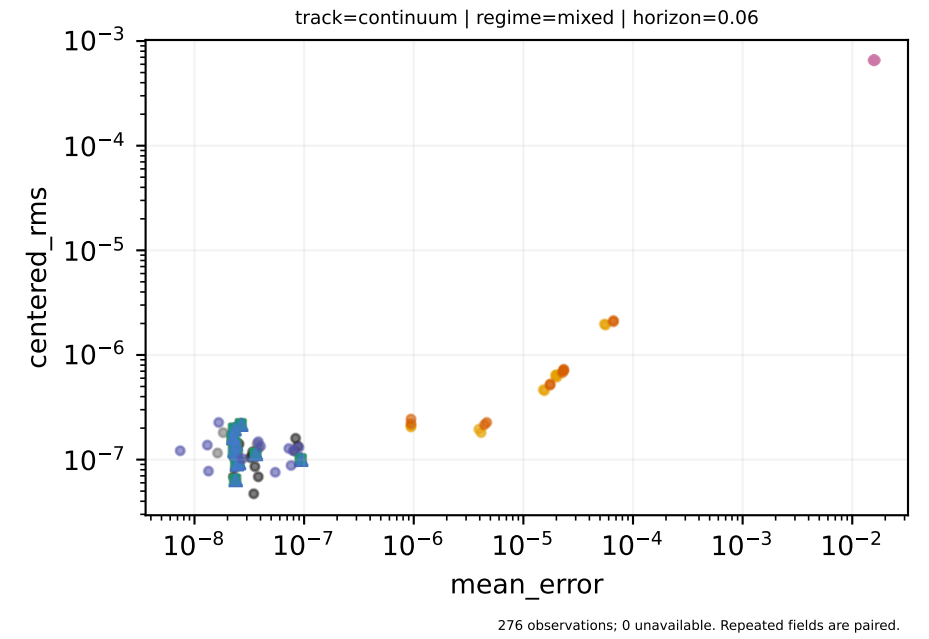
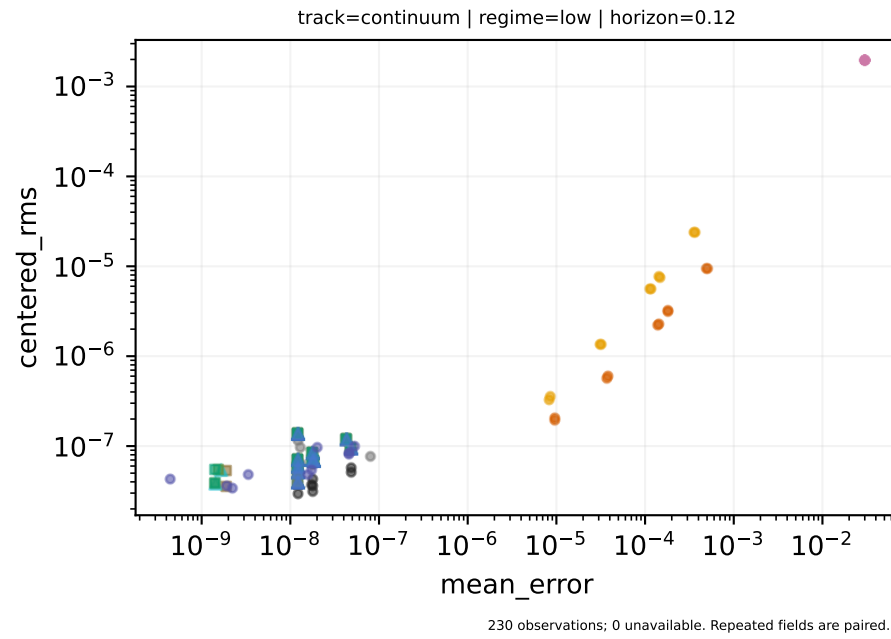
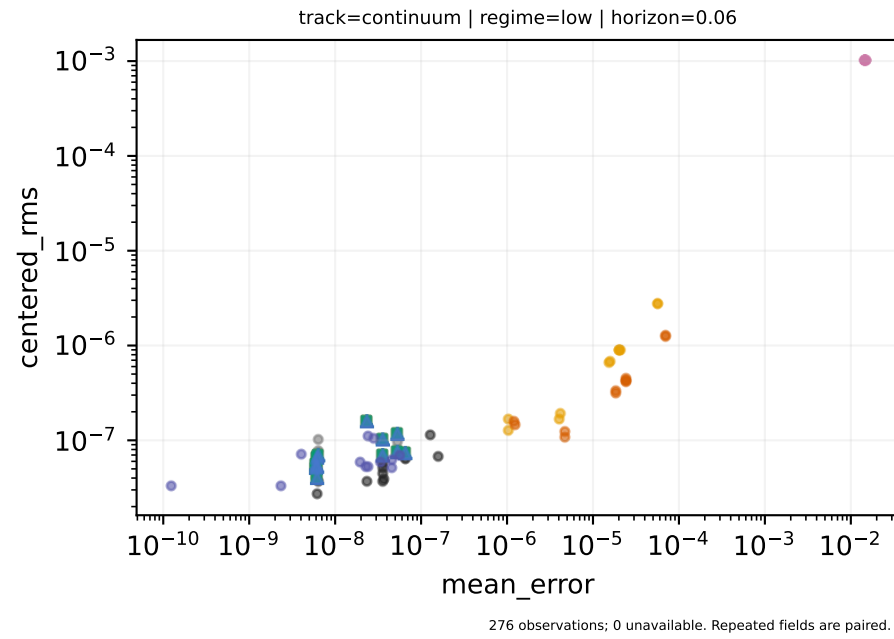
- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: historical_half | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ▲ Ours: quad2_conditioned_full | ■ Analytic control: quad4_fixed |
| ● Theirs: direct_fno | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_full |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ▲ Ours: residual_quad2 |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ● Theirs: rf |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | |

HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Mean and spatial error (1)



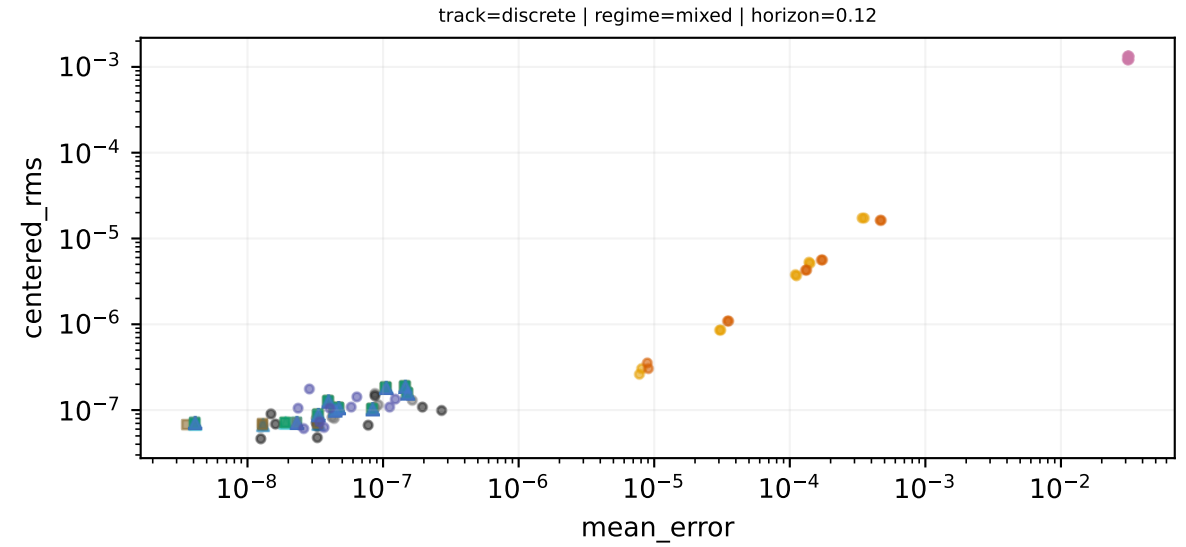
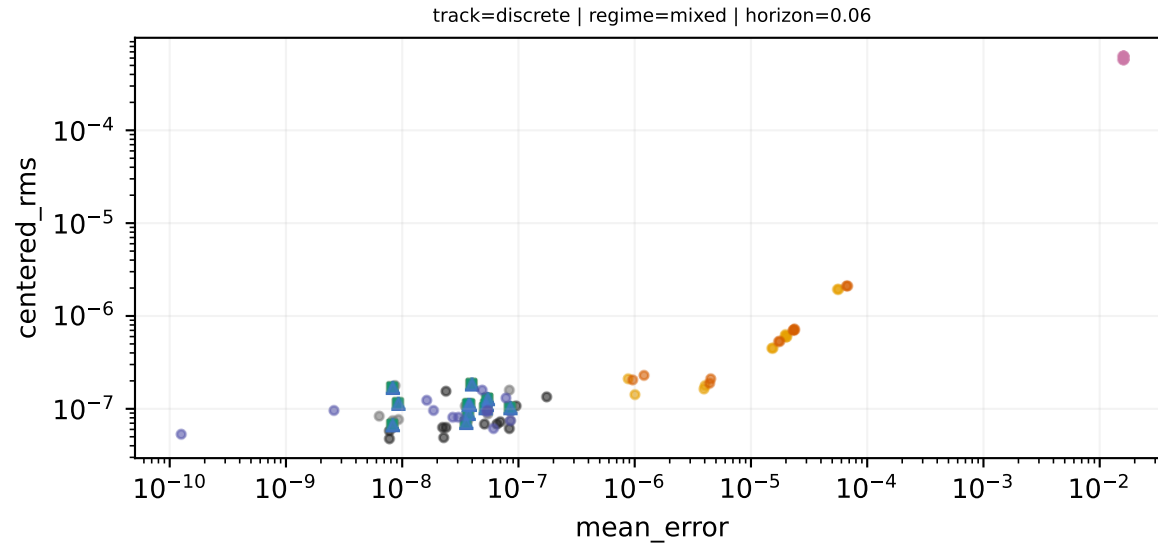
HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals.

No universal leaderboard.

Mean and spatial error (2)



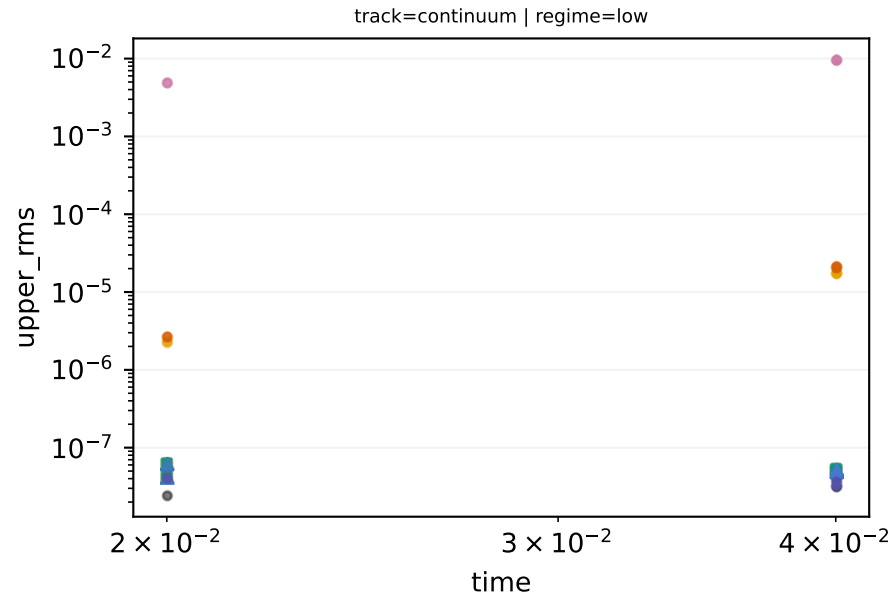
- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: historical_half | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ▲ Ours: quad2_conditioned_full | ■ Analytic control: quad4_fixed |
| ● Theirs: direct_fno | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_full |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ▲ Ours: residual_quad2 |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ● Theirs: rf |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | |

HOW TO READ: Lower-left is better: both error components are smaller on matched targets. Failures/unaccepted references stay in raw data.

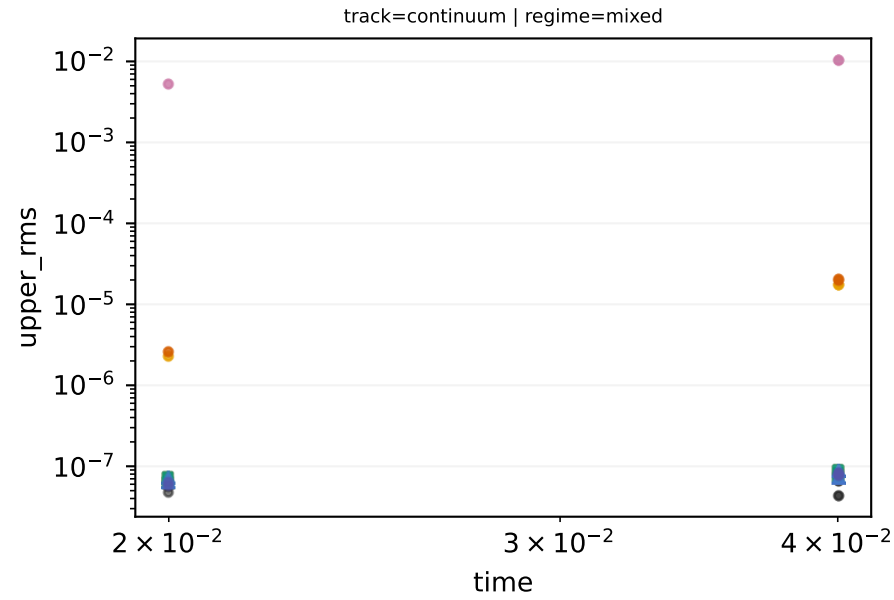
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

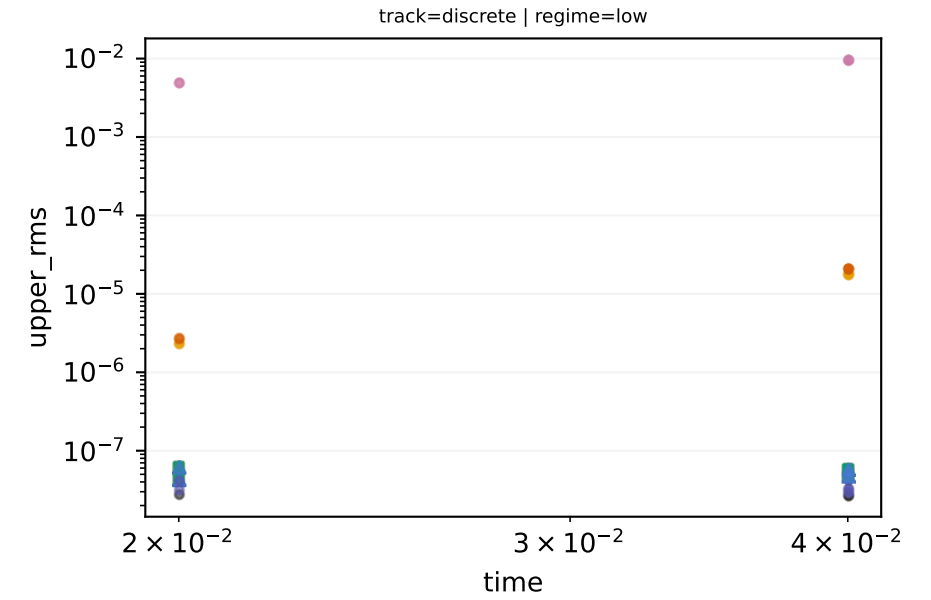
Rollout intermediate accuracy and stability



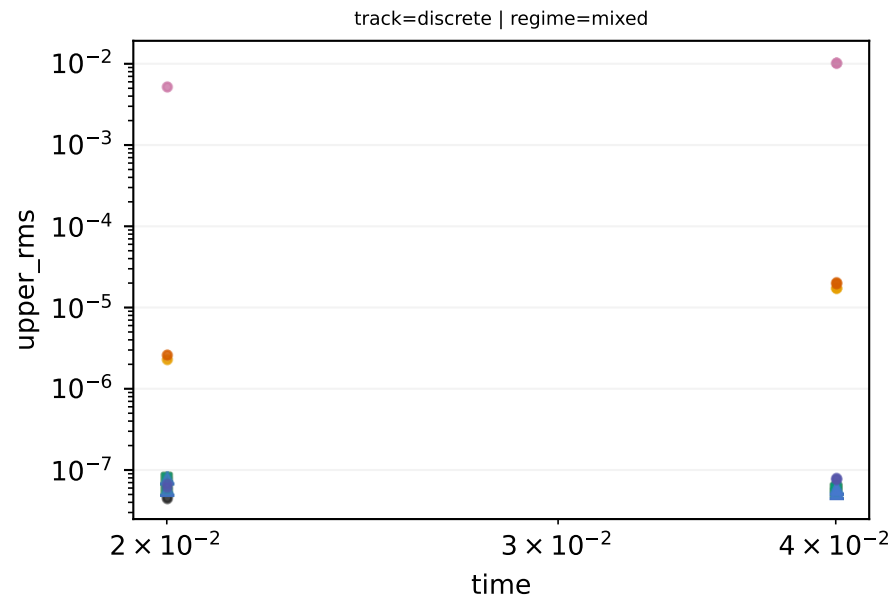
138 observations; 0 unavailable. Repeated fields are paired.



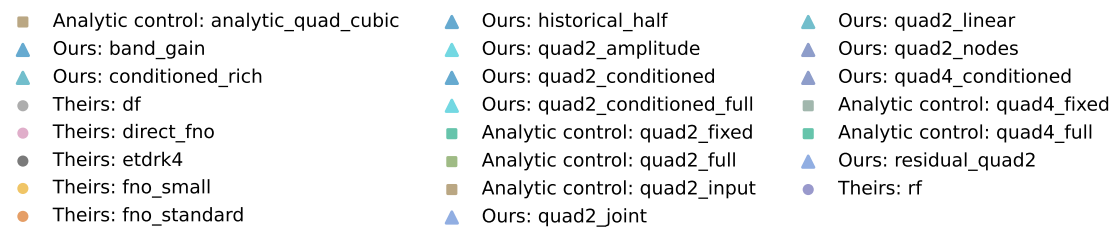
138 observations; 0 unavailable. Repeated fields are paired.



138 observations; 0 unavailable. Repeated fields are paired.

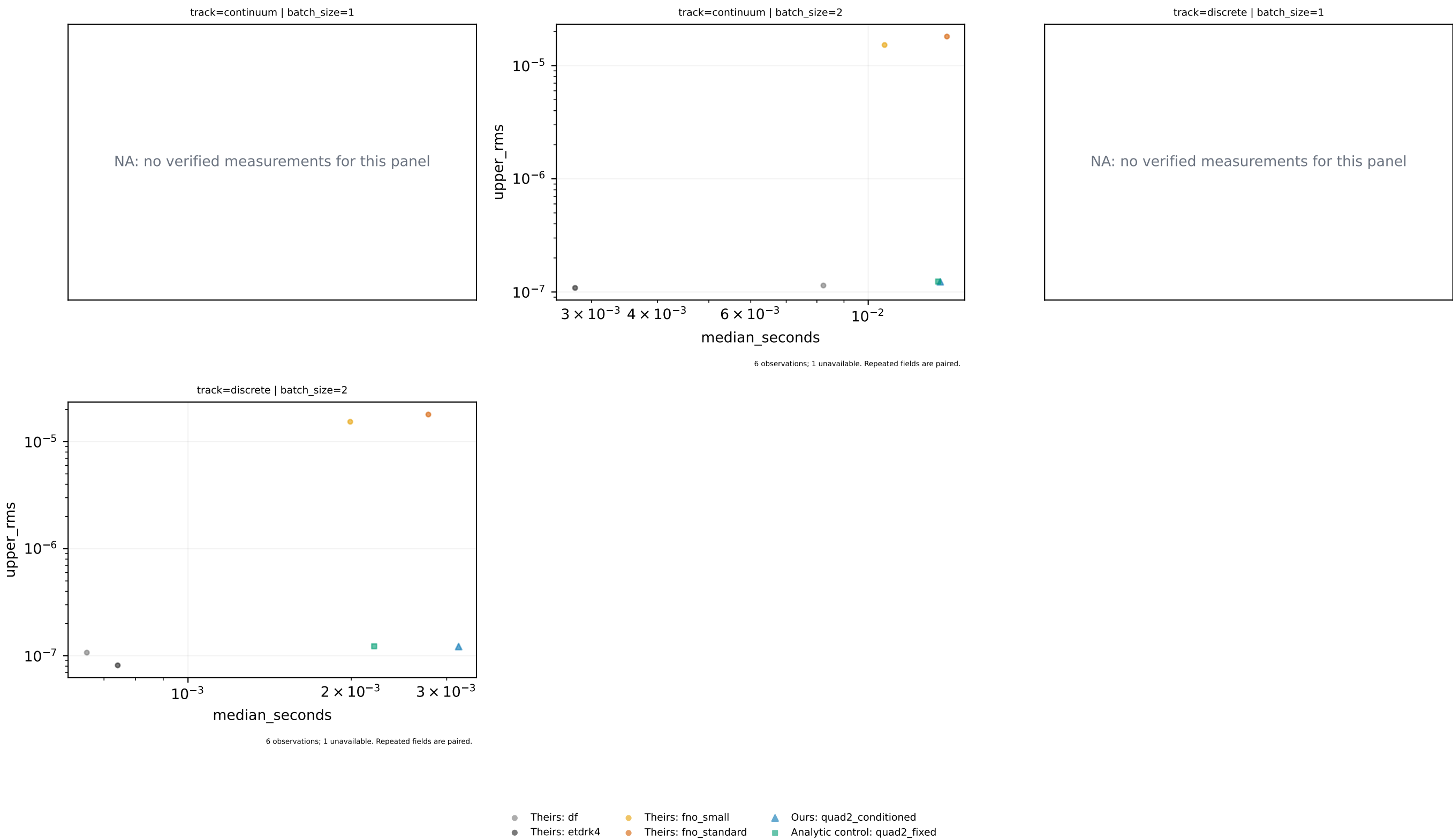


138 observations; 0 unavailable. Repeated fields are paired.



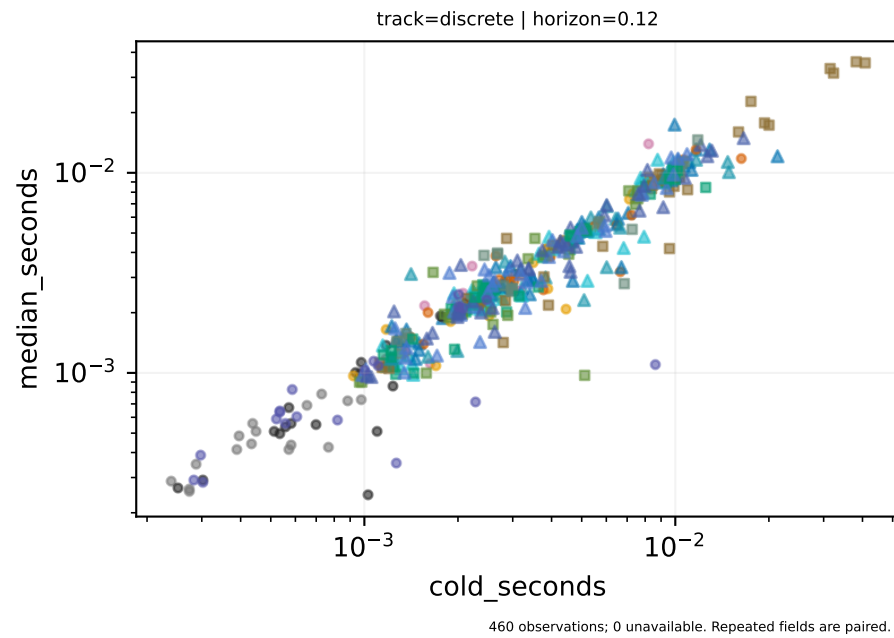
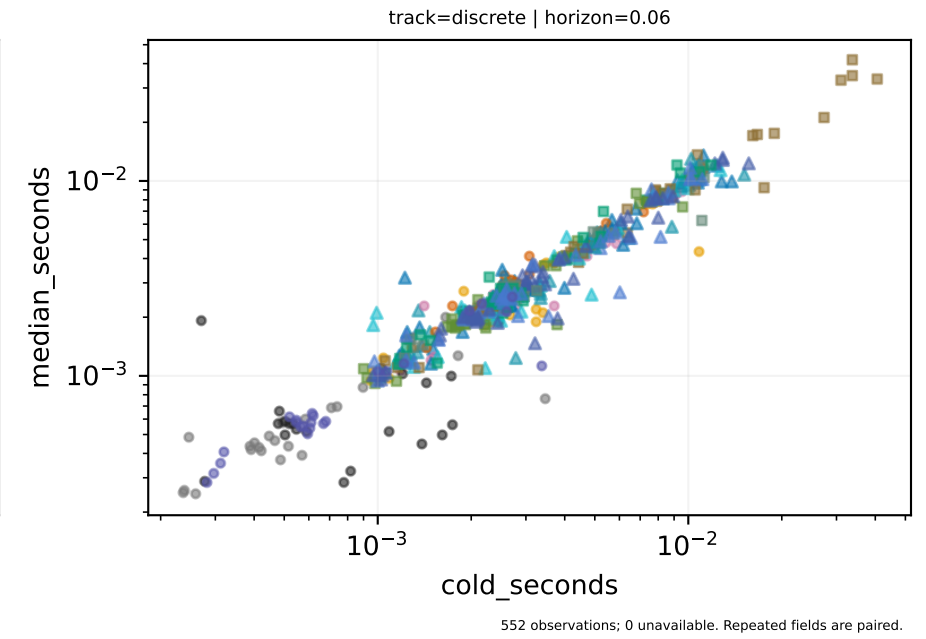
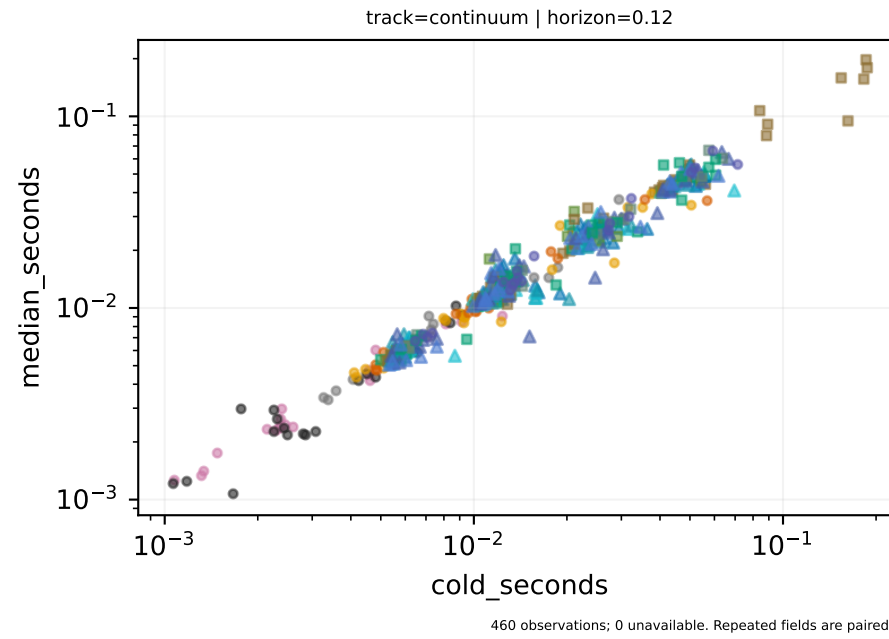
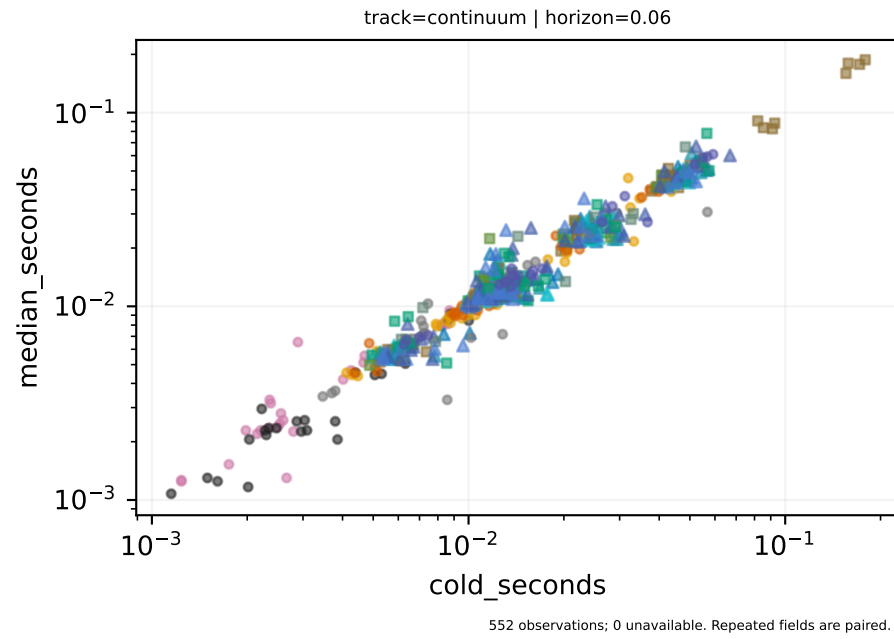
HOW TO READ: Lower error is better at the same requested time. Right is a longer horizon, not necessarily greater quality.
 METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
 SCOPE: Each intermediate truth must have an accepted reference. No interpolated teacher is invented; absent trajectory measurements remain NA.

Accuracy qualified scaling



HOW TO READ: Lower-left is faster and more accurate. Large-grid timings without qualified error are not speed wins.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals.
No universal leaderboard.

Cold versus warm latency



- | | | |
|---|---------------------------------|---------------------------------|
| ■ Analytic control: analytic_quad_cubic | ▲ Ours: historical_half | ▲ Ours: quad2_linear |
| ▲ Ours: band_gain | ▲ Ours: quad2_amplitude | ▲ Ours: quad2_nodes |
| ▲ Ours: conditioned_rich | ▲ Ours: quad2_conditioned | ▲ Ours: quad4_conditioned |
| ● Theirs: df | ▲ Ours: quad2_conditioned_full | ■ Analytic control: quad4_fixed |
| ● Theirs: direct_fno | ■ Analytic control: quad2_fixed | ■ Analytic control: quad4_full |
| ● Theirs: etdrk4 | ■ Analytic control: quad2_full | ▲ Ours: residual_quad2 |
| ● Theirs: fno_small | ■ Analytic control: quad2_input | ● Theirs: rf |
| ● Theirs: fno_standard | ▲ Ours: quad2_joint | |

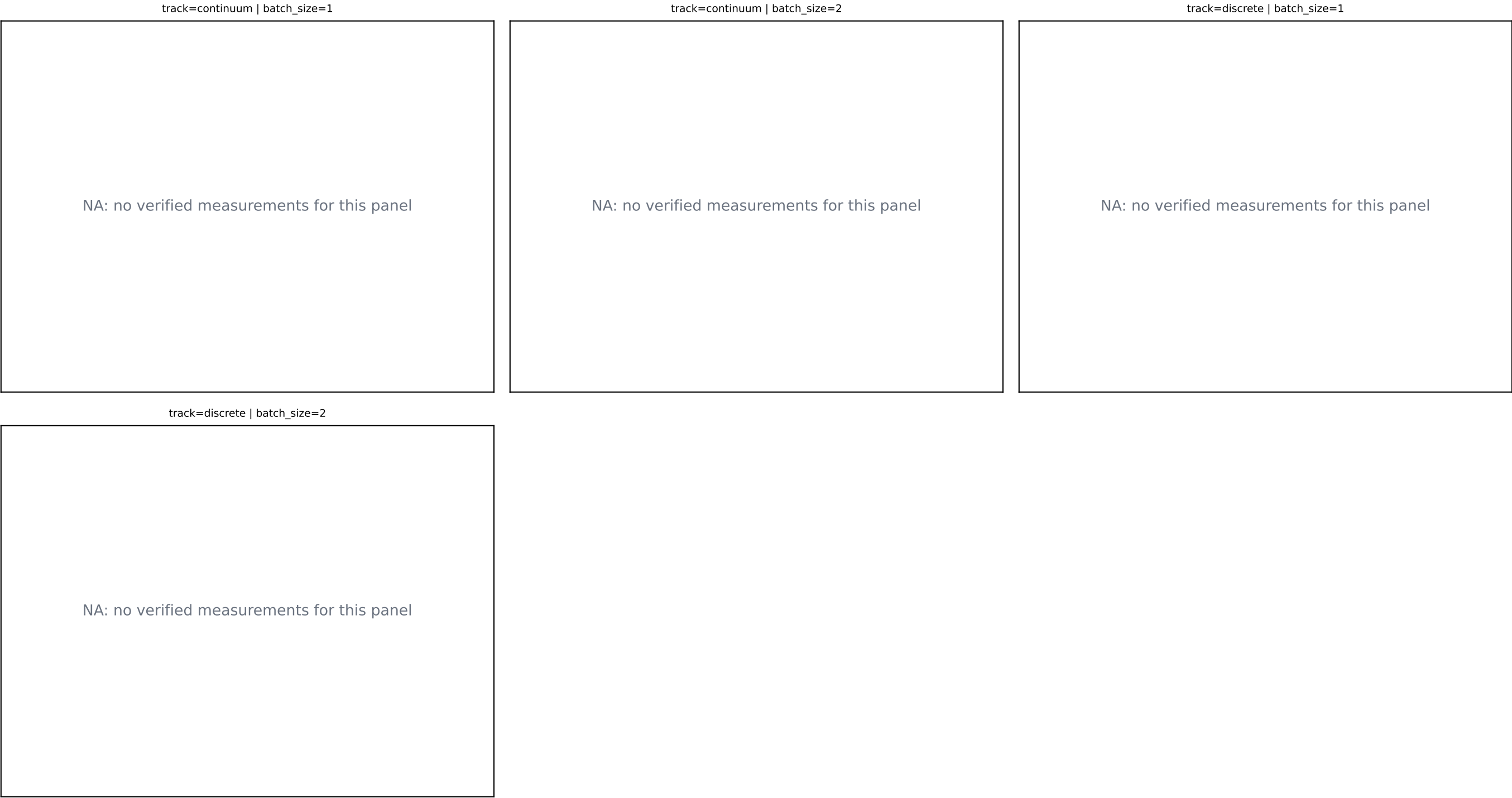
HOW TO READ: Lower-left is less first-invocation and warmed runtime. Both include the declared complete operation.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals.

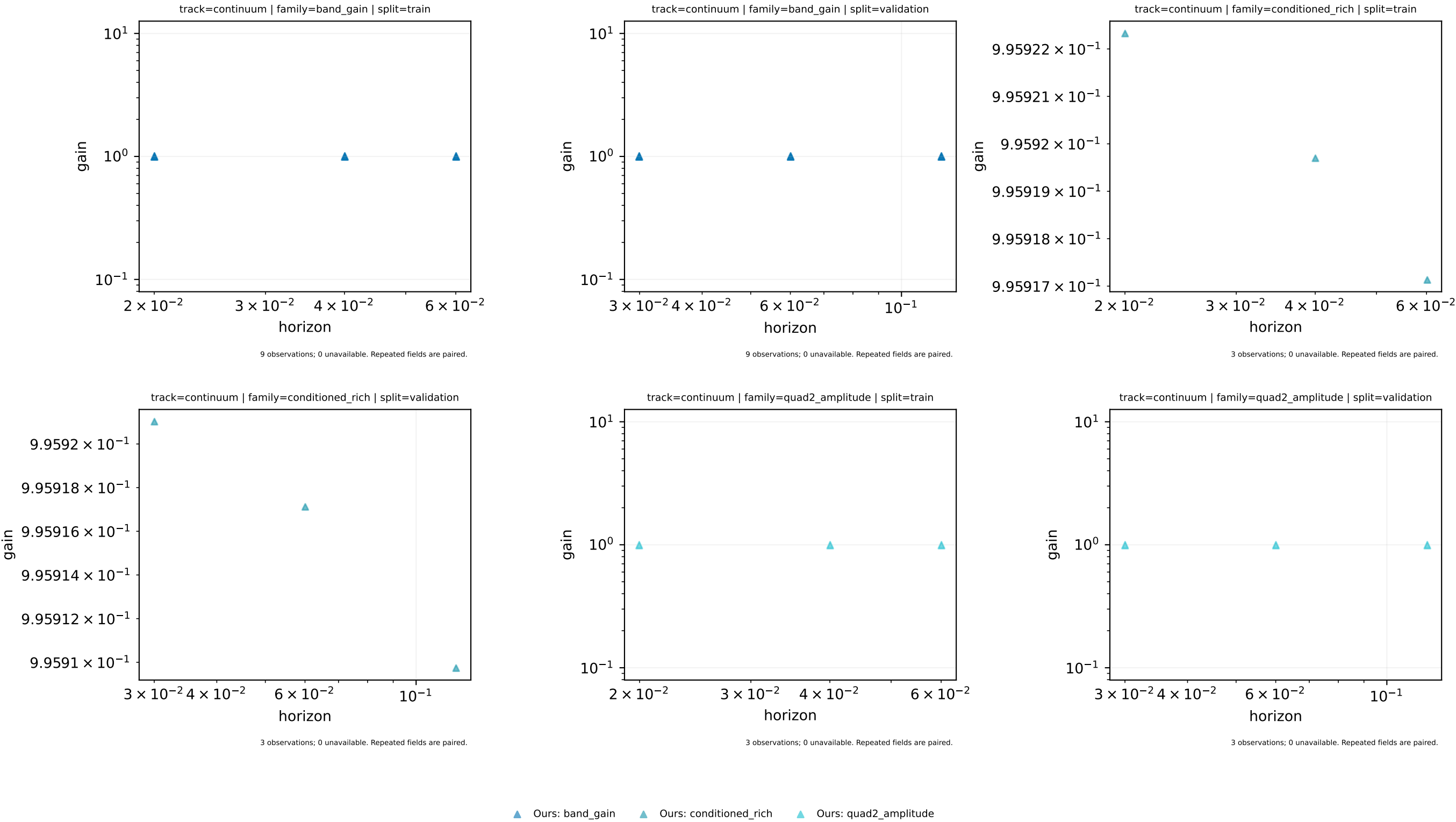
No universal leaderboard.

Peak memory and warmed cost



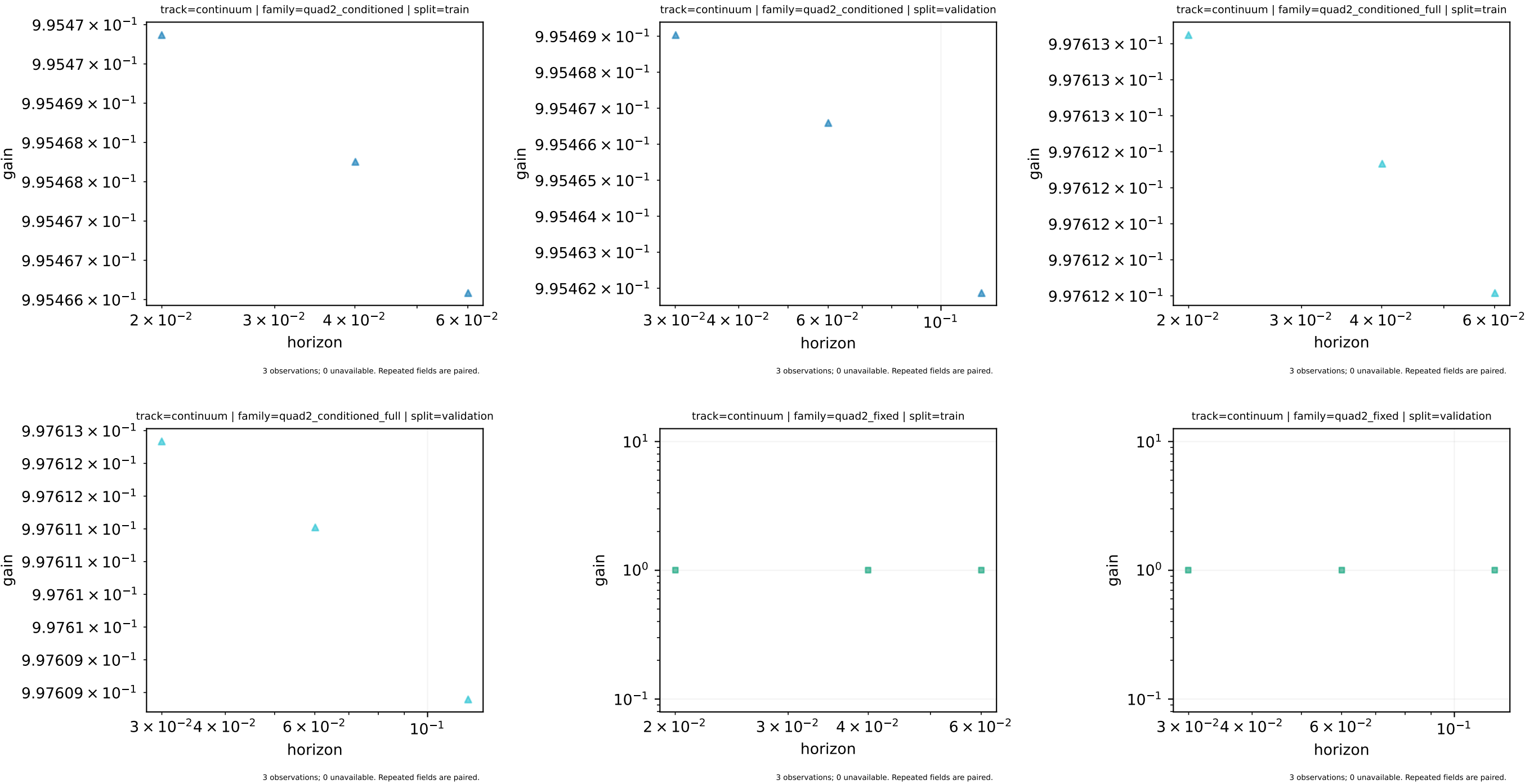
HOW TO READ: Lower-left is less observed memory and time for matched accuracy/workload. CPU process memory is not GPU VRAM.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals.
No universal leaderboard.

Learned effective gain response (1)



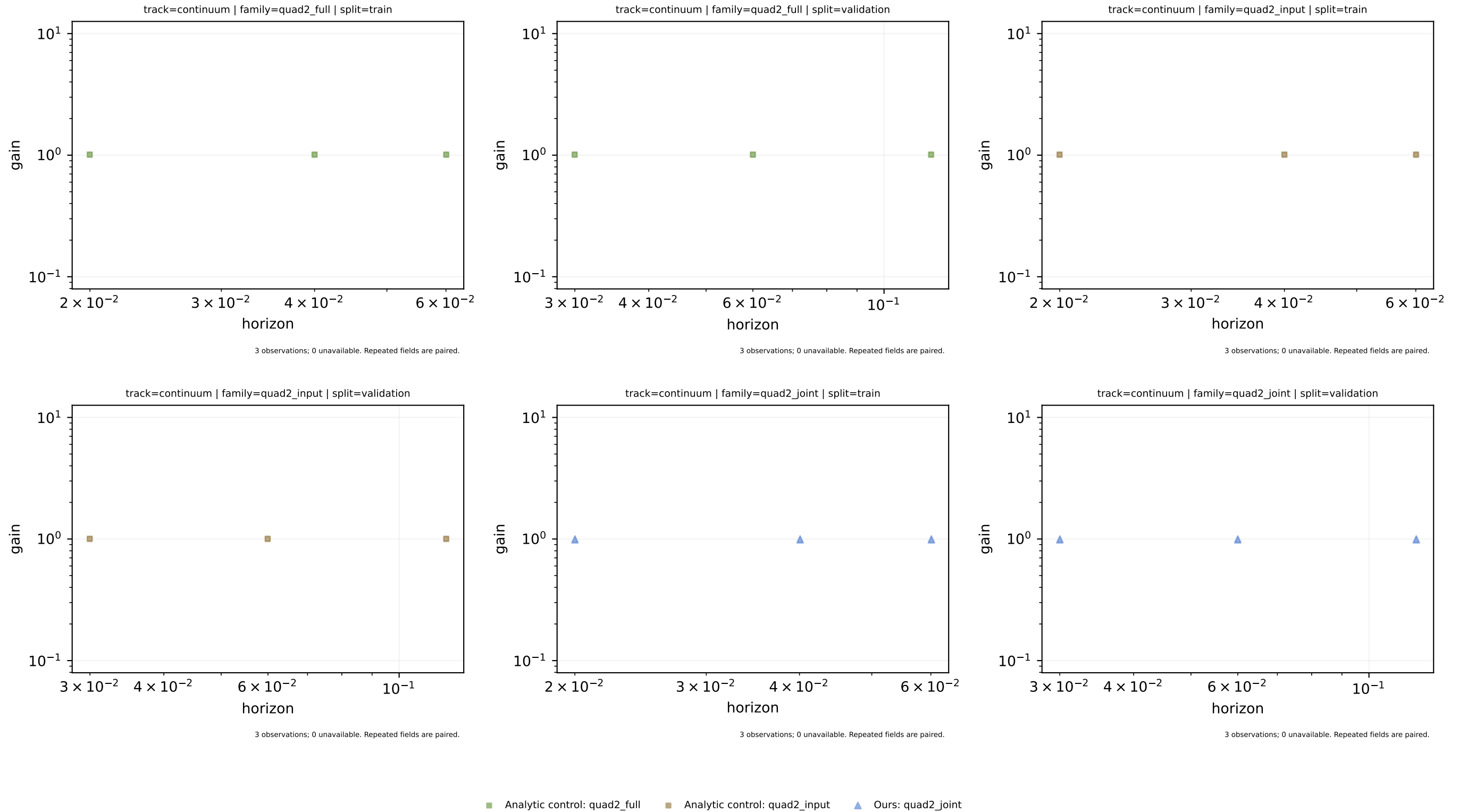
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (2)



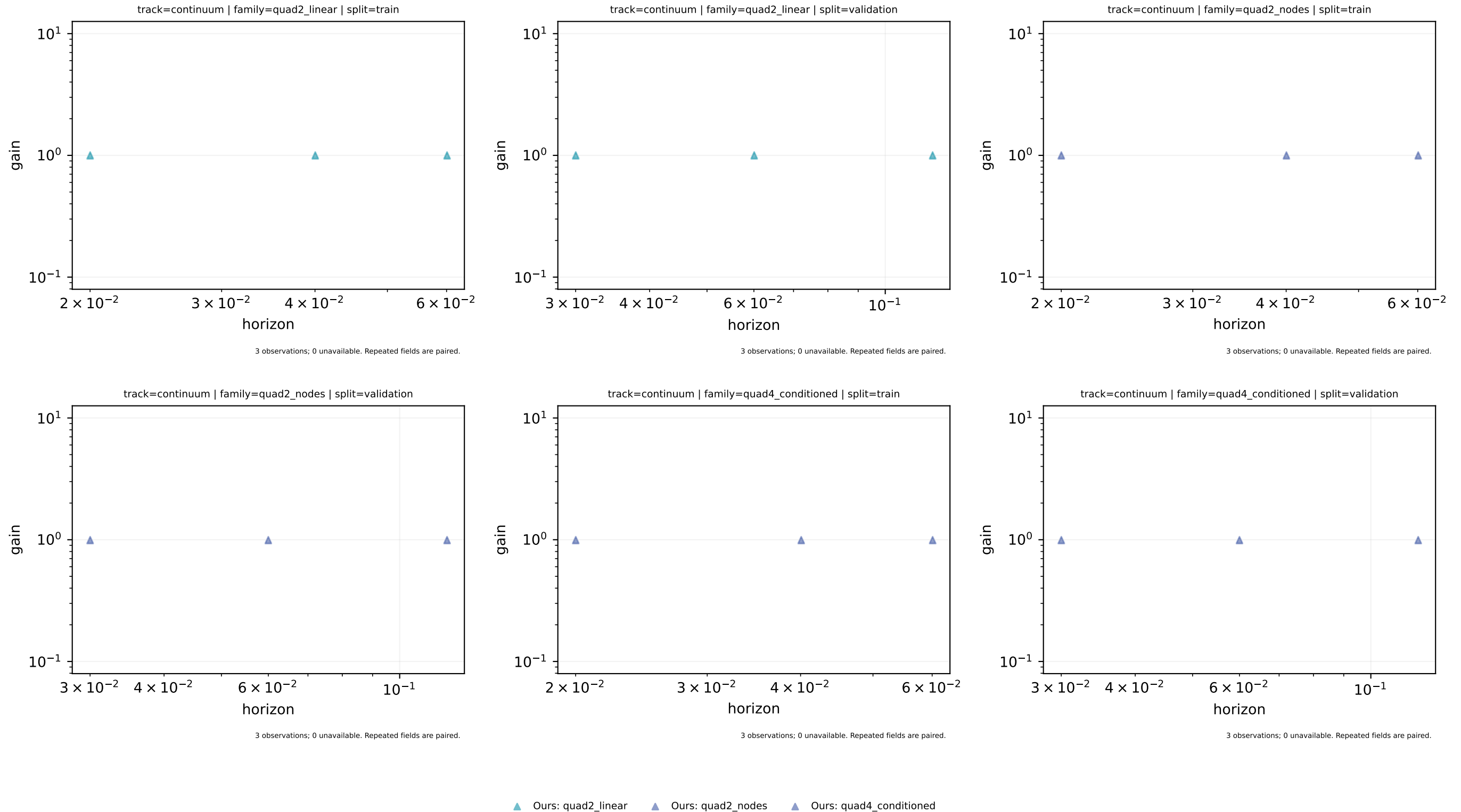
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (3)



HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
 METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
 SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
 Similar effective gains do not prove parameter identifiability.

Learned effective gain response (4)



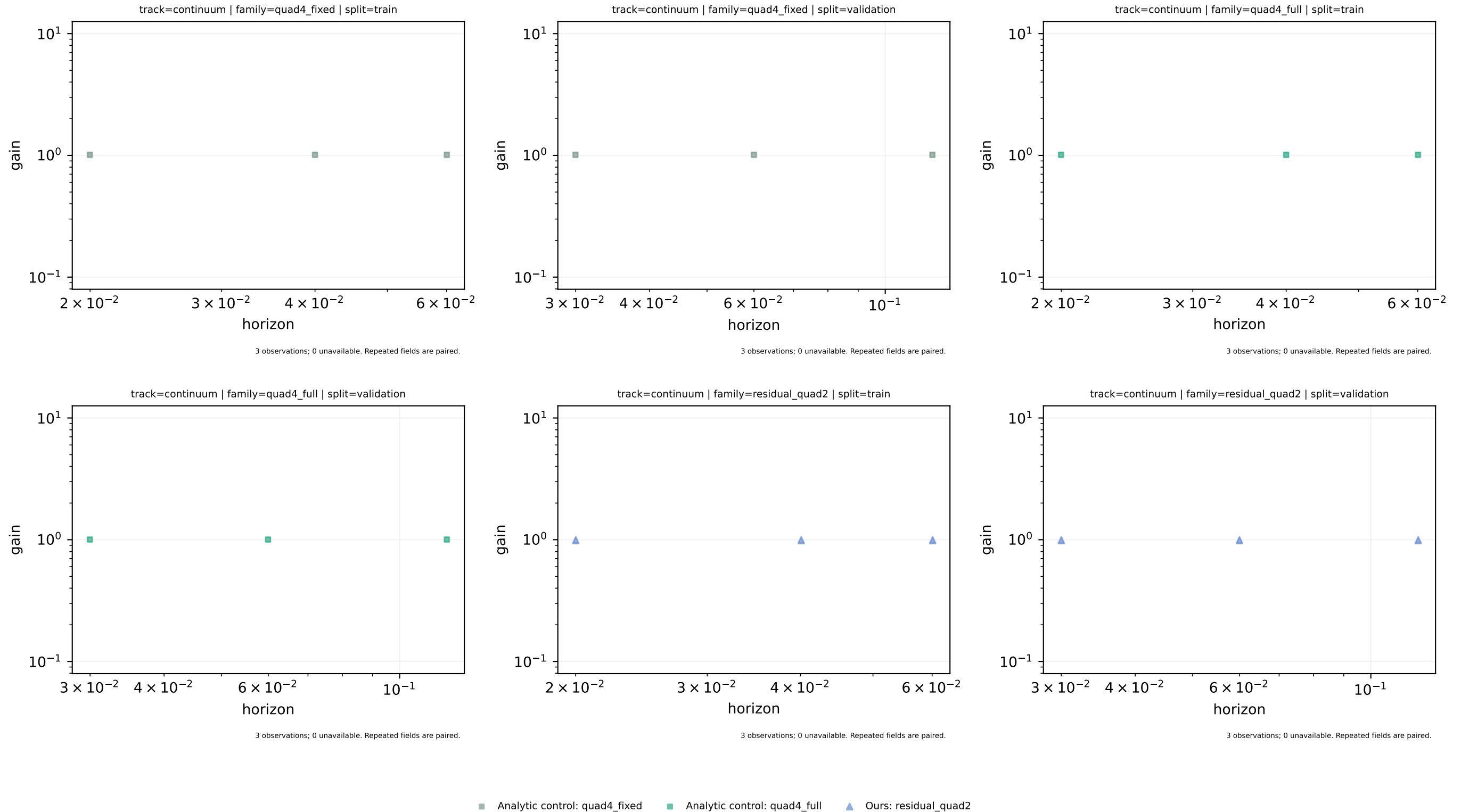
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.

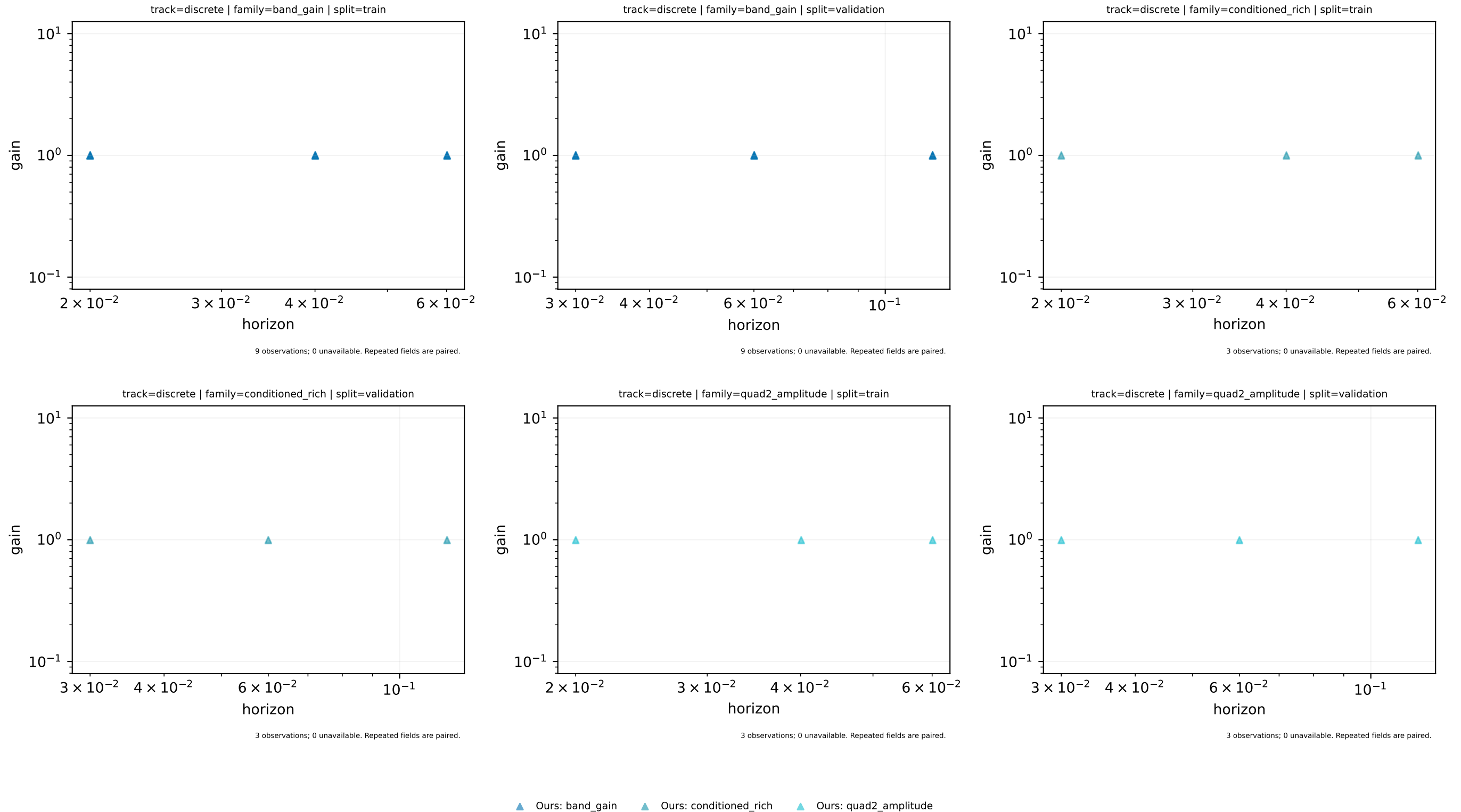
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (5)



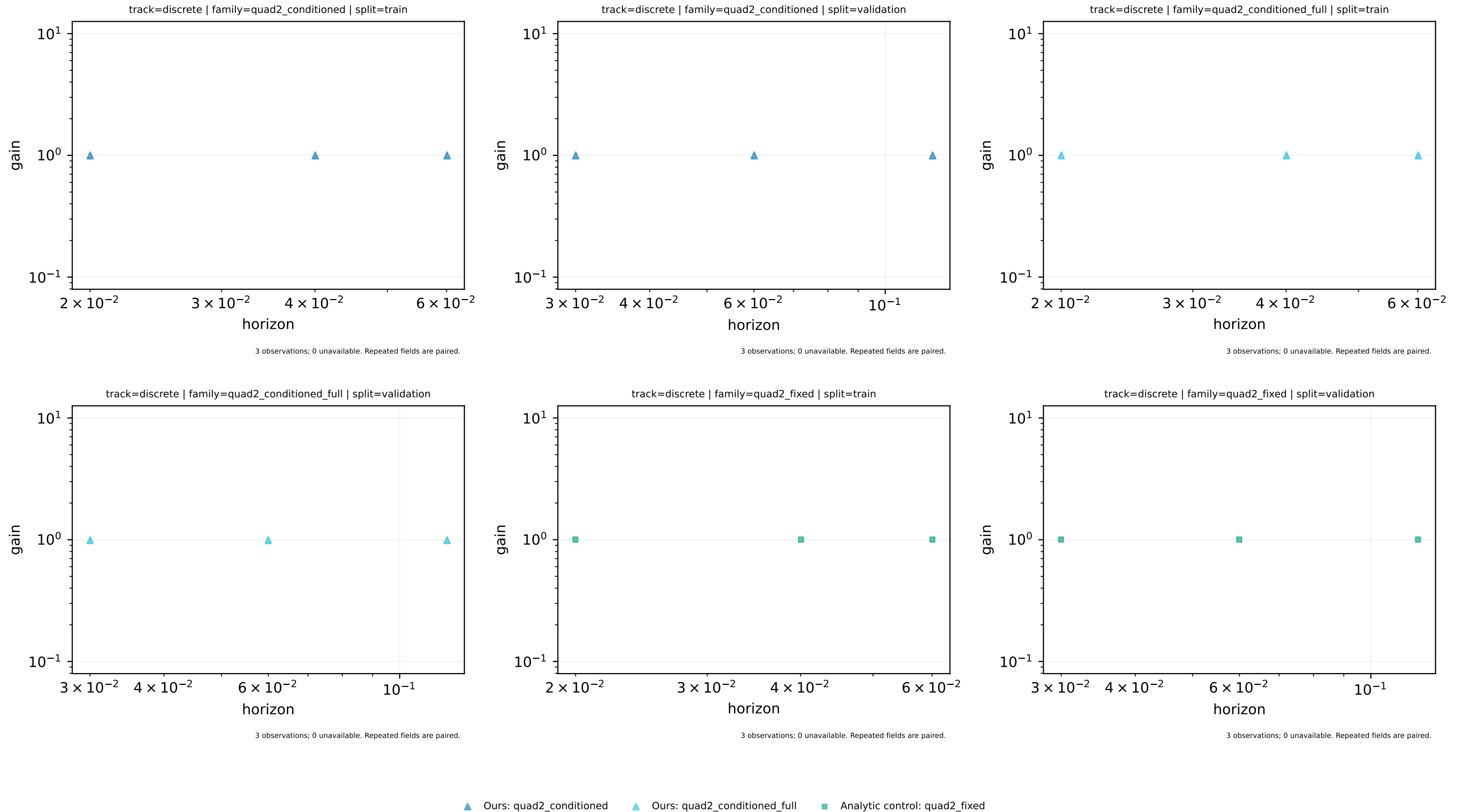
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
 METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
 SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
 Similar effective gains do not prove parameter identifiability.

Learned effective gain response (6)



HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
 METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
 SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
 Similar effective gains do not prove parameter identifiability.

Learned effective gain response (7)



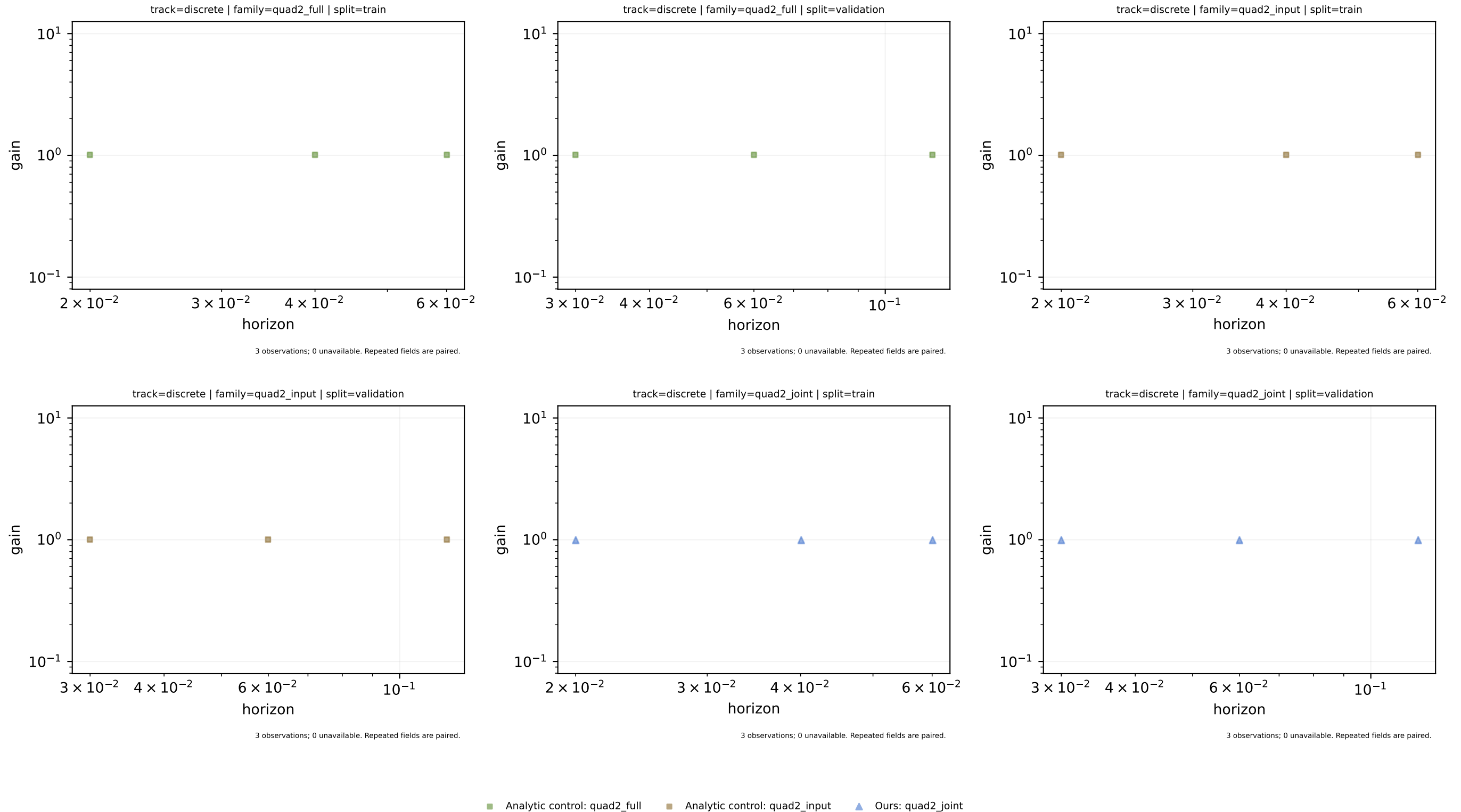
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.

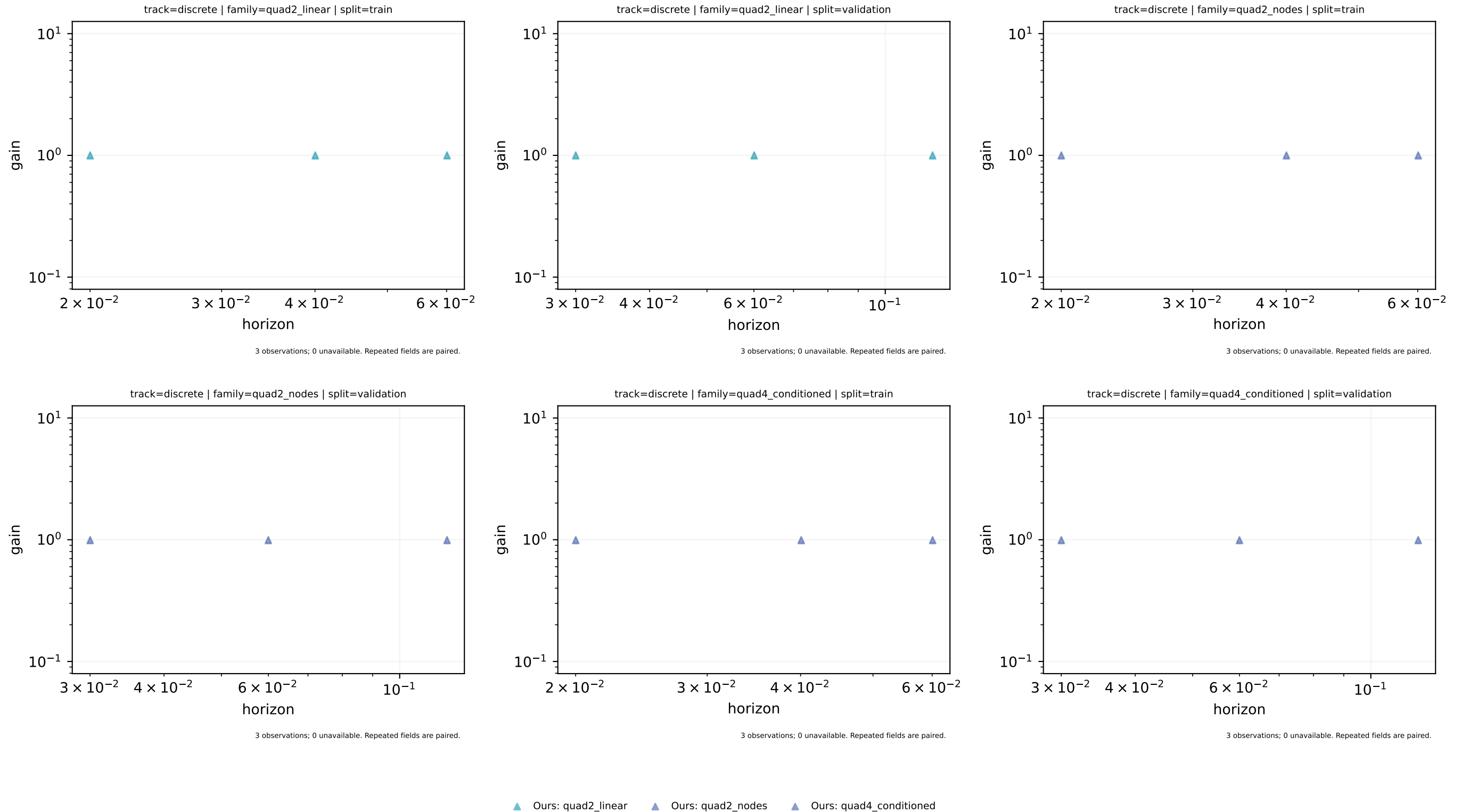
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (8)



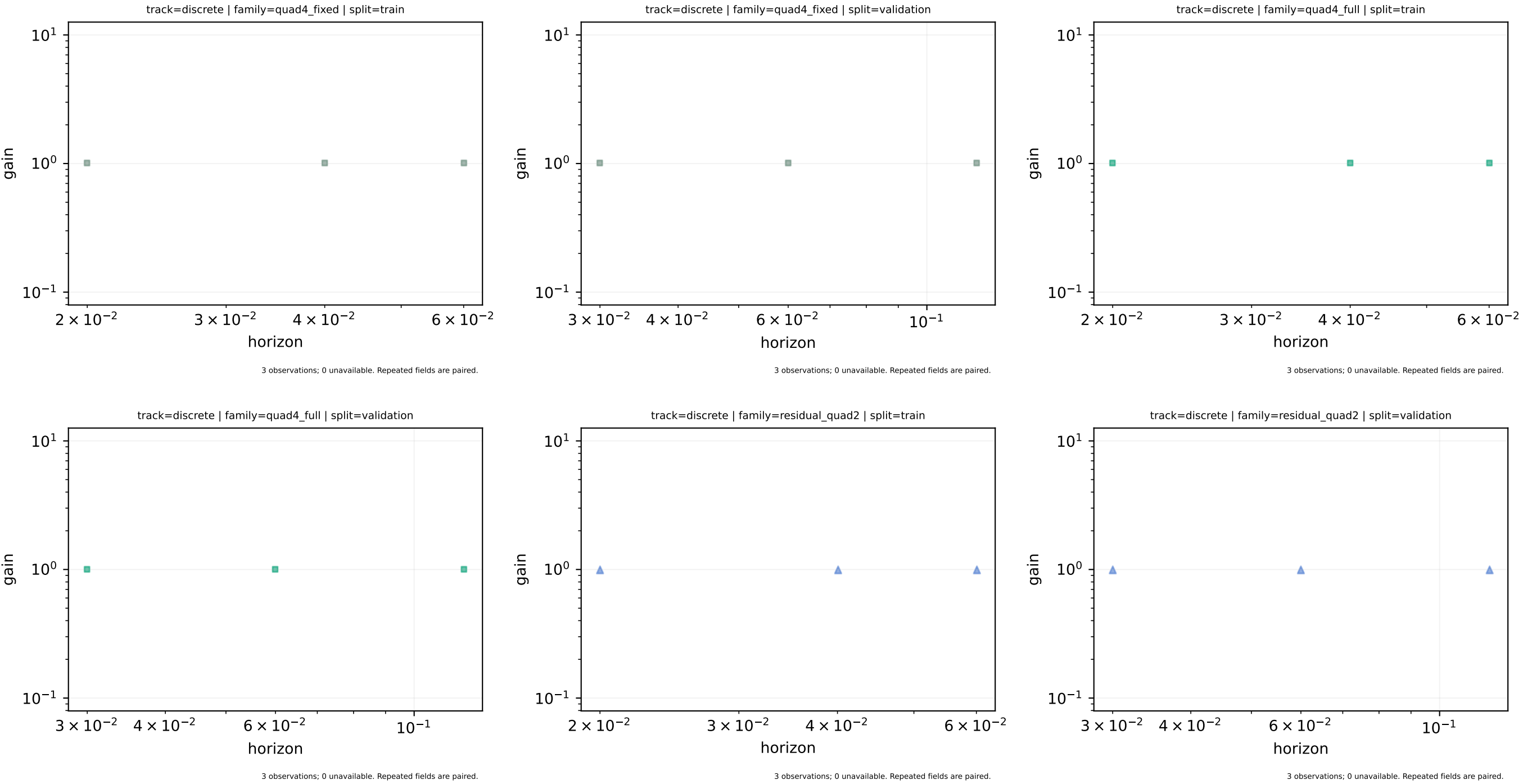
HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (9)



HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
Similar effective gains do not prove parameter identifiability.

Learned effective gain response (10)



■ Analytic control: quad4_fixed ■ Analytic control: quad4_full ▲ Ours: residual_quad2

HOW TO READ: Gain has no universal higher-is-better direction. Compare variation across times and states with the matched fitted/analytic controls.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: First declared training/validation state probes use no reference truth. Band gains remain separate observations, not an averaged gain; raw response features, nodes and weights remain source-linked.
Similar effective gains do not prove parameter identifiability.

Learned parameters and identifiability evidence

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```

All 46 exact parameter reports in learned_parameters.csv; first 18 shown here.

HOW TO READ: Parameters are not inherently better when larger or smaller. Compare effective responses and matched ablations.
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.
SCOPE: Exact learned node/weight/feature responses are retained. Several parameterizations can realize the same effective correction; no identifiability proof follows from a successful fit.

Protected high risk computational and scientific outcomes

prototype_id=X01

Method	Job	Math	Gap	Utility	Measured
exact_cached_pairs	COMPLETED	GOOD	GOOD	NA	queries=NA
uncached_pairs	COMPLETED	GOOD	GOOD	NA	queries=NA
compact_temporal_encoding	COMPLETED	GOOD	GOOD	BAD	queries=2,669
classical_dense_output	COMPLETED	GOOD	GOOD	NA	queries=NA
gauss_two	COMPLETED	GOOD	GOOD	NA	queries=NA
gauss_four	COMPLETED	GOOD	GOOD	NA	queries=NA
fft_gauss_four	COMPLETED	GOOD	GOOD	NA	queries=NA

prototype_id=X02

Method	Job	Math	Gap	Utility	Measured
instantaneous_fit	COMPLETED	GOOD	NA	NA	
memory_persistence	COMPLETED	GOOD	GOOD	NA	
memory_fitted	COMPLETED	GOOD	GOOD	NA	
full_state_instantaneous_oracle	COMPLETED	GOOD	NA	NA	

prototype_id=X03

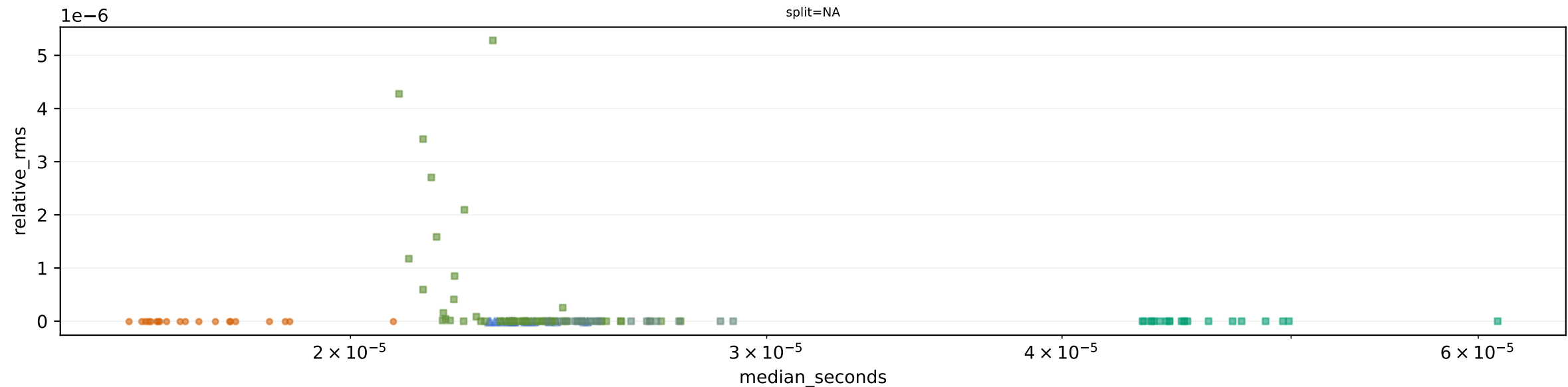
Method	Job	Math	Gap	Utility	Measured
full_pairs	COMPLETED	GOOD	GOOD	NA	error=0
deterministic_l1_tail	COMPLETED	GOOD	GOOD	BAD	error=0.00332
fitted_rank_rule	COMPLETED	GOOD	GOOD	BAD	error=2.27e-06
phase_information_audit	COMPLETED	GOOD	NA	NA	

HOW TO READ: Job COMPLETED means execution finished. Math/gap/utility GOOD means applicable checks met; BAD means a failed requirement; NA means unmeasured or inapplicable. A representation can pass while cost fails.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Every prototype and negative check remains in raw data. Break-even queries are measured build/query scenarios, not a demonstrated deployment benefit. New to TDN is not novelty in the literature; CPU diagnostics establish no GPU or full-solver superiority.

Amortized temporal query workload



140 observations; 0 unavailable. Repeated fields are paired.

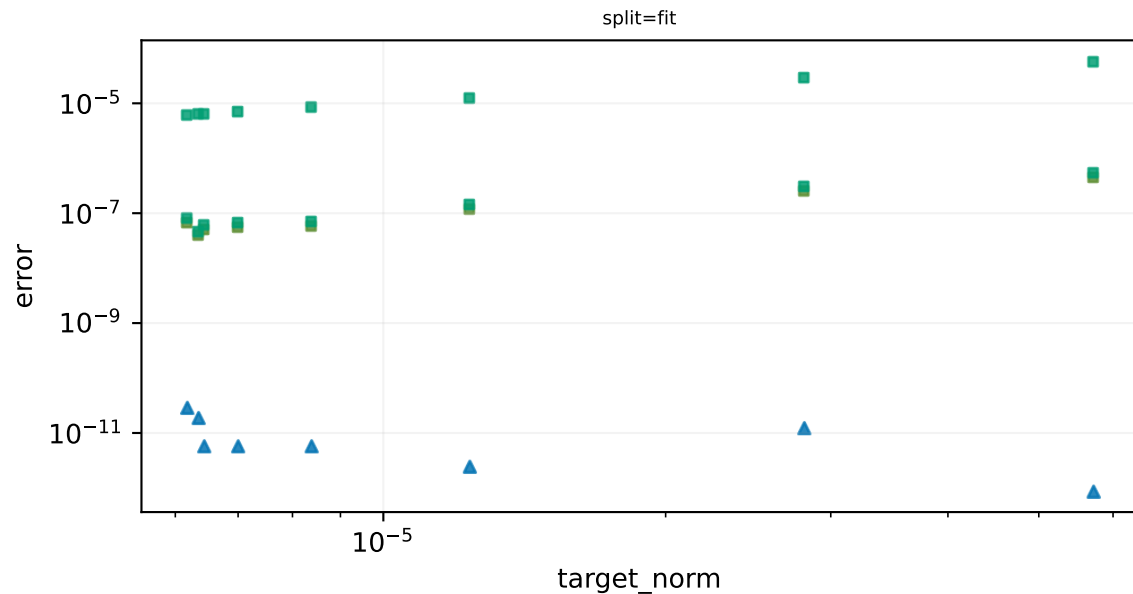
- Theirs: classical_dense_output
- ▲ Ours: compact_temporal_encoding
- Analytic control: exact_cached_pairs
- Analytic control: fft_gauss_four
- Analytic control: gauss_two
- Analytic control: gauss_four
- Analytic control: uncached_pairs

HOW TO READ: Smaller error is better; less complete work is better at equal accuracy. Feature/selection costs belong in complete time.

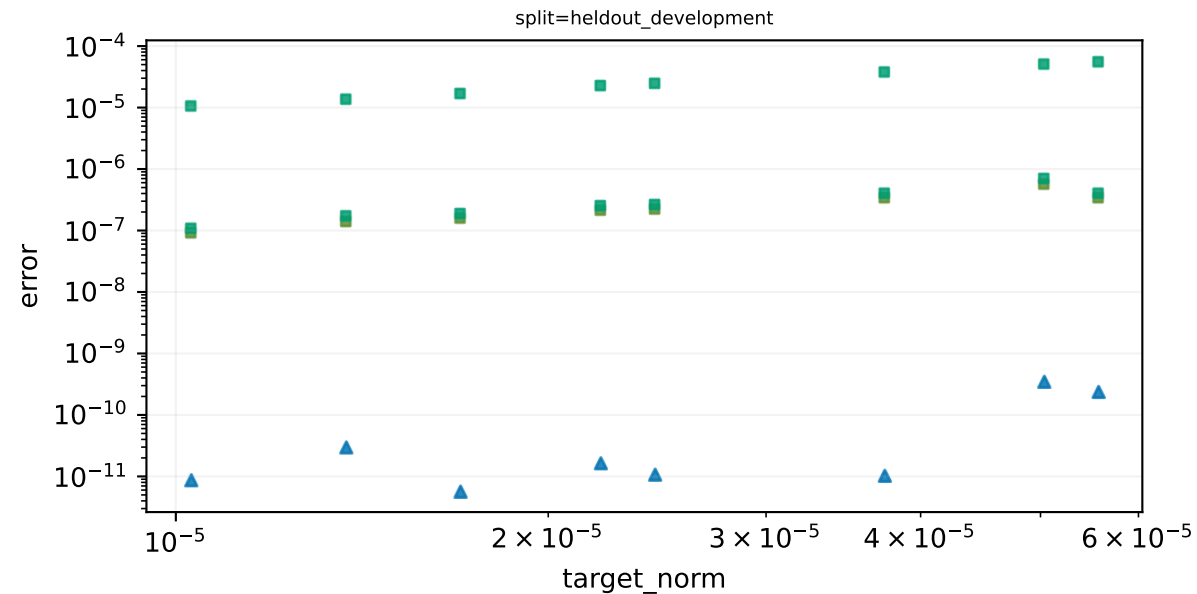
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Coarse closure and unresolved information



64 observations; 0 unavailable. Repeated fields are paired.



64 observations; 0 unavailable. Repeated fields are paired.

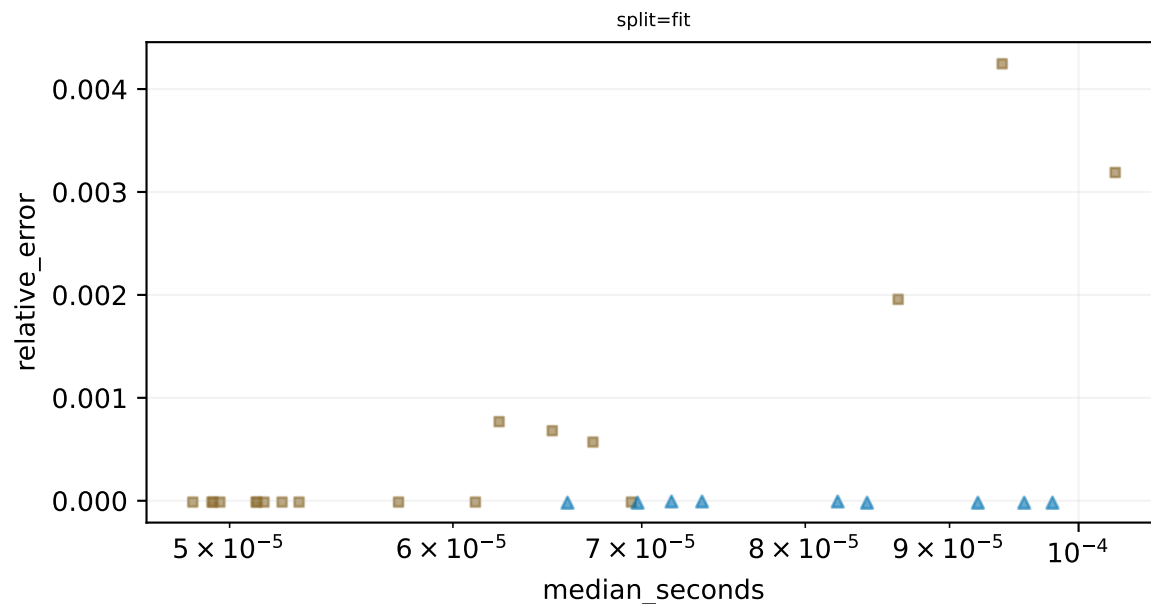
■ Analytic control: full_state_instantaneous_oracle
 ▲ Ours: memory_fitted
 ■ Analytic control: memory_persistence
■ Analytic control: instantaneous_fit

HOW TO READ: Smaller error is better; less complete work is better at equal accuracy. Feature/selection costs belong in complete time.

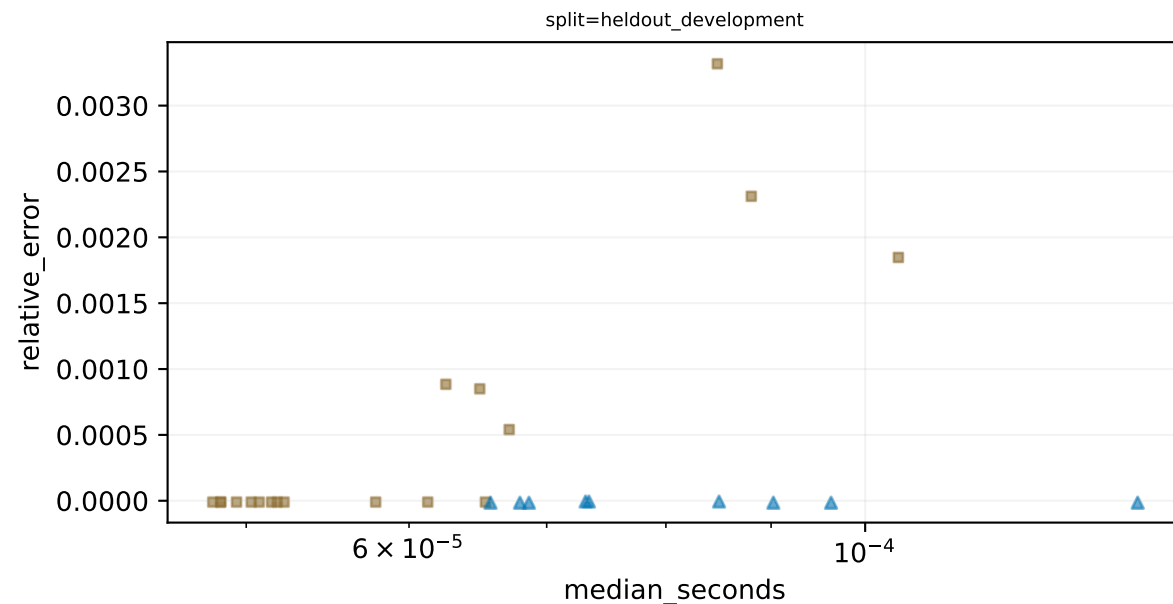
METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Selecting work before expensive computation



27 observations; 0 unavailable. Repeated fields are paired.



27 observations; 0 unavailable. Repeated fields are paired.

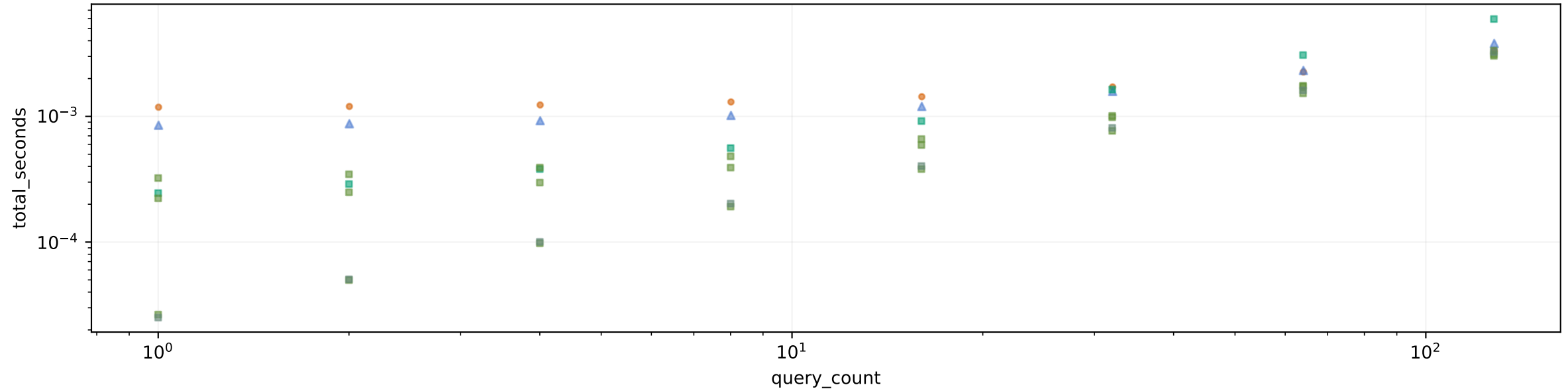
■ Analytic control: deterministic_l1_tail ▲ Ours: fitted_rank_rule ■ Analytic control: full_pairs

HOW TO READ: Smaller error is better; less complete work is better at equal accuracy. Feature/selection costs belong in complete time.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Only sealed prerequisites contribute science. Missing evidence is NA. Independent fields, not repeated schedules/grids/seeds, are statistical units. Observed learning ranges are not confidence intervals. No universal leaderboard.

Repeated query build and refresh amortization



56 observations; 0 unavailable. Repeated fields are paired.

- Theirs: classical_dense_output
- Ours: compact_temporal_encoding
- Analytic control: exact_cached_pairs
- Analytic control: fft_gauss_four
- Analytic control: gauss_four
- Analytic control: gauss_two
- Analytic control: uncached_pairs

HOW TO READ: Lower total time is better for the same number of requested horizons. Right means more queries.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Measured build plus query count times median query cost; a scenario calculation, not an observed batched execution. Reuse requires fixed state/operator/horizon range; state refresh costs remain included.

Amortization and deployment economics

NA: no positive measured matched-accuracy margin with known offline cost. No invented amortization or monetary price.

HOW TO READ: Fewer break-even queries is better if both solvers satisfy the same accuracy. No positive margin means no amortization.

METHODS: Ours uses triangles; Theirs uses circles; analytic controls use squares. Local FNO implementations are not published-paper reproductions. Colors identify methods, not quality.

SCOPE: Training-only lower bound when computable; reference generation, tuning, failures, verification, rejection and fallback can increase cost. Money remains NA without an explicit rate.